STRATOS S210 Series

S210-X22RQ

Two-Socket High Memory
2U Rackmount Server

Technical Guide

Version: 1.0.2

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Conventions

Several different typographic conventions are used throughout this manual. Refer to the following examples for common usage.

Bold type face denotes menu items, buttons and application names.

Italic type face denotes references to other sections, and the names of the folders, menus, programs, and files.

<**Enter**> type face denotes keyboard keys.

WARNING!
Warning information appears before the text it references and should not be ignored as the content may prevent damage to the device.

CAUTION!
CAUTIONS APPEAR BEFORE THE TEXT IT REFERENCES, SIMILAR TO NOTES AND WARNINGS. CAUTIONS, HOWEVER, APPEAR IN CAPITAL LETTERS AND CONTAIN VITAL HEALTH AND SAFETY INFORMATION.

Important:
Indicates information that is important to know for the proper completion of a procedure, choice of an option, or completing a task.

Note:
Highlights general or useful information and tips.

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Acronyms

TERM	DEFINITION
A/D	Analog to Digital
ACPI	Advanced Configuration and Power Interface
ASF	Alerting Standard Forum
Asserted	Active-high (positive true) signals are asserted when in the high electrical state (near power potential). Active-low (negative true) signals are asserted when in the low electrical state (near ground potential).
BIOS	Basic Input/Output System
BIST	Built-In Self Test
BMC	At the heart of the IPMI architecture is a microcontroller called the Baseboard management controller (BMC)
Bridge	Circuitry connecting one computer bus to another, allowing an agent on one to access the other
BSP	Bootstrap processor
Byte	8-bit quantity
CLI	Command Line Interface
CMOS	In terms of this specification, this describes the PC-AT compatible region of battery-backed 128 bytes of memory, which normally resides on the baseboard
CPU	Central Processing Unit
Deasserted	A signal is deasserted when in the inactive state. Active-low signal names have "_L" appended to the end of the signal mnemonic. Active-high signal names have no "_L" suffix. To reduce confusion when referring to active-high and active-low signals, the terms one/zero, high/low, and true/false are not used when describing signal states.
DTC	Data Transfer Controller
EEPROM	Electrically Erasable Programmable Read-Only Memory
EMP	Emergency Management Port
FRU	Field Replaceable Unit
GB	1024 MB.
GPIO	General Purpose Input/Out
HSC	Hot-Swap Controller
Hz	Hertz (1 cycle/second)
I^2C	Inter-Integrated Circuit bus
IANA	Internet Assigned Numbers Authority
IBF	Input buffer
ICH	I/O Controller Hub
ICMB	Intelligent Chassis Management Bus
IERR	Internal Error
IP	Internet Protocol
IPMB	Intelligent Platform Management Bus

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IPMI	Intelligent Platform Management Interface
ITP	In-Target Probe
KB	1024 bytes.
KCS	Keyboard Controller Style
KVM	Keyboard, Video, Mouse
LAN	Local Area Network
LCD	Liquid Crystal Display
LCT	Lower Critical Threshold
LED	Light Emitting Diode
LNCT	Lower Non-Critical Threshold
LNRT	Lower Non-Recoverable Threshold
LPC	Low Pin Count
LSI	Large Scale Integration
LUN	Logical Unit Number
MAC	Media Access Control
MB	1024 KB
MD2	Message Digest 2 – Hashing Algorithm
MD5	Message Digest 5 – Hashing Algorithm – Higher Security
Ms	Milliseconds
Mux	Multiplexer
NIC	Network Interface Card
NMI	Nonmaskable Interrupt

NM	Node Management
OBF	Output buffer
OEM	Original Equipment Manufacturer
Ohm	Unit of electrical resistance
PDB	Power Distribution Board
PEF	Platform Event Filtering
PEP	Platform Event Paging
PERR	Parity Error
POH	Power-On Hours
POST	Power-On Self Test
PWM	Pulse Width Modulation
RAC	Remote Access Card
RAM	Random Access Memory
RMCP	Remote Management Control Protocol
ROM	Read Only Memory
RTC	Real-Time Clock. Component of the chipset on the base-board.
RTOS	Real Time Operation System
SCI	Serial Communication Interface
SDC	SCSI Daughter Card
SDR	Sensor Data Record
SEEPROM	Serial Electrically Erasable Programmable Read-Only Memory

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SEL	System Event Log
SERR	System Error
SMBus	A two-wire interface based on the I^2C protocol. The SMBus is a low-speed bus that provides positive addressing for devices, as well as bus arbitration
SMI	Server Management Interrupt. SMI is the highest priority nonmaskable interrupt
SMM	Server Management Mode
SMS	Server Management Software
SNMP	Simple Network Management Protocol
SOL	Serial Over LAN
UART	Universal Asynchronous Receiver/Transmitter
UCT	Upper Critical Threshold
UDP	User Datagram Protocol
UNCT	Upper Non-Critical Threshold
UNRT	Upper Non-Recoverable Threshold
WDT	Watchdog Timer
Word	16-bit quantity

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Safety Information

Important Safety Instructions

Read all caution and safety statements in this document before performing any of the instructions.

Warnings

Heed safety instructions: Before working with the server, whether using this manual or any other resource as a reference, pay close attention to the safety instructions. Adhere to the assembly instructions in this manual to ensure and maintain compliance with existing product certifications and approvals. Use only the described, regulated components specified in this manual. Use of other products / components will void the UL listing and other regulatory approvals of the product and will most likely result in non-compliance with product regulations in the region(s) in which the product is sold.

System power on/off: The power button DOES NOT turn off the system AC power. To remove power from system, you must unplug the AC power cord from the wall outlet. Make sure the AC power cord is unplugged before opening the chassis, adding, or removing any components.

Hazardous conditions, devices and cables: Hazardous electrical conditions may be present on power, telephone, and communication cables. Turn off the server and disconnect the power cord, telecommunications systems, networks, and modems attached to the server before opening it. Otherwise, personal injury or equipment damage can result.

Electrostatic discharge (ESD) and ESD protection: ESD can damage drives, boards, and other parts. We recommend that you perform all procedures in this chapter only at an ESD workstation. If one is not available, provide some ESD protection by wearing an antistatic wrist strap attached to chassis ground any unpainted metal surface on the server when handling parts.

ESD and handling boards: Always handle boards carefully. They can be extremely sensitive to electrostatic discharge (ESD). Hold boards only by their edges. After removing a board from its protective wrapper or from the server, place the board component side up on a grounded, static free surface. Use a conductive foam pad if available but not the board wrapper. Do not slide board over any surface.

Installing or removing jumpers: A jumper is a small plastic encased conductor that slips over two jumper pins. Some jumpers have a small tab on top that can be gripped with fingertips or with a pair of fine needle nosed pliers. If the jumpers do not have such a tab, take care when using needle nosed pliers to remove or install a jumper; grip the narrow sides of the jumper with the pliers, never the wide sides. Gripping the wide sides can damage the contacts inside the jumper, causing intermittent problems with the function controlled by that jumper. Take care

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to grip with, but not squeeze, the pliers or other tool used to remove a jumper, or the pins on the board may bend or break.

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Revision History

Refer to the table below for the updates made to this manual.

DATE	CHAPTER	UPDATES
March, 2014	Specification	Ivy-Bridge plafform support
	BIOS	Adding iSCSI and other sections
	BMC	Adding SNMP and Quanta SMASH

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Version 1.0.2 / March, 2014

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About the Server

Chapter 1

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1.1 Introduction

This manual is written for system technicians who are responsible for troubleshooting, upgrading, and repairing the server chassis. This document provides an overview of the hardware features of the chassis, troubleshooting information, and instructions on how to add and replace components of the multi-node server series.The document also provides information on the BIOS, and Baseboard Management Controller (BMC).

For the latest version of this manual, see www.QauntaQCT.com.

System Features

The following is a list of the system main features:

- **Chipset**: Intel® C602 (Patsburg A).
- **Processor (2x)**: Intel® Xeon® E5-2600 / E5-2600 V2 series processors.
- **PCIe**:
  - (1) PCIe x16 G3 low profile riser slot (optional) or (2) PCIe x8 G3 low profile riser slots (optional), (4) PCI x8 G3 low profile riser slots and (1) PCIe x4 G3 low profile riser slot
  - (1) PCIe x8 G3 Network Mezzaine card slot and (1) PCIe x8 G3 Quanta LSISAS/RAID Mezzanie slot or (2) PCIe x8 G3 LP-MD2 riser slots

Note:
A second processor must be installed (in processor slot 2) to activate the expansion slot and to allow the mezzanine card in there to function properly.

Note:
Expansion cards descriebed in the document are optional components.

- **Memory**: Up to twenty-four DIMM slots available. DDR3 800/1066/1333/1600/1866 MHz ECC RDIMM/LR-DIMM is supported.
- **Storage**: Up to twenty-four 2.5" SATA/SAS hot-plug drives or twelve 3.5" SATA/SAS hot-plug drives. As an option, one SATADOM and two 2.5” hot plug SSD.
- **Network**: Two Intel® Powerville I350GbE RJ45 ports or (2) Intel® Twinville X540 10G BASE T RJ45 ports

Table 1-1: Specification

Chassis type	X2 (2U chassis)

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Table 1-1: Specification (Continued)

Size (L x W x H)	• 720 mm x 430 mm x 87.5 mm • 28.34" x 16.92" x 3.4"
MB size (W x L)	• 487.17mm x 419.1mm • 19.18" x 16.5"
Processor	(2) Intel® Xeon® processor E5-2600 / E5-2600 V2 family (135W)
Chipset	Intel® C602 (Patsburg A)
SAS Controller	**Intel® SAS controller <SAS 3G>** • Intel® C602 (Patsburg A) upgrade ROM #1 (optional) • Intel® C602 (Patsburg A) upgrade ROM #2 (optional) • Intel® C602 (Patsburg A) upgrade ROM #5 (optional) • Intel® C602 (Patsburg A) upgrade ROM #6 (optional) or **LSI SAS controller <SAS 6G>** • Quanta LSISAS 2008 Mezzanine card (optional) • Quanta LSISAS 2108 Mezzanine card (optional) • Quanta LSISAS 2208 Mezzanine card (optional) • Quanta LSISAS 2308 Mezzanine card (optional)
System Memory	(24) DDR3 800/1066/1333/1600/1866 MHz ECC RDIMM/LR-DIMM slots up to 768 GB

Table 1-1: Specification (Continued)

Storage	• (24) 2.5" SATA/SAS hot-plug drives or • (12) 3.5" SATA/SAS hot-plug drives • (1) SATADOM (optional) • (2) 2.5" hot-plug SSD (optional)
HDD Backplane	• 1 to 1 or • expander
PCIe Expansion Slot	• (1) PCIe x16 G3 low profile riser slot (optional) or (2) PCIe x8 G3 low profile riser slots (optional), (4) PCIe x8 G3 low profile riser slots and (1) PCIe x4 G3 low profile riser slot • (1) PCIe x8 G3 Network Mezzaine card slot and (1) PCIe x8 G3 Quanta LSISAS/RAID Mezzanie slot or (2) PCIe x8 G3 LP-MD2 riser slots

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Table 1-1: Specification (Continued)

Software RAID Options	**Intel®SW RAID** ● Intel® RSTe SATA RAID 0/1/10 (AHCI HDD) ● Intel® C602 (Patsburg A) upgrade ROM #1 RAID 0/1/10 for SCU (optional) ● Intel® C602 (Patsburg A) upgrade ROM #2 RAID 0/1/10/5 for SCU (optional) ● Intel® C602 (Patsburg A) upgrade ROM #5 RAID 0/1/10 for SCU (optional) ● Intel® C602 (Patsburg A) upgrade ROM #6 RAID 0/1/10/5 for SCU (optional) ● Intel® C602 (Patsburg A) upgrade ROM #9 SATA RAID 0/1/10/5 for SCU (optional) or **LSI SW RAID** ● Quanta LSISAS 2008 mezzanine card for RAID 0/1/10 (optional) ● Quanta LSISAS 2308 mezzanine card for RAID 0/1/10 (optional)
Hardware RAID Options	● Quanta LSISAS2108 PD-32 mezzanine card for RAID 0/1/10/5 (optional) ● Quanta LSISAS2108 PD-32 mezzanine card + LSI RAID 6 Key for RAID 0/1/10/5/6/50/60 (optional) ● Quanta LSISAS2208 PD-32 mezzanine card for RAID 0/1/10/5 (optional) ● Quanta LSISAS2208 PD-32 mezzanine card + LSI RAID 6 Key for RAID 0/1/10/5/6/50/60 (optional)

Table 1-1: Specification (Continued)

Network Interface	● (2) Intel® Powerville I350 GbE RJ45 ports or ● (2) Intel® Twinville X540 10G BASE T RJ45 ports
Management Port	(1) Dedicated GbE RJ45 management port
Integrated Graphics BMC	Aspeed AST2300 8 MB DDR3 Video memory
Rear I/O Interface	● (2) USB 2.0 ports ● (1) VGA port ● (1) RS232 serial Port ● (2) GbE or 10G BASE-T RJ45 ports ● (1) GbE RJ45 management port ● (1) ID button with LED
Power Supply	● (1) 1100W high efficiency PSU, 100-240VAC 50/60Hz ● (1) 1100W high efficiency redundant PSU, 100-240VAC 50/60Hz (optional)
TPM	Yes (optional)
RoHS	Yes
Systems Management	● IPMI v2.0 compliant ● On board “KVM over IP” support

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Package Contents

The following list includes the package components:

- (1) 2U chassis system
- (2) Processor heat sinks
- (1-2) Power supplys (Optional)
- (1-2) Power cords (Optional)
- (1) CD (user manual included)
- (1) Rail kit

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1.2 A Tour of the System

The system is available as a 2.5” or a 3.5” form factor. The following illustrations show the major components parts of these two variants.

2.5” HDD System

The system features a 2.5” hard drive arrangment as seen in the following image.

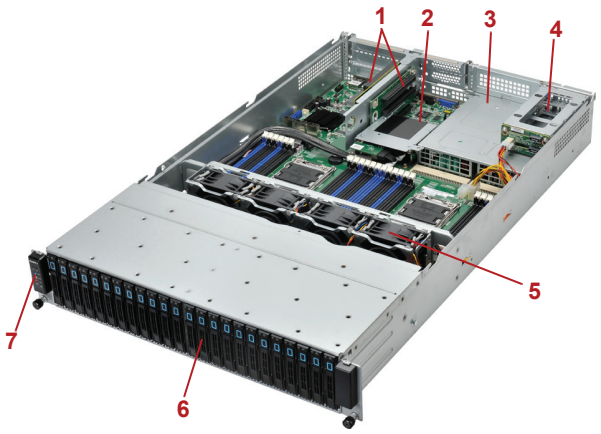

Figure 1-1. 2.5” HDD System Component Overview

Table 0-1. System Component Description

NO.	ITEM	DESCRIPTION
1.	Riser Assembly	• (1) PCIe x16 G3 low profile riser slot or (1) PCIe x8 G3 low profile riser slots, (1) PCI x8 G3 low profile riser slots and (1) PCIe x4 G3 low profile riser slots • (2) PCIe x8 G3 low Profile riser slots and (1) PCIe x4 G3 low profile riser slots • (1) PCIe x8 G3 10GbE SFP+ Mezzaine card slot and (1) PCIe x8 G3 Quanta LSISAS/RAID Mezzanie slot or (2) PCIe x8 G3 LP-MD2 riser slots
2.	LSISAS Mezzanine Module	Quanta LSISAS 2008 mezzanine module
3.	PSU Cage	(1) 1100W high efficiency PSU, 100-240VAC (Default)
4.	SSD	(2) 2.5" hot-plug SSD (2*6 Gb AHCI) (optional)
5.	Fans	(4) 80 mm fan modules
6.	HDD Cage	(24) 2.5” SATA/SAS hot plug drives
7.	Front Panel	Controls and LEDs for system management

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3.5” HDD System

The system features a 3.5” hard drive arrangment as seen in the following image.

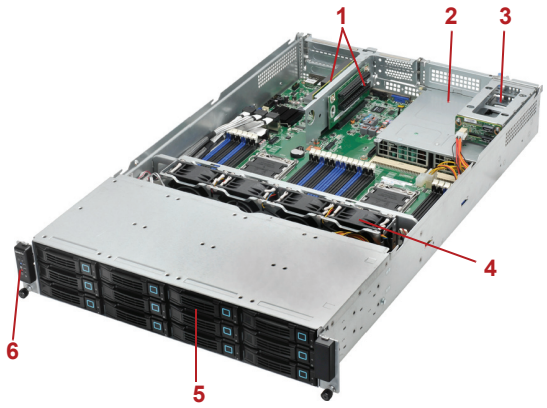

Figure 1-2. 3.5” HDD System Component Description

Table 7-1.

NO.	ITEM	DESCRIPTION
1.	Riser Assembly	• (1) PCIe x16 G3 low profile riser slot or (1) PCIe x8 G3 low profile riser slots, (1) PCI x8 G3 low profile riser slots and (1) PCIe x4 G3 low profile riser slots • (2) PCIe x8 G3 low Profile riser slots and (1) PCIe x4 G3 low profile riser slots • (1) PCIe x8 G3 10GbE SFP+ Mezzaine card slot and (1) PCIe x8 G3 Quanta LSISAS/RAID Mezzanie slot or (2) PCIe x8 G3 LP-MD2 riser slots
2.	PSU Cage	(1) 1100W high efficiency PSU, 100-240VAC (Default)
3.	SSD	(2) 2.5" hot-plug SSD (2*6 Gb AHCI) (optional)
4.	Fans	(4) 80 mm fan modules
5.	HDD Cage	(12) 3.5” SATA/SAS hot plug drives
6.	Front Panel	Controls and LEDs for system management

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System Front Features

3.5” HDD System Front Features

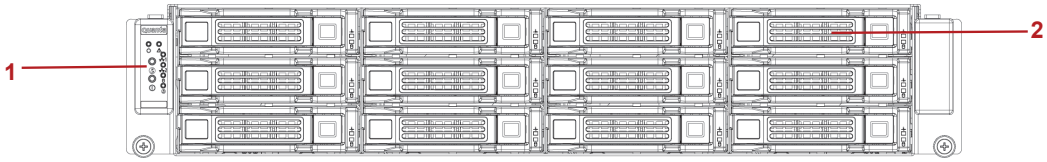

Figure 1-3. 3.5” HDD System Front Features

Table 1-2: 3.5” HDD System Component Description

NO.	ITEM	DESCRIPTION
1.	Front control panel	Node control system
2.	HDD bays	3.5” HDD array

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2.5” HDD System Front Features

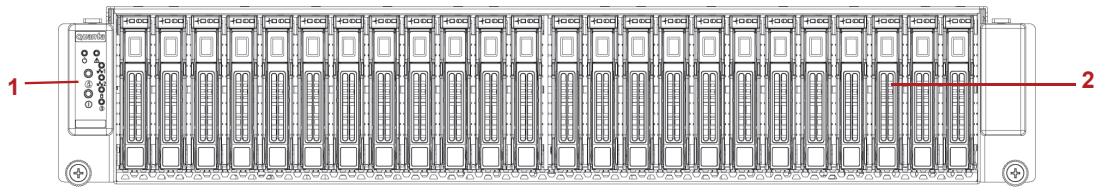

Figure 1-4. 2.5” HDD System Front Features

Table 1-3: 2.5” HDD System Component Description

NO.	ITEM	DESCRIPTION
1.	Front control panel	Node control system
2.	HDD bays	2.5” HDD array

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Front Control Panel

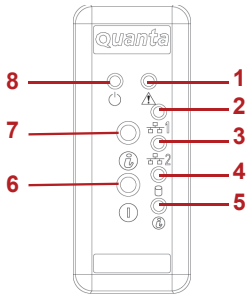

Figure 1-5. Front Control Panel Features

Table 1-4: Front Panel Definition

ITEM	ICON	NAME	DESCRIPTION
1.	⚠	Fault LED	Provides critical and non-critical failure notification
2.	品	LAN1 LED	LAN access
3.	品	LAN2 LED	LAN access
4.	🛢	HDD Activity LED	Hard disk drive access

Table 1-4: Front Panel Definition (Continued)

ITEM	ICON	NAME	DESCRIPTION
5.		ID LED	Lights for system identification
6.	⏻	Power Button	Based on System Off, Push Button to PSU and System on
			Based on System Off, Push Button to PSU and System off
7.		Identification Button	Push to activate ID LED
8.	⏻	Power LED	Green: ON, system power on OFF: system off

Table 1-5: LED Function and Behavior

NAME	COLOR	CONDITION	DESCRIPTION
Power LED	Green	ON	System power on
		OFF	System power off
Identification	Blue	ON	Unit selected for identification
		OFF	No identification requested

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Table 1-5: LED Function and Behavior (Continued)

NAME	COLOR	CONDITION	DESCRIPTION
Fault LED	Amber	Blinking	Critical Failure: critical fan, voltage, temperature state.
			Non-Critical Failure: non-critical fan, voltage, temperature state, CPU thermal trip.
		OFF	SEL Cleared
			DC Off
			Last pending warning or error has been de-asserted.
HDD Activity	Green	Blinking	Hard disk drive access (only on board SATA port)
		OFF	No access (non-SAS)
LAN1 LED	Green	ON	Link
	Green	Blinking	LAN Access (off when there is traffic)
LAN2 LED	Green	ON	Link
	Green	Blinking	LAN Access (off when there is traffic)

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System Rear View

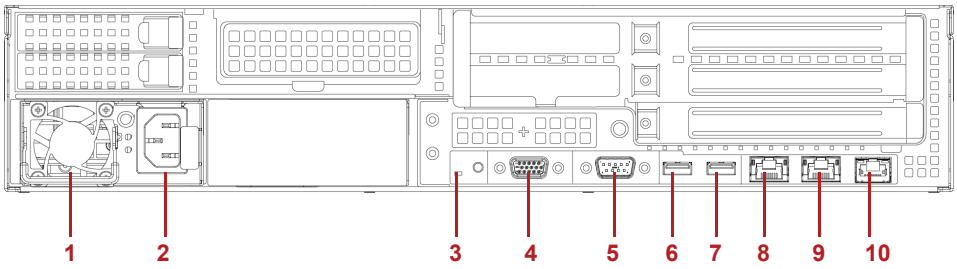

Figure 1-6. System Rear View

Table 1-6: System I/O Features

ITEM	ICON	NAME	DESCRIPTION
1.		Power Supply Unit (PSU)	Single power supply unit.
2.		PSU Connector	Insert power supply cord.
3.		ID LED	Lights for system identification
4.		VGA port	Connect a monitor to this port
5.		Console port	
6.		USB port	USB ports (2.0 compliant)

Table 1-6: System I/O Features (Continued)

ITEM	ICON	NAME	DESCRIPTION
7.		USB port	USB ports (2.0 compliant)
8.		LAN2 LED	LAN access
9.		LAN1 LED	LAN access
10.		Dedicated Management LAN Port	

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Table 1-7: Rear LED Function and Behavior

NAME		COLOR		CONDITION	DESCRIPTION
ID LED		Blue		ON	Unit selected for identification
		-		OFF	No identification requested
LAN1 LED (Right)	Link/ Act	Green		ON	LAN Link
		Green	Black	Blinking	LAN Access (off when there is traffic)
		-		OFF	Disconnect
	Speed	Green		ON	Green, link speed is 1000Mbits/sec
		Amber		ON	Amber, link speed is 100Mbits/sec
		-		OFF	OFF, link speed is 10Mbits/sec

Table 1-7: Rear LED Function and Behavior

NAME		COLOR		CONDITION	DESCRIPTION
LAN2 LED (left)	Link/ Act	Green		ON	LAN Link
		Green	Black	Blinking	LAN Access (off when there is traffic)
		-		OFF	Disconnect
	Speed	Green		ON	Green, link speed is 1000Mbits/sec
		Amber		ON	Amber, link speed is 100Mbits/sec
		-		OFF	OFF, link speed is 10Mbits/sec
Service Port (LAN3) LED	Link/ Act	Green		ON	LAN Link
		Green	Black	Blinking	LAN Access (off when there is traffic)
		-		OFF	Disconnect
	Speed	Green		ON	Green, link speed is 1000Mbits/sec
		Amber		ON	Amber, link speed is 100Mbits/sec
		-		OFF	OFF, link speed is 10Mbits/sec

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Installing Hardware

Chapter 2

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2.1 Safety Measures

WARNING!
Always ask for assistance to move or lift the system.

WARNING!
Only perform troubleshooting as authorized by the product documentation, or as directed by a service and support team. Repairs not authorized by warranty may void the warranty and damage the system.

WARNING!
Always make sure to disconnect the system from the electrical source. Powering down the system DOES NOT insure the system is safe for repairs.

WARNING!
Server components and circuit boards are easily damaged by discharges of static electricity. Working on servers that are connected to a power supply can be extremely dangerous.

WARNING!
Always disconnect the server from the power outlet whenever you are working inside the server case.

WARNING!
Wear a grounded wrist strap. If none are available, discharge any personal static electricity by touching the bare metal chassis of the server case, or the bare metal body of any other grounded device.

WARNING!
Humid environments tend to have less static electricity than dry environments. A grounding strap is warranted whenever danger of static electricity exists.

WARNING!
Do not touch the components on the boards unless it is necessary to do so. Do not flex or stress circuit boards.

WARNING!
Leave all replacement components inside their static-proof packaging until you are ready to use them.

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2.2 Hard Drives

WARNING!
Do not operate the system without all hard drive trays inserted into the chassis. All hard drive bays must be occupied by either a hard drive or an empty hard drive tray. Hard drives may be removed while the system is operational but should be immediately replaced by another hard drive or an empty hard drive tray.

Removing a 2.5” Hard Drive

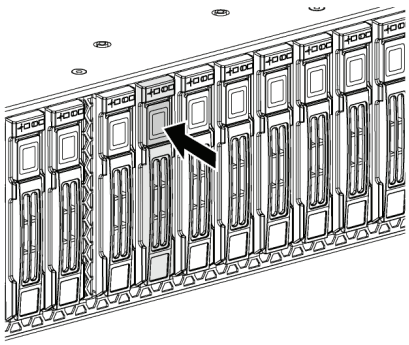

Figure 2-1. Opening the 2.5” Hard Drive Tray Handle

1. Press the tray handle button.

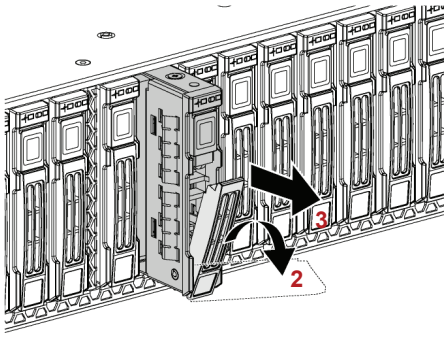

Figure 2-2. Removing a 2.5” Hard Drive Tray

2. Pull the HDD tray handle completely open.

WARNING!
Make sure the tray handle is in the fully open position.

3. Grasp the tray handle and pull the tray out of the system.

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Installing a 2.5” Hard Drive into a Hard Drive Tray

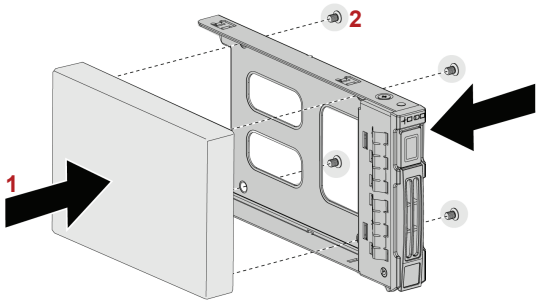

Figure 2-3. Installing a 2.5” HDD into a Tray

1. Place the hard drive into the hard drive tray.
2. Secure the hard drive with four (4) screws.

Installing a 2.5” Hard Drive

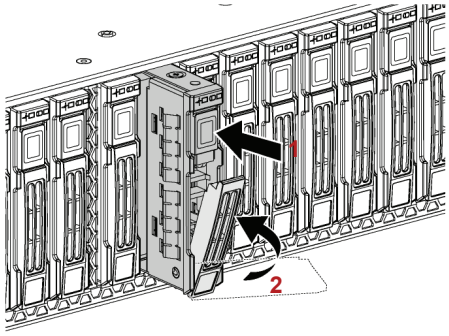

Figure 2-4. Installing a 2.5” HDD Drive

1. Insert the HDD tray into the system. Make sure that the tray is fully inserted.
2. Close the HDD tray handle.

WARNING!
Do not force the tray handle closed. If resistance is encountered check the hard drive is properly inserted and the hard drives on either side are properly inserted.

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Removing a 3.5" Hard Drive

Hard drives are grouped together so that each group of hard drives is fixed to a specific node.

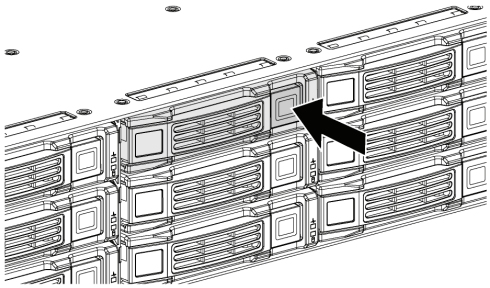

Figure 2-5. Opening the Hard Drive Tray Handle

1. Press the tray handle button.

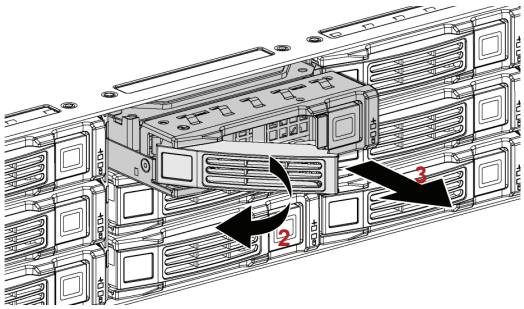

Figure 2-6. Removing the 3.5" Hard Drive Tray

2. Pull the hard drive tray handle completely open.

WARNING!
Make sure the hard drive tray handle is in the fully open position.

3. Grasp the tray handle and pull the tray from the system.

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Installing a 3.5” Hard Drive into a Hard Drive Tray

1. Place a 3.5” hard drive into the tray.
2. Replace four (4) screws.

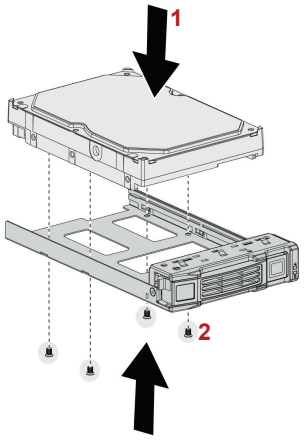

Figure 2-7. Installing a 3.5” HDD into a Tray

Installing a 3.5” Hard Drive

WARNING!
Do not force the tray handle closed. If resistance is encountered, check the hard drive is properly inserted and the hard drives on either side are properly inserted.

1. Insert the hard drive into the system. Make sure the hard drive is fully inserted.
2. Push the tray handle closed.

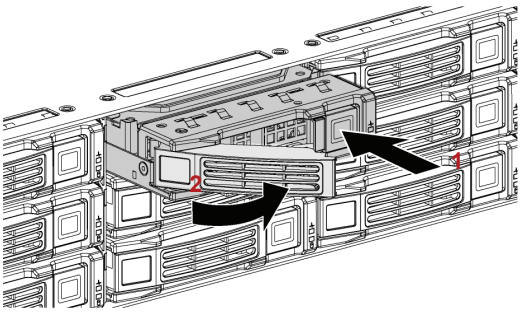

Figure 2-8. Installing a 3.5” Hard Drive

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2.3 Power Supply Unit

The server models support both single and redundant power supplies as shown in the following illustrations.

CAUTION!
DISCONNECT THE POWER SUPPLY UNIT FROM THE POWER SOURCE BEFORE REMOVING PSU. FAILURE TO DO SO COULD RESULT IN DAMAGE TO THE EQUIPMENT OR PERSONAL INJURY.

Note:
The redundant power supply unit can be replaced without shutting down the system.

Removing a Power Supply Unit

1. Pull the PSU handle up to the open position.

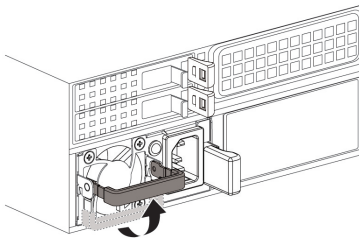

Figure 2-9. Opening the PSU Handle

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2. Press and hold the locking latch lever.
3. Pull the PSU from the system.

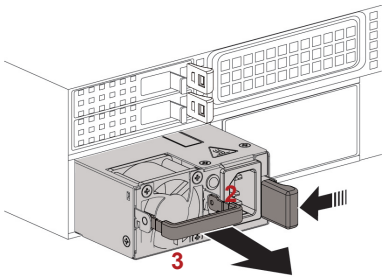

Figure 2-10. Removing the PSU

Installing a Power Supply Unit

Insert the power supply unit (PSU) into the system.

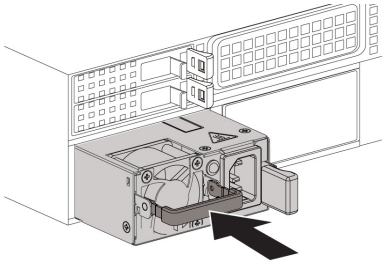

Figure 2-11. Installing a Power Supply Unit

Make sure the PSU is flush with the system and the side lever is locked in place.

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2.4 Top Cover

Removing the Top Cover

1. Turn off the system and any attached peripherals.
2. Unplug the AC power cables and disconnect all peripherals, LAN lines and any other cables.
3. Remove one (1) screw from the top cover.
4. Press the top cover release button and slide the top cover off the chassis (Figure 2-12).
5. Lift the top cover off the chassis.

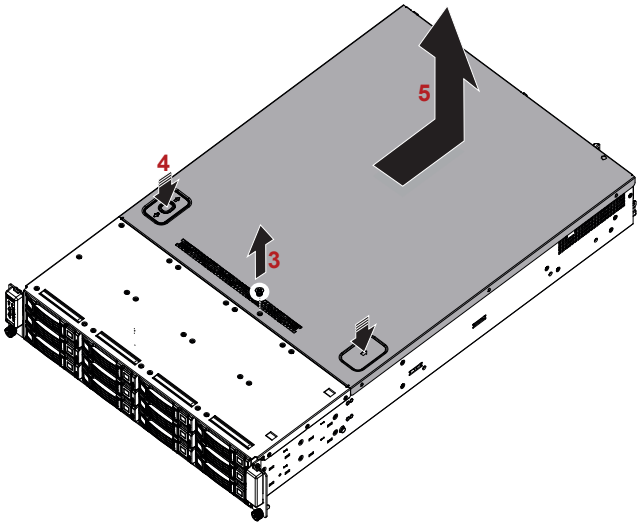

Figure 2-12. Removing the Top Cover

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Replacing the Top Cover

1. Place the top cover on the chassis.
2. Slide the top cover toward in place (Figure 2-13).
3. Secure the top cover with one (1) screw.

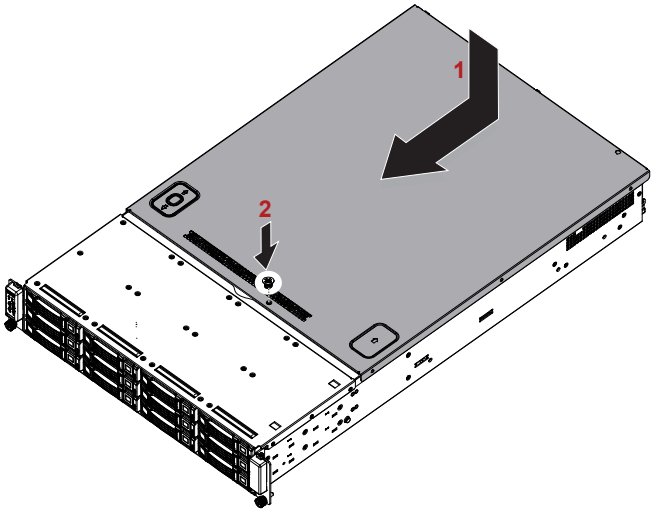

Figure 2-13. Closing the Top Cover

4. Attach peripherals, and plug in the AC power cables, LAN lines and any remaining cables.

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2.5 Fan Modules

Removing a Fan Module

1. Remove the top cover. See *Removing the Top Cover* on page 2-8.

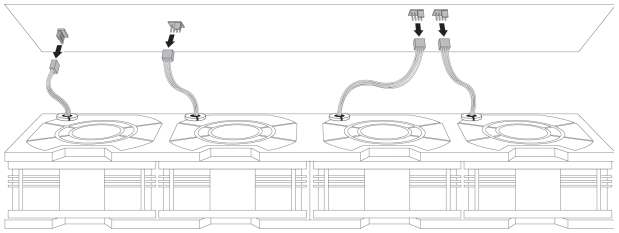

Figure 2-14. Disconnecting Fan Cables from Expander Board

2. Disconnect the four (4) fan module cables from the connectors on the HDD backplane.

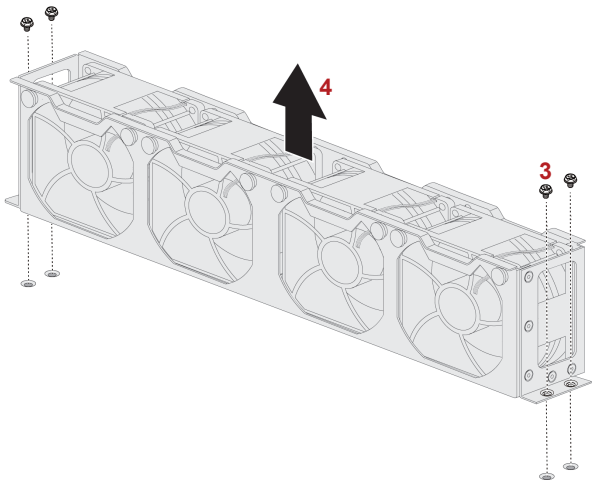

Figure 2-15. Removing the Fan Assembly

3. Loosen four (4) screws on the sides of the fan assembly bracket.
4. Remove the fan assembly from the chassis.

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Installing Fan Assembly

1. Remove the top cover. See *Removing the Top Cover* on page 2-8.
2. With module cables facing the HDD backplane, align the fan assembly to the screw holes on the chassis and install.
3. Lower the assembly in place.
4. Secure the assembly to the chassis with four (4) screws on both sides of the assembly bracket.

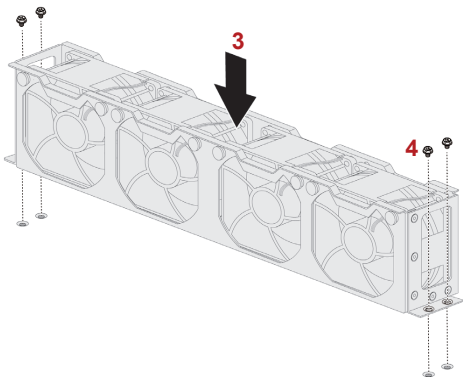

Figure 2-16. Installing the Fan Assembly

5. Connect the four (4) fan module cables to the connectors on the HDD backplane.

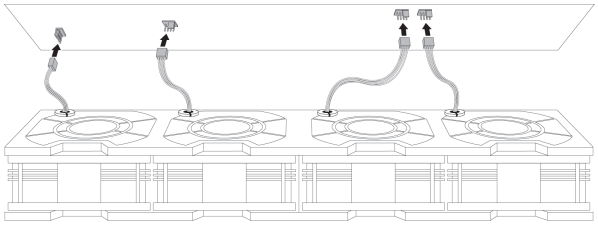

Figure 2-17. Connecting Fan Cables to Expander Board

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2.6 Processor Heat Sinks

Removing a Heat Sink

WARNING!
The heatsink remains hot after the system has been powered down. Allow sufficient time to cool before handling system components.

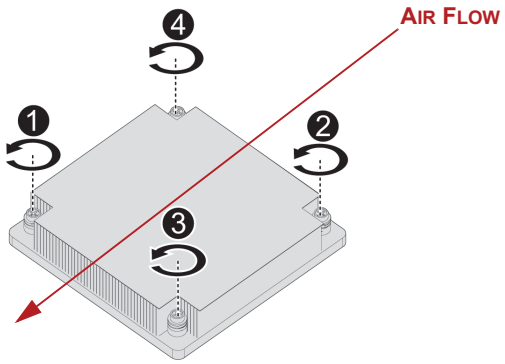

Figure 2-18. Removing the Heat Sink

1. Remove the top cover. See *Removing the Top Cover* on page 2-8.
2. Loosen the four (4) captive screws of the heat sink in sequence.
3. Remove the heat sink.

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Installing a Heat Sink

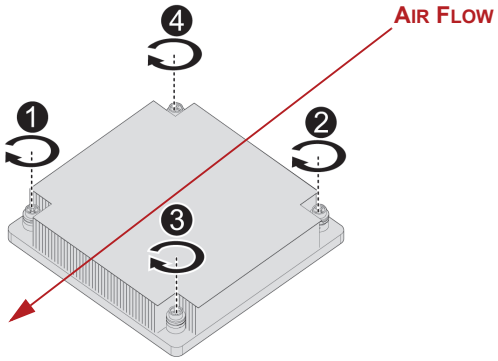

Figure 2-19. Installing the Heat Sink

1. Remove the top cover. See *Removing the Top Cover* on page 2-8.
2. Place the heat sink on the processor.
3. Secure the heat sink with four (4) screws in seqence.

Note:
The air flows over the heat sink in the direction shown.

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2.7 Processors

Removing a Processor

WARNING!
The processor remains hot after the system has been powered down. Allow sufficient time to cool before handling system components.

Prerequisite:

Remove the top cover. See *Removing the Top Cover* on page 2-8.

1. Carefully press down and outwards on the right processor locking lever to release it.

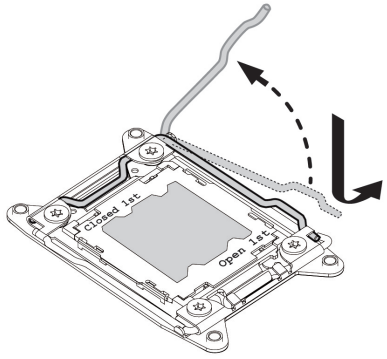

Figure 2-20. Releasing the Right Processor Locking Lever

CAUTION!
DO NOT TRY TO PULL THE RIGHT-HAND PROCESSOR LOCKING LEVER FULLY OPEN.

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2. Carefully press down and outwards on the left processor locking lever, pulling it fully open.

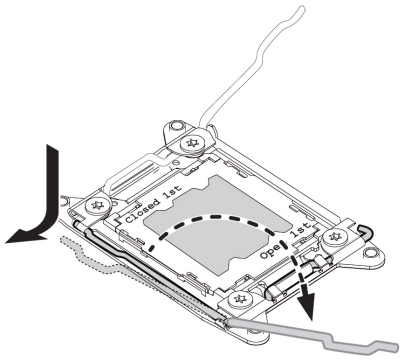

Figure 2-21. Releasing the Left Processor Locking Lever

CAUTION!
THE LOCKING LEVER IS HELD UNDER CONSIDERABLE FORCE AND MAY SPRING UP UNEXPECTEDLY.

3. Press down on the right processor locking lever to lift the processor cover part way and then lift the processor load plate to the fully open position.

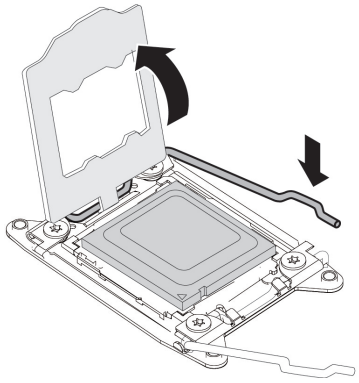

Figure 2-22. Opening the Processor Cover

4. Lift the processor out of the socket.

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CAUTION!
AVOID CONTACT WITH THERMAL GREASE ON THE PROCESSOR.

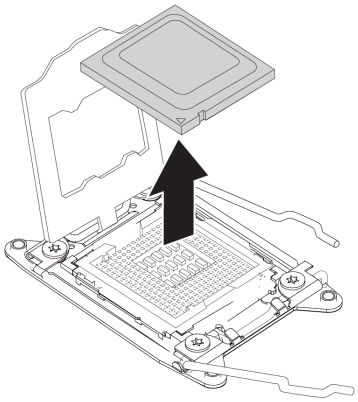

Figure 2-23. Removing the Processor

Installing a Processor

Prerequisite:

Remove the top cover. See *Removing the Top Cover* on page 2-8.

Remove the CPU heat sink. See *Removing a Heat Sink* on page 2-12.

1. Align the gold triangle identifying pin 1 of the processor with the triangular cutout of the processor, and the four (4) indents on processor with the four (4) tabs on socket.

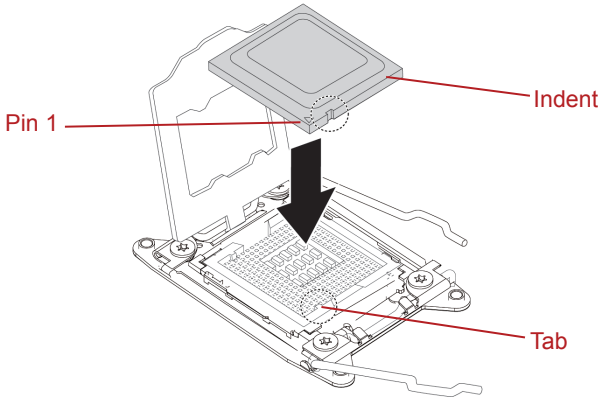

Figure 2-24. Installing the Processor

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2. Insert the processor into the socket, ensuring the four keys on the socket fit into the corresponding keys on the processor.

WARNING!
The processor should fit easily into the socket. Do not force the processor in place.

3. Lower the processor load plate over the processor ❶.

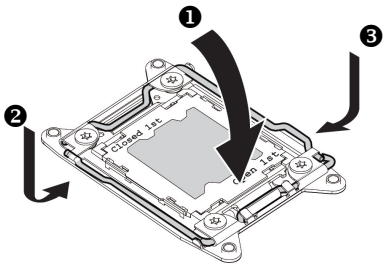

Figure 2-25. Closing the Processor Cover

4. Push the left processor locking lever down and latch it into the locked position ❷. The protective plastic cover pops out as the latch is engaged.
5. Push the right processor locking lever down and latch it into the locked position ❸.

Note:
Use the socket cover to protect the socket when the socket is empty.

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2.8 Memory Modules

WARNING!
The mainboard is supplied with all DIMM slots populated with memory 9 dummies for proper air flow. When installing and replacing memory modules, only remove those dummies that are to be directly replaced. All DIMM slots must be occupied at all times by either a memory module or dummy cover.

General Guidelines

All multi-node servers have specific rules for the population of memory on the individual mainboards that must be obeyed. Refer to the following individual server rules for information on how to populate the particular server required.

Removing a Memory Module

WARNING!
Memory modules remain hot after the system is powered down. Allow sufficient time for the memory modules to cool before handling system components.

Prerequisite:

Remove the top cover. See *Removing the Top Cover* on page 2-8.

1. Press down on the two memory module slot levers. The memory module partially ejects.

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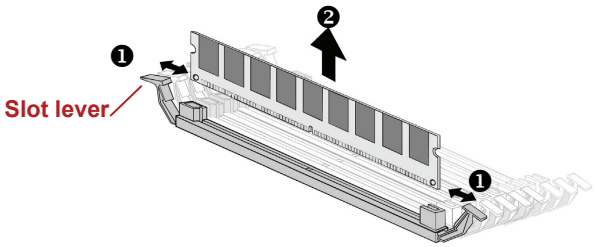

Figure 2-26. Removing Memory Modules

2. Lift out the memory module.

CAUTION!
HANDLE MEMORY MODULES BY THE EDGES AT ALL TIMES.

Installing a Memory Module

Prerequisite:

Remove the top cover. See *Removing the Top Cover* on page 2-8.

CAUTION!
HANDLE THE MEMORY MODULE BY THE EDGES AT ALL TIMES.

1. Align the notch on the memory module with the obstruction on the slot.

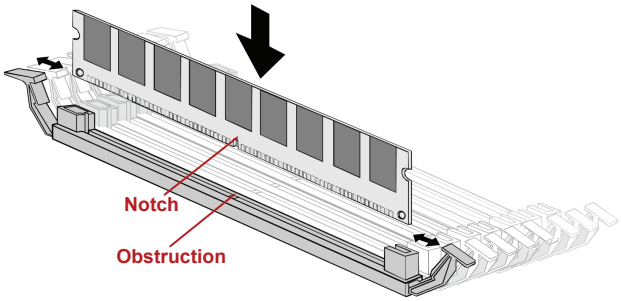

Figure 2-27. Installing Memory Modules

2. Push the memory module firmly into the memory module slot. The locking latches automatically lock in place.

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Memory Population Rules

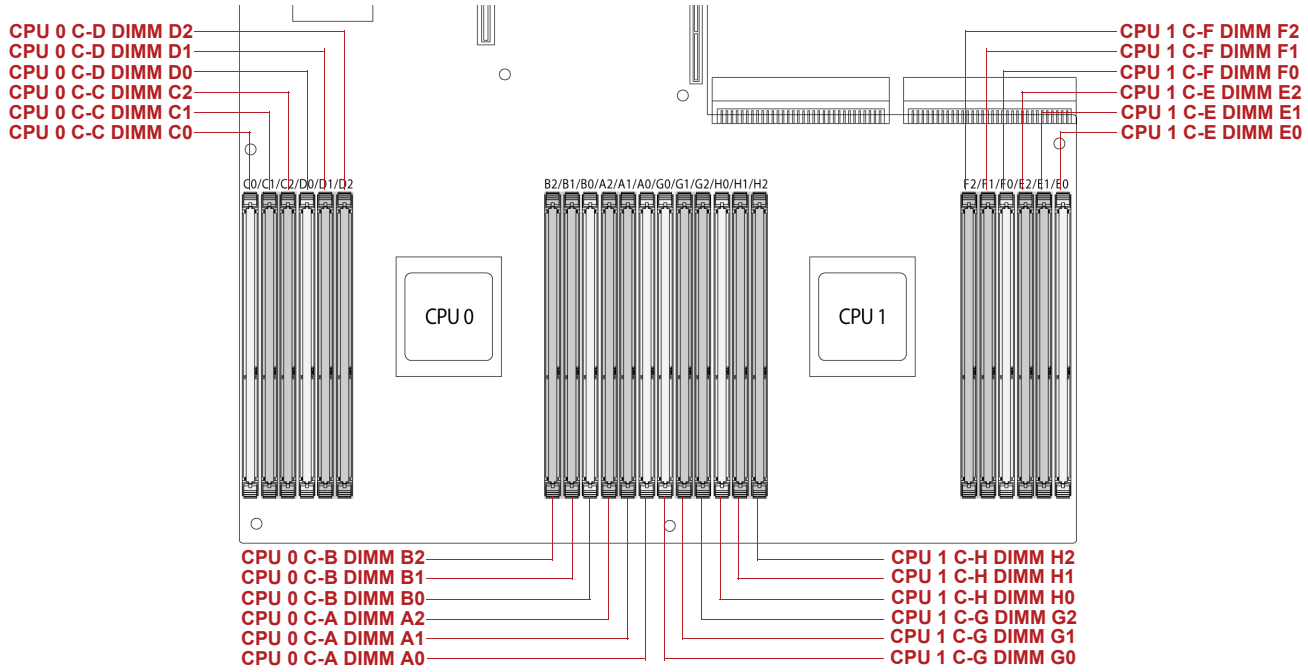

Figure 2-28. Memory Population Rules

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Table 2-1: RDIMM Support (Reduced Length)

RANKS PER DIMM & DATA WIDTH	MEMORY CAPACITY PER DIMM (GB)[1*]			SPEED (MT/S) & VOLTAGE VALIDATED FOR SHORT LENGTH PDG BY SLOT PER CHANNEL (SPC) & DIMM PER CHANNEL (DPC)[2,3*]											
				1 Slot per Channel		2 Slots per Channel				3 Slots per Channel					
				1DPC (V)		1 DPC (V)		2 DPC (V)		1DPC (V)		2DPC (V)		3DPC (V)	
				1.35	1.5	1.35	1.5	1.35	1.5	1.35	1.5	1.35	1.5	1.35	1.5
SR x 8	1[a*]	2[b*]	4[b*]	1066, 1333, 1600	1066, 1333, 1600, 1866	1066, 1333	1066, 1333, 1600, 1866	1066, 1333	1066, 1333, 1600	1066, 1333	1066, 1333, 1600, 1866	1066, 1333	1066, 1333, 1600	800	800, 1066
DR x 8	2[a*]	4[b*]	8[b*]	1066, 1333, 1600	1066, 1333, 1600, 1866	1066, 1333	1066, 1333, 1600, 1866	1066, 1333	1066, 1333, 1600	1066, 1333	1066, 1333, 1600, 1866	1066, 1333	1066, 1333, 1600	800	800, 1066
SR x 4	2[a*]	4[a*]	8[a*]	1066, 1333, 1600	1066, 1333, 1600, 1866	1066, 1333	1066, 1333, 1600, 1866	1066, 1333	1066, 1333, 1600	1066, 1333	1066, 1333, 1600, 1866	1066, 1333	1066, 1333, 1600	800	800, 1066
DR x 4	4[a*]	8[b*]	16[b*]	1066, 1333, 1600	1066, 1333, 1600, 1866	1066, 1333	1066, 1333, 1600, 1866	1066, 1333	1066, 1333, 1600	1066, 1333	1066, 1333, 1600, 1866	1066, 1333	1066, 1333, 1600	800	800, 1066
QR x 4	4[a*]	8[c*]	32[c*]	800	800, 1600	800	800	800	800	800	800, 1066	800	800	n/a	n/a
QR x 8	8[a*]	16[c*]	16[c*]	800	800, 1600	800	800	800	800	800	800, 1066	800	800	n/a	n/a

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Important:

a*: Supported but not validated.

b*: Supported and validated.

c*: Supported with limited validation

Note:

1*: Supported DRAM densities are 1Gb, 2Gb, and 4 Gb. Only 2 Gb and 4 Gb are validated by Intel.

2*: Command Address Timing is 1N.

3*: QR RDIMM are supported but only validated by Intel/PMO in a homogenous environment. The coverage will have limited system level testing, no signal integrity testing, and no interoperability testing The passing QR RDIMMs will be web posted.

Table 2-2: LR-DIMM Support (Reduced Length)

RANKS PER DIMM & DATA WIDTH	MEMORY CAPACITY PER DIMM (GB)[1, 2*]		SPEED (MT/S) & VOLTAGE VALIDATED BY SLOT PER CHANNEL (SPC) & DIMM PER CHANNEL (DPC)[3,4,5*]											
			1 Slot per Channel		2 Slots per Channel				3 Slots per Channel					
			1 DPC (V)		1 DPC (V)		2 DPC (V)		1DPC (V)		2DPC (V)		3DPC (V)	
			1.35	1.5	1.35	1.5	1.35	1.5	1.35	1.5	1.35	1.5	1.35	1.5
QR x 4 (DDP)[6*]	16[a*]	32[a*]	1066, 1333, 1600	1066, 1333, 1600, 1866	1066, 1333, 1600	1066, 1333, 1600, 1866	1066, 1333, 1600	1066, 1333, 1600,	1066, 1333, 1600	1066, 1333, 1600, 1866	1066, 1333, 1600	1066, 1333, 1600	1066	1066
8R x 4 (QDP)[6*]	32[a*]	64[a*]	1066	1066	1066	1066	1066	1066	1066	1066	1066	1066	1066	1066

a*: Supported and Validated
1*: Physical Rank is used to calculate DIMM Capacity.
2*: Supported and validated DRAM Densities are 2Gb and 4Gb.
3*: Command Address Timing is 1N.
4*: The speeds are estimated targets and will be verified through simulation.
5*: QR: For 3DPC – Rank Multiplication (RM) = 2
8R: For 2DPC and 3DPC – Rank Multiplication (RM) = 4
6*: QDP – Quad Die Package DRAM stacking.

DDP – Dual Die Package DRAM stacking.
P – Planer monolithic DRAM Die.

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2.9 Mainboard Module

Removing a Mainboard Module

Prerequisite:

Remove the top cover. See *Removing the Top Cover* on page 2-8.

Disconnect all cables from mainboard.

1. Remove eight (8) screws securing mainboard to chassis.

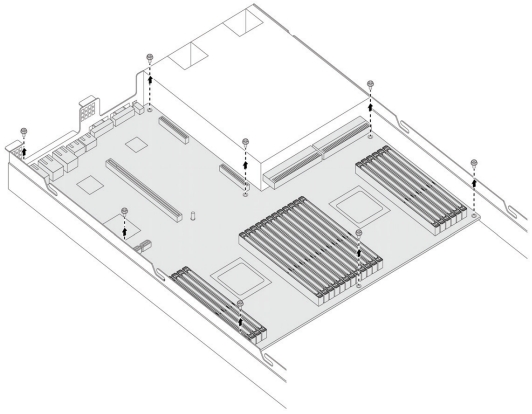

Figure 2-29. Removing the Mainboard Module Screws

2. Lift the front end of the mainboard until free of the chassis guide pin.

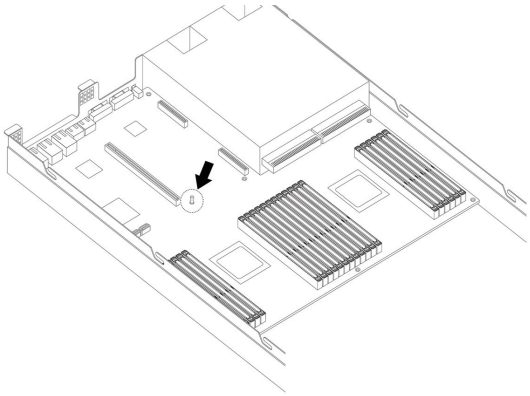

Figure 2-30. Aligning to the Chassis Guide Pin

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3. Lift mainboard in an angle to release the connectors from the slots on rear panel.

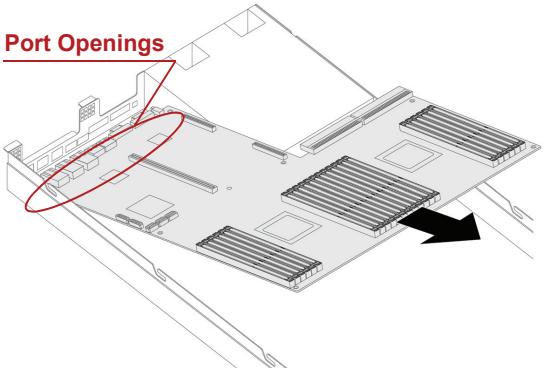

Figure 2-31. Removing the Mainboard

4. Remove the mainboard from the chassis.

Installing a Mainboard Module

Prerequisite:

Remove the top cover. See *Removing the Top Cover* on page 2-8.

1. Place the mainboard on the chassis.
2. Align the mainboard with the port openings on the rear of the chassis.
3. Align the mainboard opening with the chassis guide pin.
4. Install the mainboard on the chassis.

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5. Secure nine (8) screws to the mainboard.

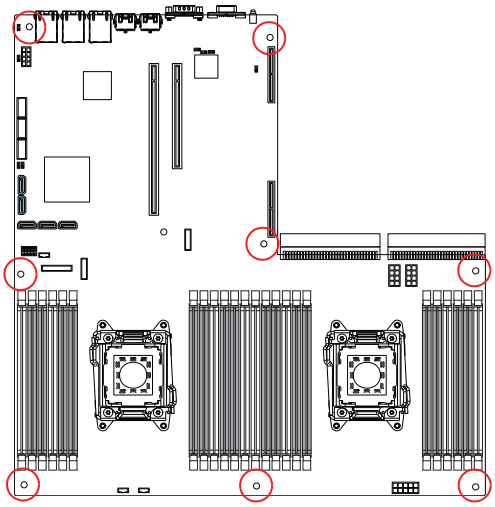

Figure 2-32. Removing the Mainboard Module Screws

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2.10 Expansion Cards (Optional)

Installing a Riser Assembly

1. Remove one (1) screw from the chassis.
2. Pull the riser assembly out of the chassis.

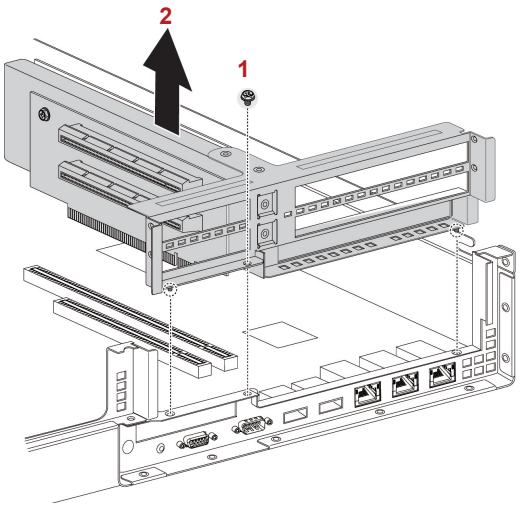

Figure 2-33. Removing the PCIe Bracket

3. Insert the PCIe expansion card into the PCIe riser.
4. Secure the PCIe expansion card with the single screw.
5. Repeat steps 3 and 4 for the additional PCIe expansion cards.

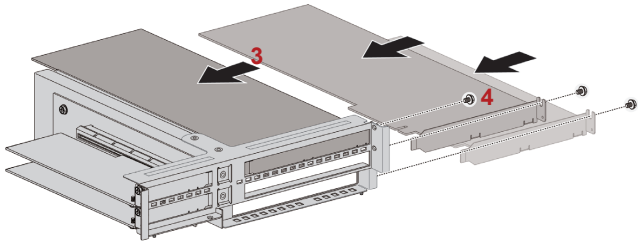

Figure 2-34. Installing a PCIe Expansion Card

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6. Align the PCIe assembly with the guide pin hole in the chassis.
7. Insert the PCIe assembly into the PCIe slots on the mainboard.
8. Secure the PCIe assembly to the chassis with one (1) screw.

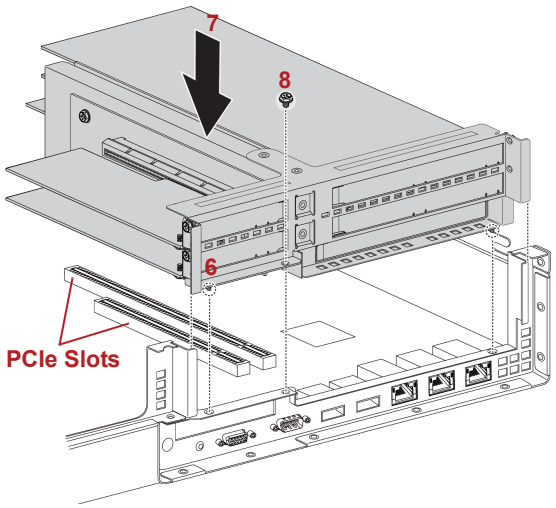

Figure 2-35. Installing the PCIe Bracket

Removing the PCIe Expansion Card

1. Remove one (1) screw from the chassis.

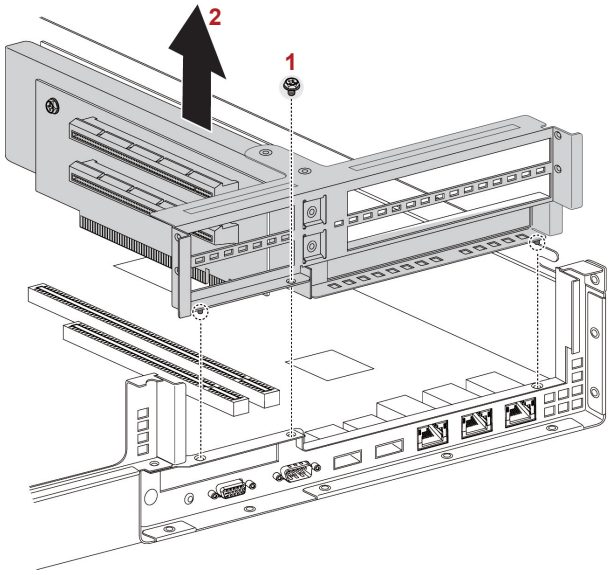

Figure 2-36. Removing the PCIe Expansion Card

2. Pull the PCIe bracket out of the chassis.

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3. Remove one (1) screw from the PCIe expansion card.
4. Disconnect the PCIe expansion card from the PCIe riser.

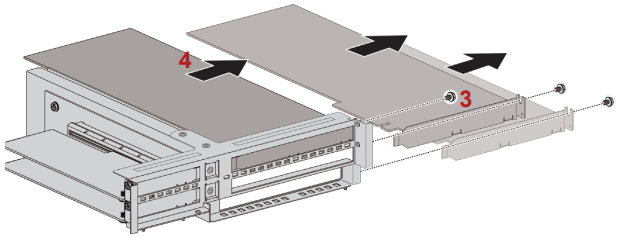

Figure 2-37. Installing a PCIe Expansion Card

5. Repeat steps 3 and 4 to remove the additional PCIe expansion cards.
6. Align the PCIe bracket with the guide pin hole in the chassis.
7. Insert the PCIe bracket into the PCIe slots on the mainboard.
8. Secure the PCIe bracket to the chassis with one (1) screw.

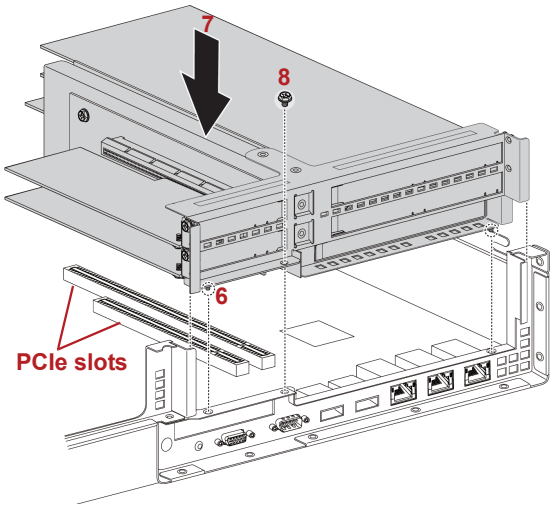

Figure 2-38. Installing the PCIe Bracket

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2.11 Mezzanine Card

Installing a LAN Mezzanine Card

Prerequisite:

Remove the PCIe riser assembly. See *Removing the PCIe Expansion Card* on page 2-27.

1. Align the module with the screw holes on the module support plate.

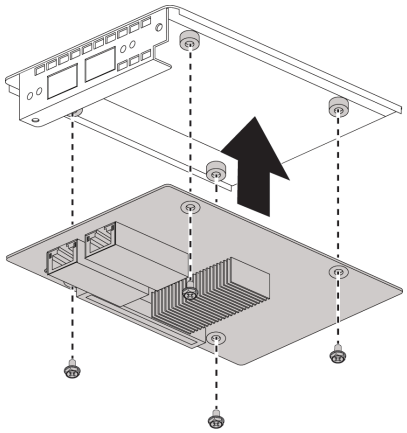

Figure 2-39. ModuleAssembly Screws

2. Secure four (4) screws to the module support plate.
3. Insert the linking board into the module connector.

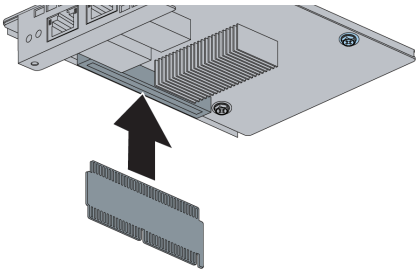

Figure 2-40. Linking Board and Module Connector

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4. Remove the screw securing the dummy bracket to the chassis.

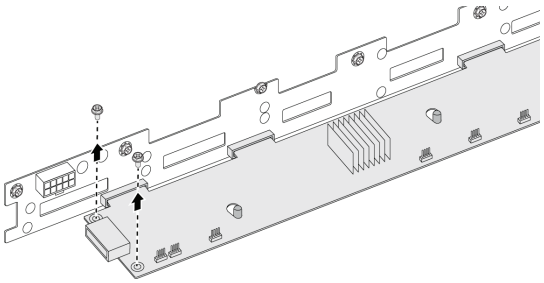

Figure 2-41. Dummy Bracket

5. Remove the dummy bracket from the chassis.

6. Align the linking board with the mainboard connector.

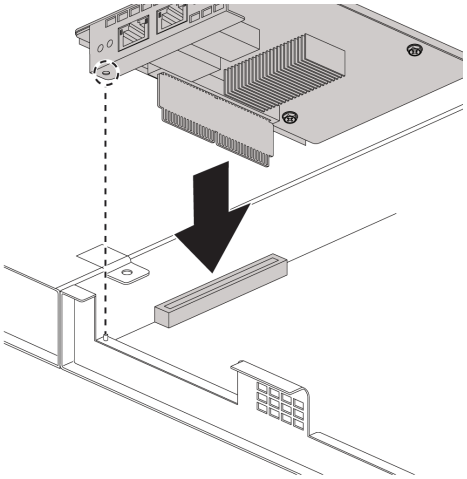

Figure 2-42. Linking Board and Mainboard Connector

7. Insert the linking board into the mainboard connector.

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8. Secure the module assembly to the chassis with one (1) screw.

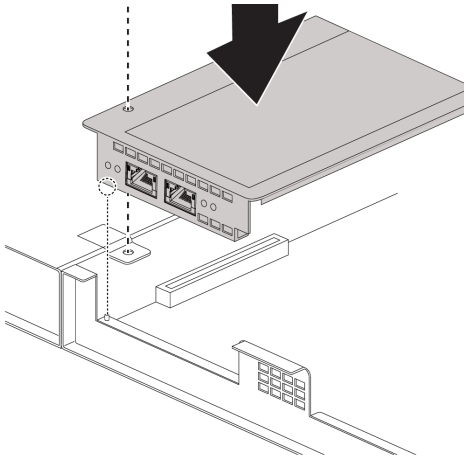

Figure 2-43. Module Screw

Removing a LAN Mezzanine Card

Prerequisite:

Remove the top cover.

Remove the PCIe riser assembly. See *Removing the PCIe Expansion Card* on page 2-27.

1. Remove one (1) screw securing the module assembly to the chassis.

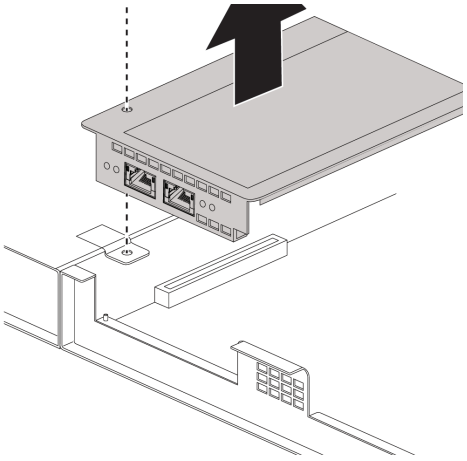

Figure 2-44. Module Assembly Screw

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2. Disconnect the module linking board from the mainboard connector.

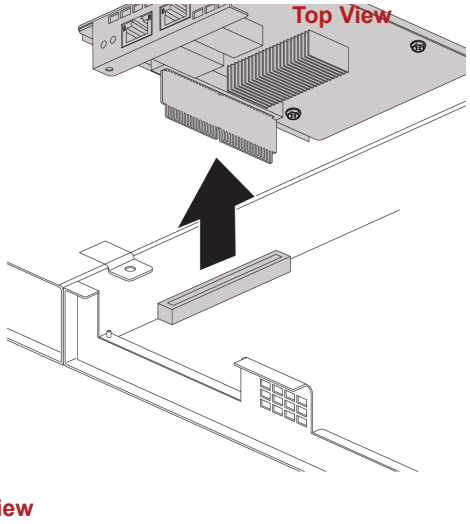

Figure 2-45. Linking Board and Mainboard Connector

3. Remove the module assembly from the chassis.

4. Disconnect the linking board from the module connector.

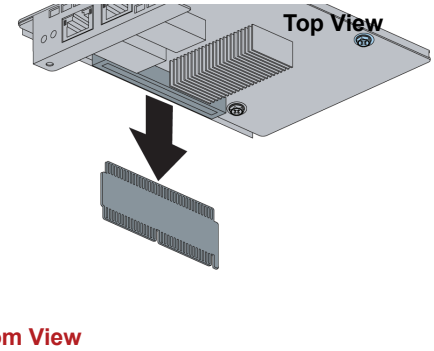

Figure 2-46. Linking Board and Module Connector

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5. Remove the four (4) screws from the module support plate.

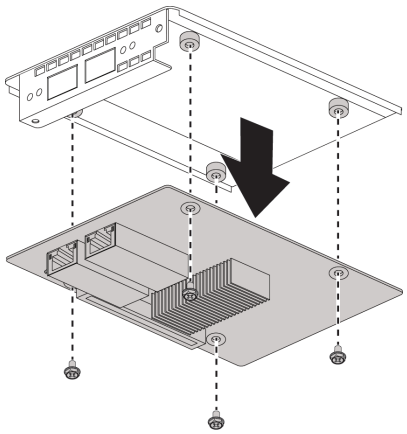

Figure 2-47. Module Screws

6. Remove the module from the support plate.

Installing a SAS 6G Mezzanine Card

Prerequisite:

Remove the top cover.

Remove the PCIe riser assembly. See *Removing the PCIe Expansion Card* on page 2-27.

Remove memory modules:

CPU 0 C-A DIMM A0 ~ CPU 0 C-A DIMM A2,

CPU 0 C-B DIMM B0 ~ CPU 0 C-B DIMM B2,

CPU 1 C-G DIMM G0 ~ CPU 1 C-G DIMM G2,

CPU 1 C-H DIMM H0 ~ CPU 0 C-H DIMM H2,

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1. Align the module with the screw holes on the module support plate.

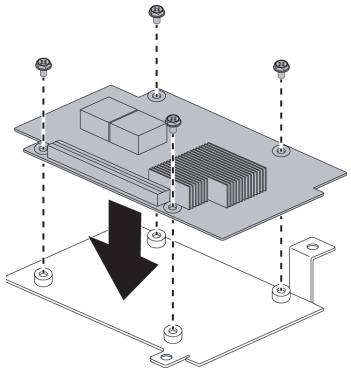

Figure 2-48. 6G Module Screws

2. Secure four (4) screws to the module support plate.

3. Connect the linking board to the module connector.

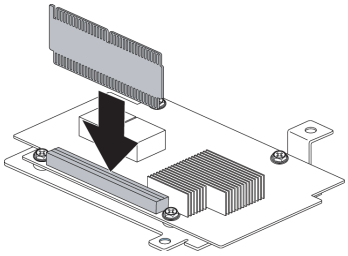

Figure 2-49. 6G Linking Board and Module Connector

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4. Remove nine (9) memory modules.
5. Connect the SAS cables to the ports on the module.

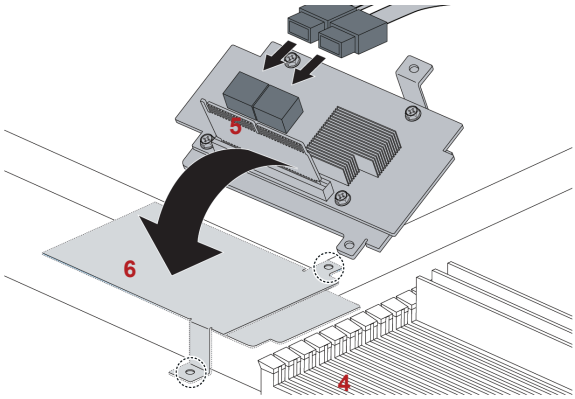

Figure 2-50. 6G SAS Cables and Module Ports

6. Turn the module assembly so that the linking board is facing the mainboard.
7. Align the linking board with the mainboard connector.

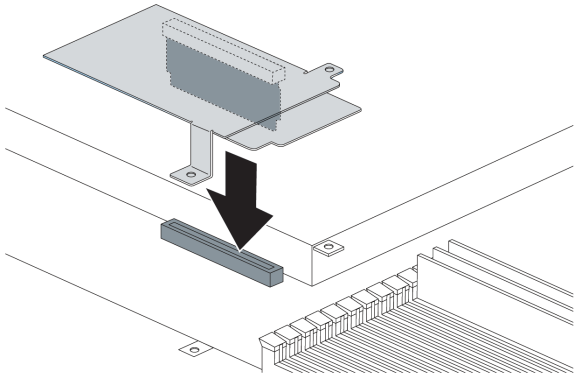

Figure 2-51. 6G Linking Board and Mainboard Connector

8. Connect the linking board to the mainboard connector.

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9. Secure the module assembly to the chassis with two (2) screws.

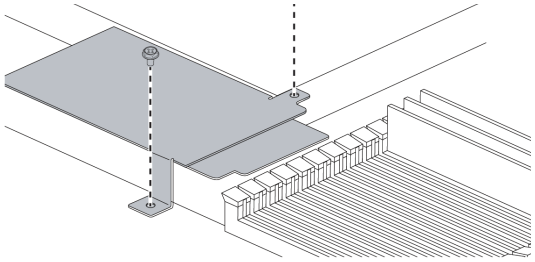

Figure 2-52. 6G Module Assembly Screws

10. Route the SAS cables along the side of the chassis.

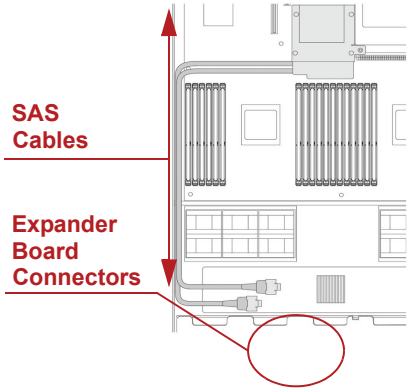

Figure 2-53. 6G SAS Cables and Expander Board Connectors

11. Connect the SAS cables to the connectors on the expander board.

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Removing a SAS 6Gb Mezzanine Card

Prerequisite:

Remove the PCIe riser assembly. See *Removing the PCIe Expansion Card* on page 2-27.

Remove memory modules:

CPU 0 C-A DIMM A0 ~ CPU 0 C-A DIMM A2,

CPU 0 C-B DIMM B0 ~ CPU 0 C-B DIMM B2,

CPU 1 C-G DIMM G0 ~ CPU 1 C-G DIMM G2,

CPU 1 C-H DIMM H0 ~ CPU 0 C-H DIMM H2,

See *Memory Population Rules* on page 2-22 for location. See *Removing a Memory Module* on page 2-20.

1. Disconnect the SAS cables from the connectors on the expander board.

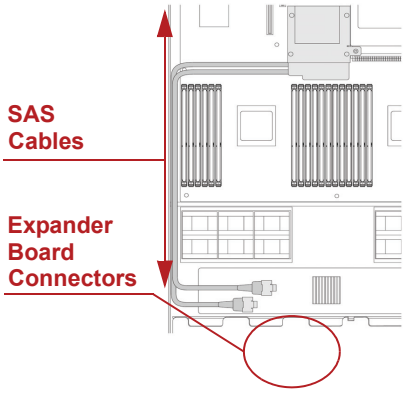

Figure 2-54. 6G SAS Cables and Expander Board Connectors

2. Remove the SAS cables from along the side of the chassis.

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3. Remove two (2) screws securing the module assembly to the chassis.

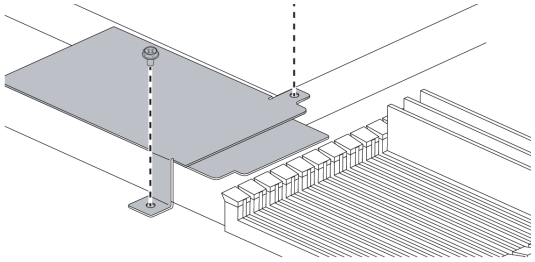

Figure 2-55. 6G Module Assembly Screws

4. Disconnect the module linking board from the mainboard connector.

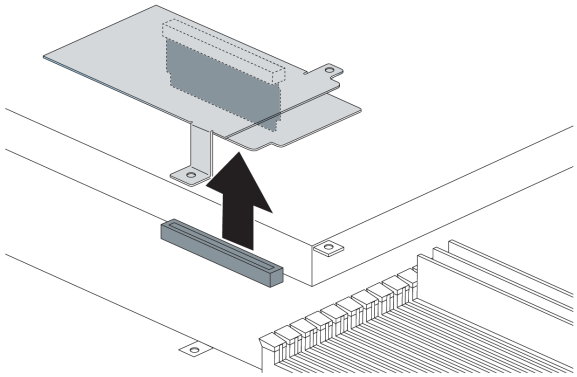

Figure 2-56. 6G Module Linking Board

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5. Remove the module assembly from the chassis.

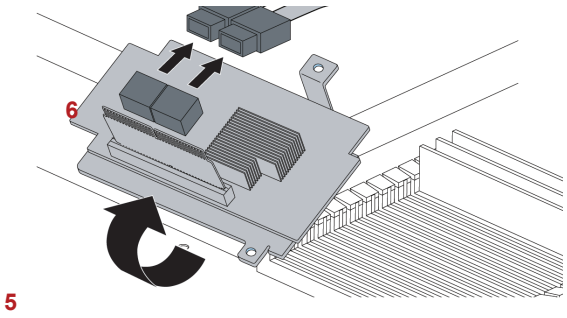

Figure 2-57. 6G Module Assembly and SAS Cables

6. Disconnect the SAS cables from the ports on the module.

7. Disconnect the linking board from the module connector.

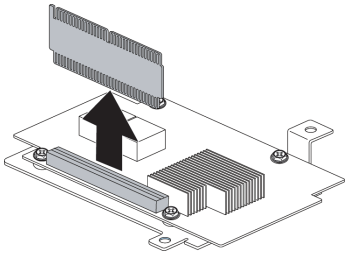

Figure 2-58. 6G Linking Board and Module Connector

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8. Remove the four (4) screws from the module support plate.

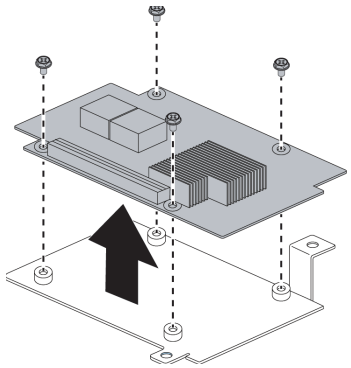

Figure 2-59. 6G Module Screws and Support Plate

9. Remove the module from the support plate.

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2.12 Intelligent Battery Backup Unit

A single iBBU provides backup functionality for a single module. To provide backup functionality for multiple modules, each module must be connected to an iBBU module.

Installing an iBBU

Prerequisite:

Remove the top cover. See *Removing the Top Cover* on page 2-8.

1. Remove the two screws securing the iBBU bracket (Figure 2-60).
2. Remove the iBBU bracket (Figure 2-60).

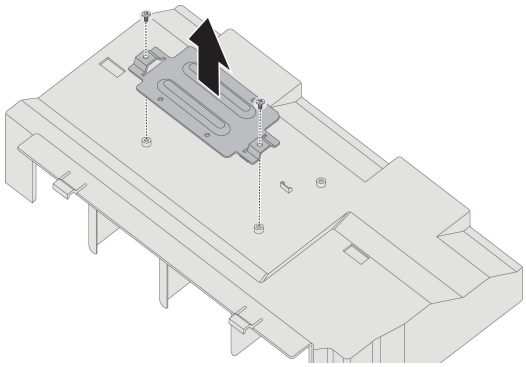

Figure 2-60. Installing the iBBU Holder and Battery

3. Align the bracket and iBBU battery and secure the assembly with three (3) screws.

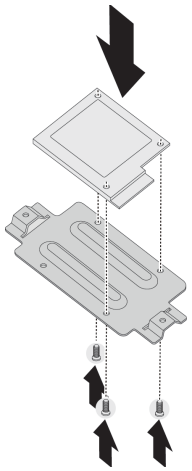

Figure 2-61. Installing the iBBU Battery in the Bracket

4. Place the iBBU assembly over the air duct.

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5. Secure the iBBU holder to the module with four (4) screws.

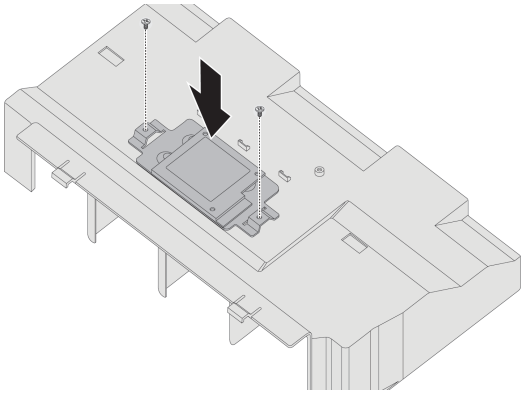

Figure 2-62. Installing the iBBU Holder to the Module

6. Connect the iBBU cable to the battery.

Note:
The iBBU can be connected to either a mezzanine card or a PCI-E expansion card.

7. Connect the iBBU cable to the expansion card.

Removing an iBBU

Prerequisite:

Remove the top cover. See *Removing the Top Cover* on page 2-8.

1. Disconnect the iBBU cable from the card.
2. Disconnect the iBBU cable from the battery.
3. Remove two (2) screws securing the iBBU assembly to the air duct.

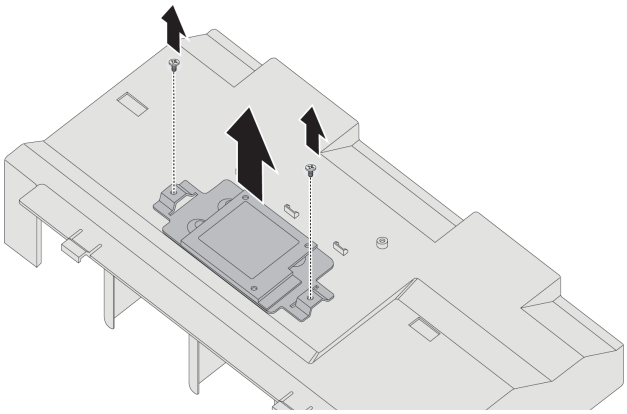

Figure 2-63. Removing the iBBU Assembly

4. Remove the iBBU battery holder from the module.

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5. Remove three (3) screws from the iBBU assembly to seperate the bracket.
6. Remove the iBBU battery from the holder.

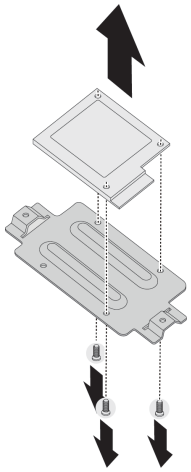

Figure 2-64. Removing the iBBU Battery from the Bracket

Note:
The iBBU can be connected to either a mezzanine card or a PCI-E expansion card.

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2.13 Air Duct

Removing the Air Duct

Prerequisite:

Remove the top cover. See *Removing the Top Cover* on page 2-8.

Remove the air duct from the chassis.

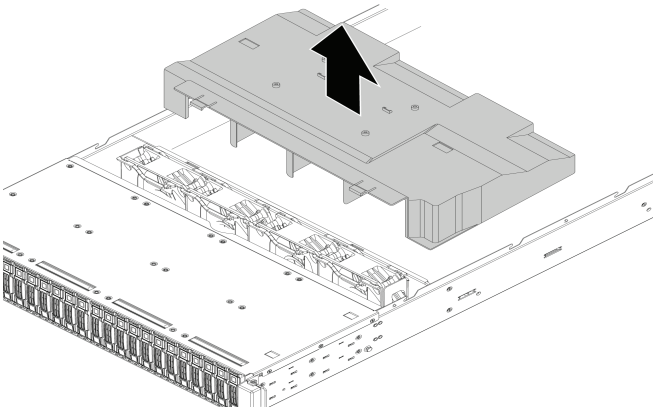

Figure 2-65. Removing the Air Duct

Installing the Air Duct

Remove the top cover. See *Removing the Top Cover* .

1. Align the air duct with the middle bracket (Figure 2-66).
2. Insert the plastic tabs into the wells in the middle bracket.

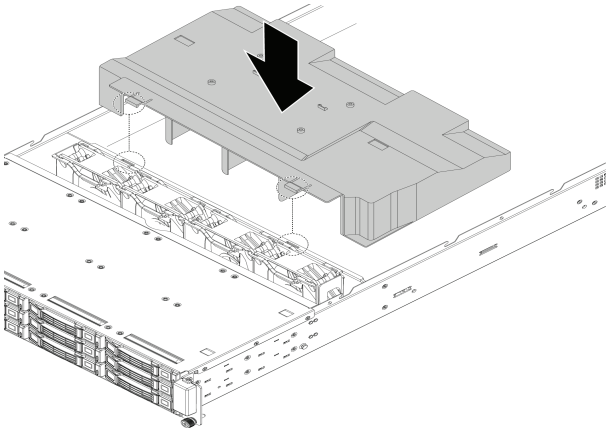

Figure 2-66. Installing the Air Duct

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2.14 Solid State Disks

Removing a Solid State Disk Cage

Prerequisite:

Remove the top cover. See *Removing the Top Cover* on page 2-8.

1. Remove the single screw from Solid State Disk (SSD) cage.
2. Slide the cage towards the fan assembly (Figure 2-68).
3. Remove the SSD cage from the chassis.

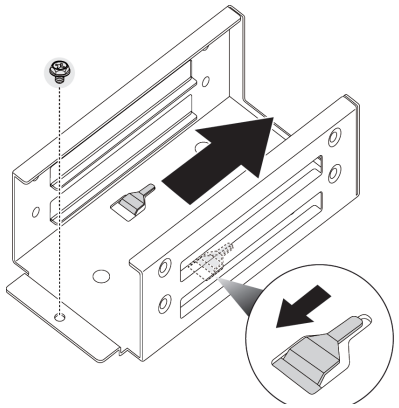

Figure 2-67. Removing the SSD Cage

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Installing a Solid State Disk Cage

Prerequisite:

Remove the SSD cage. See *Removing a Solid State Disk Cage* on page 2-45.

1. Align the Solid State Disk (SSD) cage with the chassis (Figure 2-68).
2. Slide the cage towards the rear of the chassis and align with the securing screw well.
3. Secure with the single screw.

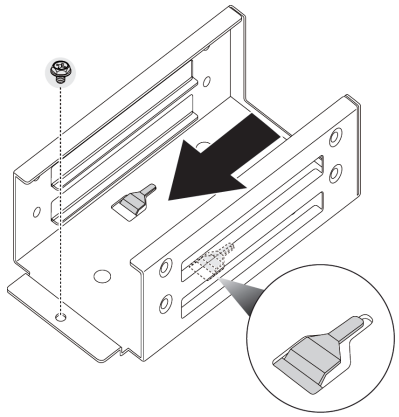

Figure 2-68. Installing the SSD Cage

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BIOS

Chapter 3

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3.1 BIOS Setup Utility

The BIOS Setup utility is provided to perform system configuration changes and to display current settings and environment information.

The BIOS Setup utility stores configuration settings in system non-volatile storage. Changes affected by BIOS Setup will not take effect until the system is rebooted. The BIOS Setup Utility can be accessed during POST by using the <**DEL**> or <**F2**> key.

The following sections describe the look and behavior for platform Setup.

Operation

BIOS Setup has the following features:

- The server board BIOS will only be available in English.
- BIOS Setup is functional via console redirection over various terminal emulation standards. This may limit some functionality for compatibility, e.g., usage of colors, some keys or key sequences, or support of pointing devices.

Setup Page

The setup page layout is sectioned into functional areas. Each occupies a specific area of the screen and has dedicated functionality. The following table lists and describes each functional area.

Table 3-1: BIOS Setup Page Layout

FUNCTIONAL AREA	DESCRIPTION
Title Bar	The title bar is located at the top of the screen and displays the title of the form (page) the user is currently viewing. It may also display navigational information.
Setup Item List	The Setup Item List is a set of controllable and informational items. Each item in the list occupies the left column of the screen. A Setup Item may also open a new window with more options for that functionality on the board.
Item Specific Help Area	The Item Specific Help area is located on the right side of the screen and contains help text for the highlighted Setup Item. Help information may include the meaning and usage of the item, allowable values, effects of the options, etc.
Keyboard Command Bar	The Keyboard Command Bar is located at the bottom right of the screen and continuously displays help for keyboard special keys and navigation keys.

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Entering BIOS Setup

BIOS Setup is started by pressing <**DEL**> or <**F2**> during boot time when the OEM logo is displayed.

When Quiet Boot is disabled, the message "press <**DEL**> or <**F2**> to enter setup" will be displayed on the diagnostics screen.

Keyboard Commands

The bottom right portion of the Setup screen provides a list of commands that are used to navigate through the Setup utility. These commands are displayed at all times.

Each Setup menu page contains a number of features. Except those used for informative purposes, each feature is associated with a value field. This field contains user-selectable parameters. Depending on the security option chosen and in effect by the password, a menu feature's value may or may not be changeable. If a value is non-changeable, the feature's value field is inaccessible and displays as grayed out.

Table 3-2: BIOS Setup: Keyboard Command

KEY	OPTION	DESCRIPTION
<**Enter**>	Execute Command	The <**Enter**> key is used to activate sub-menus when the selected feature is a sub-menu, or to display a pick list if a selected option has a value field, or to select a sub-field for multi-valued features like time and date. If a pick list is displayed, the <**Enter**> key will select the currently highlighted item, undo the pick list, and return the focus to the parent menu.
<**Esc**>	Exit	The <**Esc**> key provides a mechanism for backing out of any field. When the <**Esc**> key is pressed while editing any field or selecting features of a menu, the parent menu is re-entered.
		When the <**Esc**> key is pressed in any sub-menu, the parent menu is re-entered. When the <**Esc**> key is pressed in any major menu, the exit confirmation window is displayed and the user is asked whether changes can be discarded. If "No" is selected and the <**Enter**> key is pressed, or if the <**Esc**> key is pressed, the user is returned to where he/she was before <**Esc**> was pressed, without affecting any existing any settings. If "Yes" is selected and the <**Enter**> key is pressed, setup is exited and the BIOS returns to the main System Options Menu screen.

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Table 3-2: BIOS Setup: Keyboard Command (Continued)

KEY	OPTION	DESCRIPTION
-	Select Item	The up arrow is used to select the previous value in a pick list, or the previous option in a menu item's option list. The selected item must then be activated by pressing the <**Enter**> key.
↓	Select Item	The down arrow is used to select the next value in a menu item's option list, or a value field's pick list. The selected item must then be activated by pressing the <**Enter**> key.
↔	Select Menu	The left and right arrow keys are used to move between the major menu pages. The keys have no affect if a sub-menu or pick list is displayed.
<**Tab**>	Select Field	The <**Tab**> key is used to move between fields. For example, <**Tab**> can be used to move from hours to minutes in the time item in the main menu.
-	Change Value	The minus key on the keypad is used to change the value of the current item to the previous value. This key scrolls through the values in the associated pick list without displaying the full list.
+	Change Value	The plus key on the keypad is used to change the value of the current menu item to the next value. This key scrolls through the values in the associated pick list without displaying the full list. On 106-key Japanese keyboards, the plus key has a different scan code than the plus key on the other keyboard, but will have the same effect.

Table 3-2: BIOS Setup: Keyboard Command (Continued)

KEY	OPTION	DESCRIPTION
<**F8**>	Previous Values	Pressing <**F8**> makes the following message to appear: Load Previous Values? Yes No If *Yes* is highlighted and <**Enter**> is pressed, all Setup fields are set to their previous values. If *No* is highlighted and <**Enter**> is pressed, or if the <**Esc**> key is pressed, the user is returned to where they were before <**F8**> was pressed without affecting any existing field values.
<**F9**>	Setup Defaults	Pressing <**F9**> makes the following message to appear: Load Optimized Defaults? Yes No If *Yes* is highlighted and <**Enter**> is pressed, all Setup fields are set to their default values. If *No* is highlighted and <**Enter**> is pressed, or if the <**Esc**> key is pressed, the user is returned to where they were before <**F9**> was pressed without affecting any existing field values.

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Table 3-2: BIOS Setup: Keyboard Command (Continued)

KEY	OPTION	DESCRIPTION
<**F10**>	Save and Exit	Pressing <**F10**> makes the following message to appear: Save configuration and exit? Yes No If *Yes* is highlighted and <**Enter**> is pressed, all changes are saved and Setup is exited. If *No* is highlighted and <**Enter**> is pressed, or the <**Esc**> key is pressed, the user is returned to where they were before <**F10**> was pressed without affecting any existing values.

Menu Selection Bar

The Menu Selection Bar is located at the top of the BIOS Setup Utility screen. It displays the major menu selections available to the user. By using the left and right arrow keys, the user can select the menus listed here.

Server Platform Setup Utility Screens

The sections below describe the screens available for the configuration of a server platform. In these sections, tables are used to describe the contents of each screen. These tables follow the following guidelines:

- The text and values in the Setup Item, Options, and Help columns in the tables are displayed on the BIOS Setup screens.
- **Bold text** in the Options column of the tables indicates default values. These values are not displayed in bold on the setup screen. The bold text in this document is to serve as a reference point.
- The Comments column provides additional information where it may be helpful. This information does not appear in the BIOS Setup screens.
- Information in the screen shots that is enclosed in brackets (< >) indicates text that varies, depending on the option(s) installed. For example <Current Date> is replaced by the actual current date.
- Information that is enclosed in square brackets ([]) in the tables indicates areas where the user needs to type in text instead of selecting from a provided option.

Whenever information is changed (except Date and Time) the systems requires a save and reboot to take place. Pressing <**ESC**> will discard the changes and boot the system according to the boot order set from the last boot.

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Main Screen

The Main screen is the screen that is first displayed when BIOS Setup is entered, unless an error has occurred. If an error has occurred, the Error Manager screen will be displayed instead.

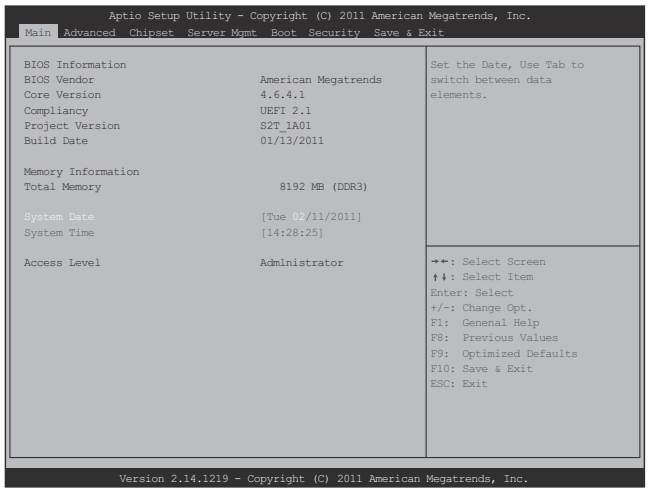

Figure 3-1. Main Screen

Table 3-3: Main Screen Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
BIOS Vendor			Information only. Displays the BIOS Vendor.
Core Version			Information only. Displays the AMI BIOS Core version.
Compliancy			Information only. Displays the BIOS compliancy.
Project Version			Information only. Displays the Project version.
Build Date			Information only. Displays the BIOS build date.
Total Memory			Information only. Displays the Total System Memory Size.
System Data	[Day of week MM/DD/ YYYY]	Set the Date. Use Tab to switch between Date elements.	
System Time	[HH:MM:SS]	Set the Time. Use Tab to switch between Time elements.	

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Table 3-3: Main Screen Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Access Level			Information only. Displays the Access Level.

Advanced Screen

The Advanced screen provides an access point to configure several options. On this screen, the user selects the option that is to be configured. Configurations are performed on the selected screen, not directly on Advanced screen.

To access this screen from Main screen, press the right arrow until Advanced screen is chosen.

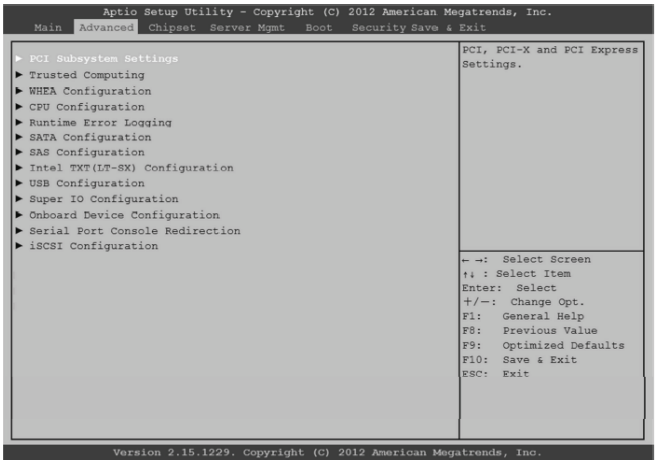

Figure 3-2. Advanced Screen

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Table 3-4: Advanced Screen Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
PCI Subsystem Settings		PCI, PCI-X and PCI Express Settings.	
Trusted Computing		Trusted Computing Settings	
WHEA Configuration		General WHEA Configuration settings.	
CPU Configuration		CPU Configuration Parameters.	
Runtime Error Logging		Runtime Error Logging Support Setup Option	
SATA Configuration		SATA Devices Configuration.	
SAS Configuration		SAS Devices Configuration.	
Intel TXT (LT-SX) Configuration		Intel Trusted Execution Technology Configuration	
USB Configuration		USB Configuration Parameters.	

Table 3-4: Advanced Screen Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Super IO Configuration		System Super IO Chip Parameters.	
Onboard Device Configuration		Onboard Device Parameters	
Serial Port Console Redirection		Serial Port Console Redirection.	
iSCSI Configuration		Configure iSCSI parameter	

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PCI Screen

The PCI Screen provides fields to configure PCI add-in cards, the onboard NIC controllers, and video options. To access this screen from the Main screen, select Advanced | PCI.

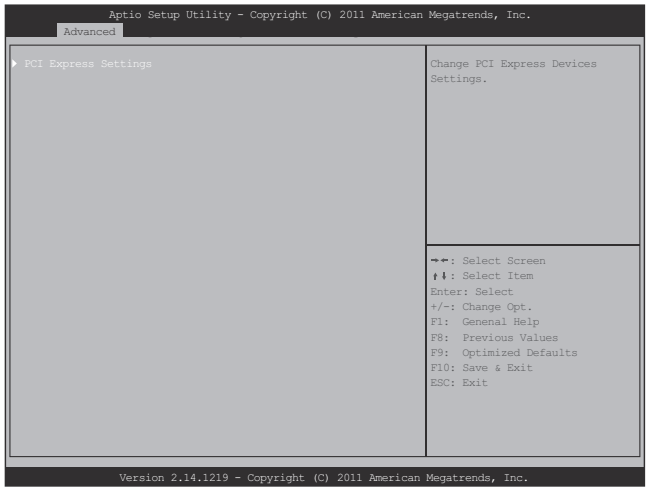

Figure 3-3. PCI Subsystem Settings Screen

Table 3-5: PCI Subsystem Settings Screen Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
PCI Express Settings		Change PCI Express Devices Settings.	

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PCI Express Settings Screen

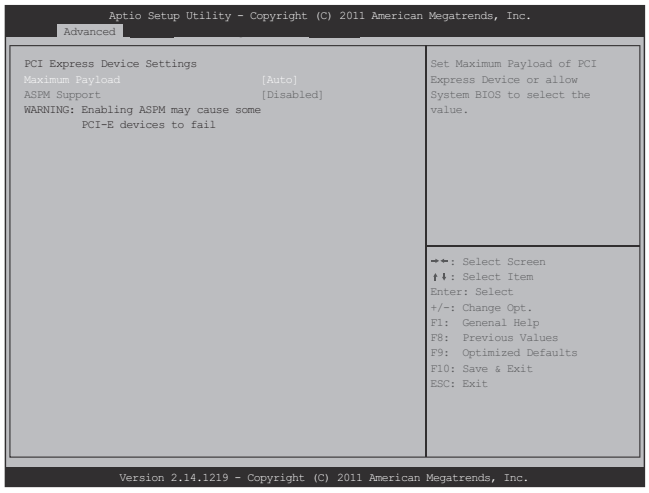

Figure 3-4. PCI Express Settings Screen

Table 3-6: PCI Express Settings Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Maximum Payload	[**Auto**] [128 Bytes] [256 Bytes] [512 Bytes] [1024 Bytes] [2048 Bytes] [4096 Bytes]	Set Maximum Payload of PCI Express Device or allow System BIOS to select the value.	
ASPM Support	[**Disabled**] [Auto] [Force L0s]	Set the ASPM Level: Force L0s - Force all links to L0s State: AUTO - BIOS auto configure: DISABLE - Disables ASPM.	

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Trusted Computing Screen

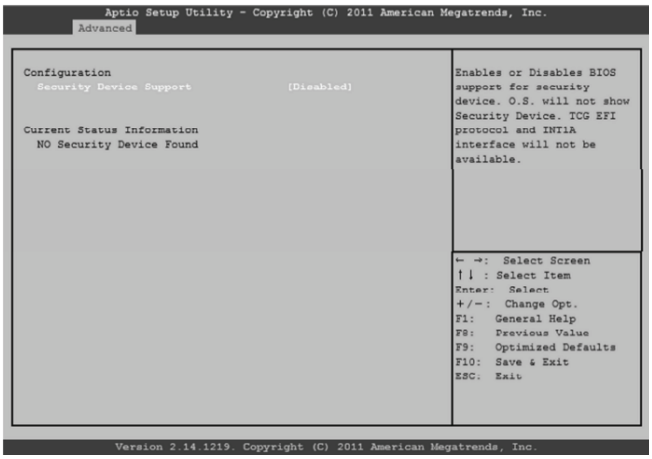

Figure 3-5. Trusted Computing Screen

Table 3-7: Trusted Computing Screen Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Security Device Support	[**Disabled**] [Enabled]	Enables or Disables BIOS support for security device. O.S. will not show Security Device. TCG EFI protocol and INT1A interface will not be available.	
Current Status Information			Information only. Displays the Current Status Information

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WHEA Screen

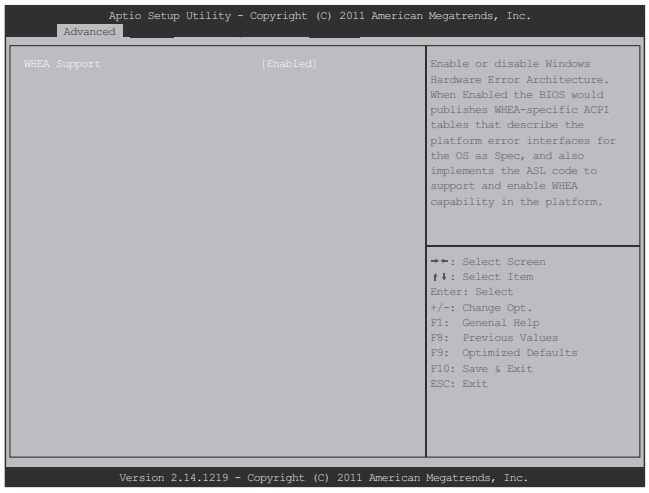

Figure 3-6. WHEA Configuration Screen

Table 3-8: WHEA Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
WHEA Support	[Disabled] [**Enabled**]	Enable or disable Windows Hardware Error Architecture. When Enabled the BIOS would publishes WHEA-specific ACPI tables that describe the platform error interfaces for the OS as Spec, and also implements the ASL code to support and enable WHEA capability in the platform.	

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Processor Configuration Screen

The Processor screen provides a place for the user to view the processor core frequency, system bus frequency, and enable or disable several processor options. The user can also select an option to view information about a specific processor.

Figure 3-7. Processor Configuration Screen

Table 3-9: Processor Configuration Fields

SETUP ITEM	Options	HELP TEXT	COMMENTS
Socket 0 CPU Information		Socket specific CPU Information.	
Socket 1 CPU Information		Socket specific CPU Information.	
CPU Speed			Information only. Displays the speed of the processor.
64-bit			Information only. Displays 64-t supported or not.
Hyper-threading	[Disabled] [**Enabled**]	Enabled for Windows XP and Linux (OS optimized for Hyper-Threading Technology) and Disabled for other OS (OS not optimized for Hyper-Threading Technology). When Disabled only one thread per enabled core is enabled.	

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Table 3-9: Processor Configuration Fields (Continued)

SETUP ITEM	Options	HELP TEXT	COMMENTS
Active Processor Cores	[**All**] [1] [2] [3] [4] [5] [6] [7]	Number of cores to enable in each processor package.	
Execute Disable Bit	[Disabled] [**Enabled**]	XD can prevent certain classes of malicious buffer overflow attacks when combined with a supporting OS (Windows Sever 2003 SP1, Windows XP SP2, SuSE Linux 9.2, RedHat Enterprise 3 Update 3).	
Hardware Prefetcher	[Disabled] [**Enabled**]	Enable the Mid Level Cache (L2) Streamer Prefetcher.	
Adjacent Cache Line Prefetch	[Disabled] [**Enabled**]	Enable the Mid Level Cache (L2) prefetching of adjacent cache lines.	

Table 3-9: Processor Configuration Fields (Continued)

SETUP ITEM	Options	HELP TEXT	COMMENTS
DCU Streamer Prefetcher	[Disabled] [**Enabled**]	Enable prefetcher of next L1 Data line based upon multiple loads in same cache line.	
DCU IP Prefetcher	[Disabled] [**Enabled**]	Enable prefetcher of next L1 line based upon sequential load history.	
Intel Virtualization Technology	[Disabled] [**Enabled**]	When enabled, a VMM can utilize the additional hardware capabilities provided by Vanderpool Technology.	
CPU Power Management Configuration		CPU Power Management Configuration Parameters	

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Socket 0 CPU Information Screen

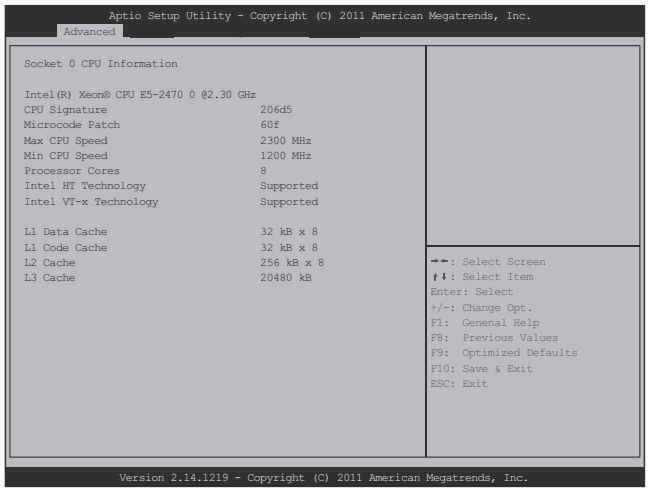

Figure 3-8. Socket 0 CPU Information Screen

Table 3-10: Socket 0 CPU Information Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
CPU Signature			Information only. Displays the CPU Signature.
Microcode Patch			Information only. Displays the Microcode Patch.

Table 3-10: Socket 0 CPU Information Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Max CPU Speed			Information only. Displays the Max CPU Speed.
Min CPU Speed			Information only. Displays the Min CPU Speed.
Processor Cores			Information only. Displays the number of Processor Cores.
Intel HT Technology			Information only. Displays Intel HT Technology supported or not.
Intel VT-x Technology			Information only. Displays Intel VT-x Technology supported or not.
L1 Data Cache			Information only. Displays the size of L1 Data Cache.
L1 Code Cache			Information only. Displays the size of L1 Code Cache.
L2 Cache			Information only. Displays the size of L2 Cache.

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Table 3-10: Socket 0 CPU Information Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
L3 Cache			Information only. Displays the size of L3 Cache.

CPU Power Management Configuration Screen

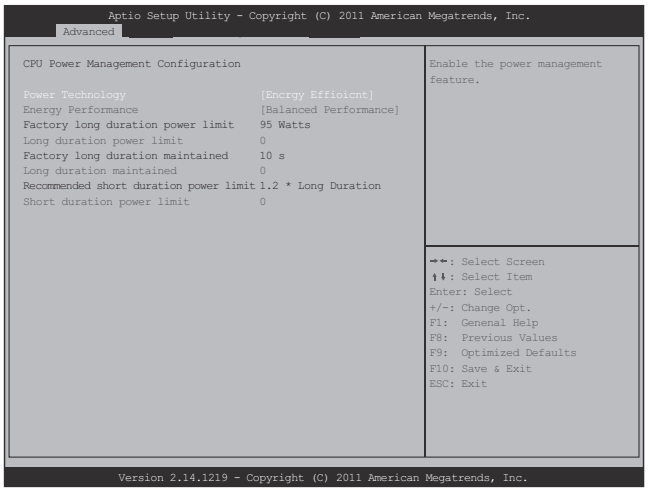

Figure 3-9. CPU Power Management Configuration Screen

Table 3-11: CPU Power Mgmt Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Power Technology	**[Disabled]** [Energy Efficient] [Custom]	Enable the power management features.	

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Table 3-11: CPU Power Mgmt Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
EIST	[Disabled] **[Enabled]**	Enable/Disable Intel SpeedStep.	Only appears when Power Technology is set to [Custom].
Turbo Mode	[Disabled] **[Enabled]**	Turbo Mode.	Only appears when Power Technology is set to [Custom].
CPU C3 Report	**[Disabled]** [Enabled]	Enable/Disable CPU C3(ACPI C2) report to OS.	Only appears when Power Technology is set to [Custom].
CPU C6 Report	[Disabled] **[Enabled]**	Enable/Disable CPU C6(ACPI C3) report to OS.	Only appears when Power Technology is set to [Custom].
Energy Performance	[Performance] [**Balanced Performance**] [Balanced Energy] [Energy Efficient]	Optimize between performance and power savings. Windows 2008 and later Oses overrides this value according to its power plan.	
Factory long duration power limit			Information only. Displays the Factory long duration power limit.

Table 3-11: CPU Power Mgmt Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Long duration power limit		Long duration power limit in Watts.	
Factory long duration maintained			Information only. Displays the Factory long duration maintained.
Long duration maintained		Time window which the long duration power is maintained.	
Recommended short duration power			Information only. Displays the Recommended short duration power.
Short duration power limit		Short duration power limit in Watts.	

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Runtime Error Logging Screen

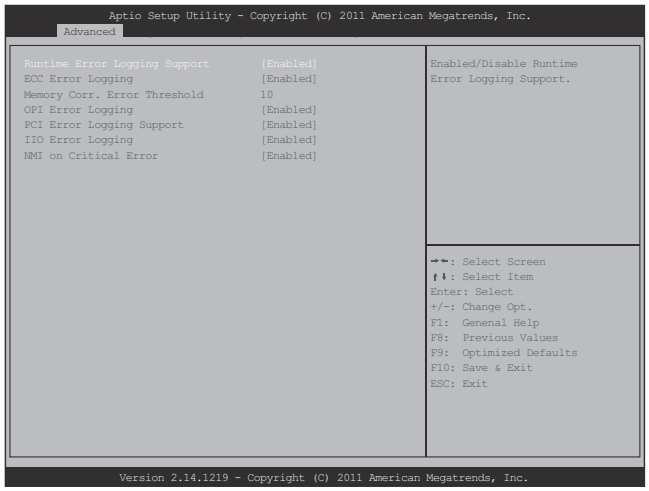

Figure 3-10. Runtime Error Logging Screen

Table 3-12: Runtime Error Logging Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Runtime Error Logging Support	[Disabled] **[Enabled]**	Enable/Disable Runtime Error Logging Support.	
ECC Error Logging	[Disabled] **[Enabled]**	Enable/Disable ECC Error Logging.	

Table 3-12: Runtime Error Logging Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Memory Corr. Error Threshold		Enter the Memory Correctable Error Threshold value	
QPI Error Logging	[Disabled] **[Enabled]**	Enable/Disable QPI Error Logging	
PCI Error Logging Support	[Disabled] **[Enabled]**	Enable/Disable PCI Error Logging Support	
IIO Error Logging	[Disabled] **[Enabled]**	Enable/Disable IIO Error Logging	
NMI on Critical Error	[Disabled] **[Enabled]**	Enable/Disable NMI generation on fatal or uncorrectable error	

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SATA Controller Screen

The ATA Controller screen provides fields to configure SATA hard disk drives. It also provides information on the hard disk drives that are installed.

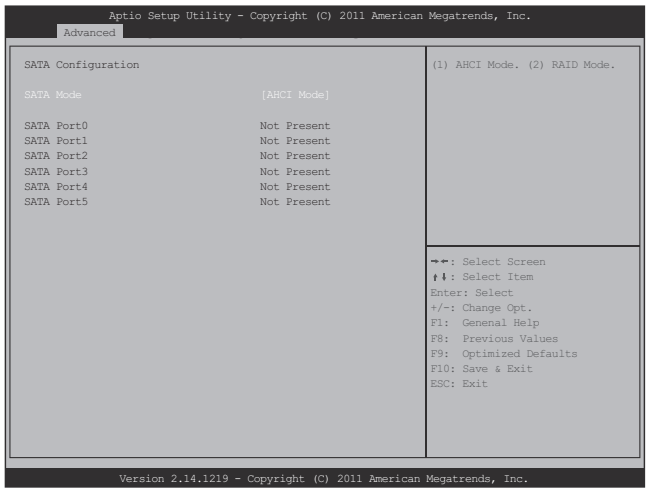

Figure 3-11. ATA Controller Configuration Screen

Table 3-13: ATA Controller Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
SATA Mode	[Disabled] [**AHCI Mode**] [RAID Mode]	(1) AHCI Mode. (2) RAID Mode.	Select SATA Type for onboard SATA ports.
SATA Port0			Information only. Displays the device on Port0.
SATA Port1			Information only. Displays the device on Port1.
SATA Port2			Information only. Displays the device on Port2.
SATA Port3			Information only. Displays the device on Port3.
SATA Port4			Information only. Displays the device on Port4.
SATA Port5			Information only. Displays the device on Port5.

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SAS Configuration Screen

BIOS will display device information only when user select Intel RSTe OPROM.

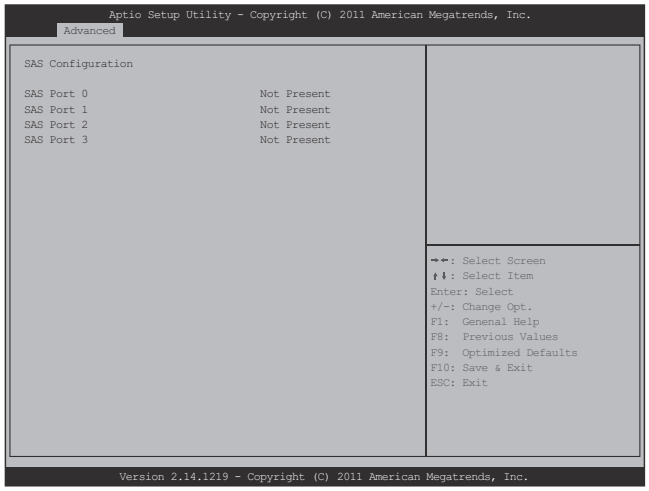

Figure 3-12. Setup Utility - SAS Configuration Screen

Table 3-14: SAS Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
SAS Port 0			Information only. Displays the device on SAS Port 0.

Table 3-14: SAS Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
SAS Port 1			Information only. Displays the device on SAS Port 1.
SAS Port 2			Information only. Displays the device on SAS Port 2.
SAS Port 3			Information only. Displays the device on SAS Port 3.
SAS Port 4			Information only. Displays the device on SAS Port 4.
SAS Port 5			Information only. Displays the device on SAS Port 5.
SAS Port 6			Information only. Displays the device on SAS Port 6.
SAS Port 7			Information only. Displays the device on SAS Port 7.

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Intel TXT(LT-SX) Screen

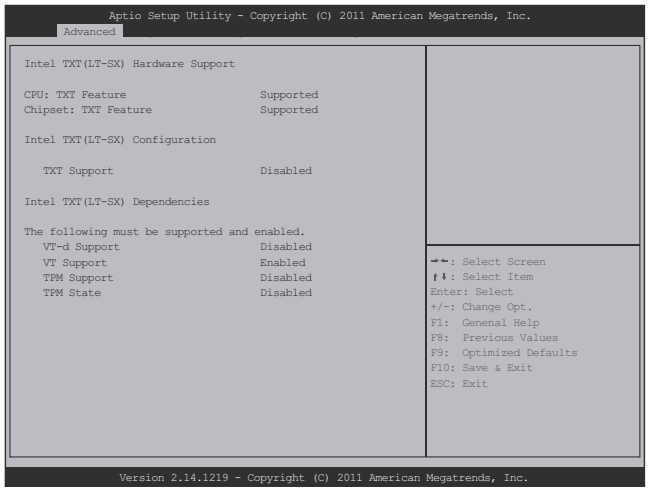

Figure 3-13. Intel TXT(LT-SX) Configuration Screen

Table 3-15: Intel TXT (LT-SX) Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
CPU: TXT Feature			Information only. Displays the CPU: TXT Feature

Table 3-15: Intel TXT (LT-SX) Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Chipset: TXT Feature			Information only. Displays the Chipset: TXT Feature
TXT Support	**[Disabled]** [Enabled]	TXT requires a properly configured TPM, LT-SX enabled CPU, and the below dependencies be met.	Only selectable when all other options are supported.
TXT Support			Information only. Displays the TXT Support
VT-d Support			Information only. Displays the VT-d Support
VT Support			Information only. Displays the VT Support
TPM Support			Information only. Displays Intel TPM Support
TPM State			Information only. Displays Intel TPM State

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Table 3-15: Intel TXT (LT-SX) Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
CPU: TXT Feature			Information only. Displays the CPU: TXT Feature

USB Configuration Screen

The USB Configuration screen provides fields to configure the USB controller options.

To access this screen from the Main screen, select Advanced | USB Configuration.

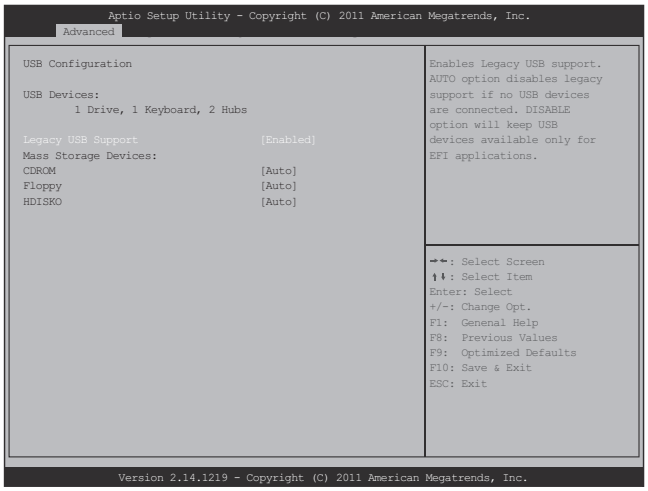

Figure 3-14. USB Configuration Screen

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Table 3-16: USB Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
USB Devices:			Information only. Display all of the USB devices attached.
Legacy USB Support	**[Enabled]** [Disabled] [Auto]	Enables Legacy USB support. AUTO option disables legacy support if no USB devices are connected, DISABLE option will keep USB devices available only for EFI applications.	
Mass Storage Devices:	[Auto] [Floppy] [Forced FDD] [Hard Disk] [CD-ROM]	Mass storage device emulation type. 'AUTO' enumerates devices according to their media format. Optical drives are emulated as 'CDROM', drives with no media will be emulated according to a drive type.	Information only. Display only when the type of the USB devices attached.

Super I/O Configuration Screen

The Serial Ports screen provides fields to configure the Serial Port [COM Port].

To access this screen from the Main screen, select Advanced | Super IO Configuration.

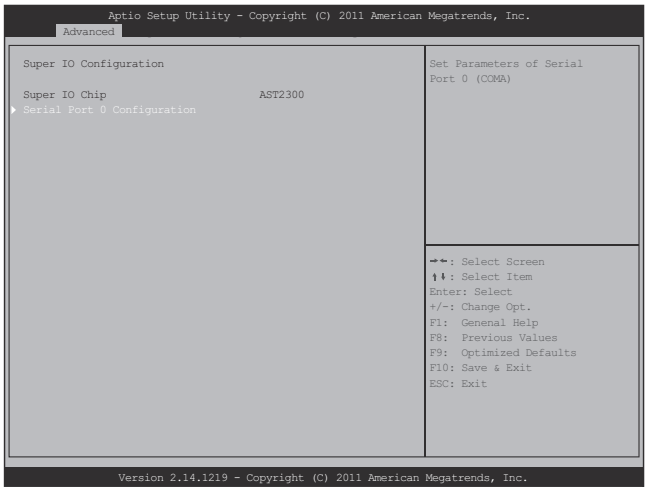

Figure 3-15. Super I/O Configuration Screen

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Table 3-17: Super I/O Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Super IO Chip			Information only. Display Super IO Chip.
Serial Port 0 Configuration		Set Parameters of Serial Port 0 (COMA).	

Serial Port 0 Configuration Screen

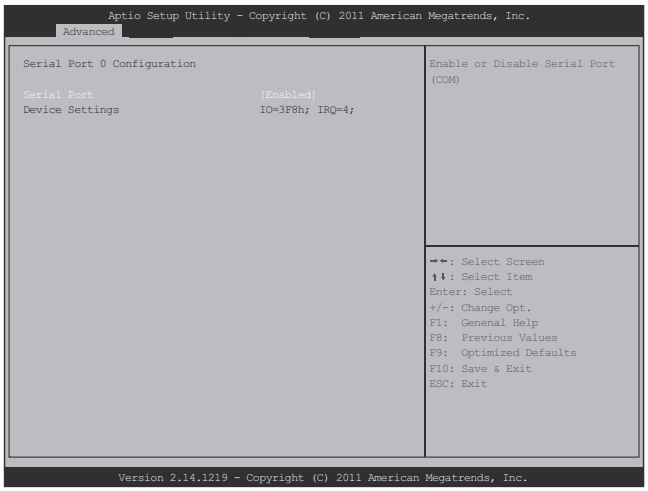

Figure 3-16. Serial Port 0 Configuration Screen

Table 3-18: Serial Port 0 Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Serial Port	[Disabled] **[Enabled]**	Enable or Disable Serial Port (COM).	
Device Settings			Information only. Display Device Settings.

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Onboard Device Configuration Screen

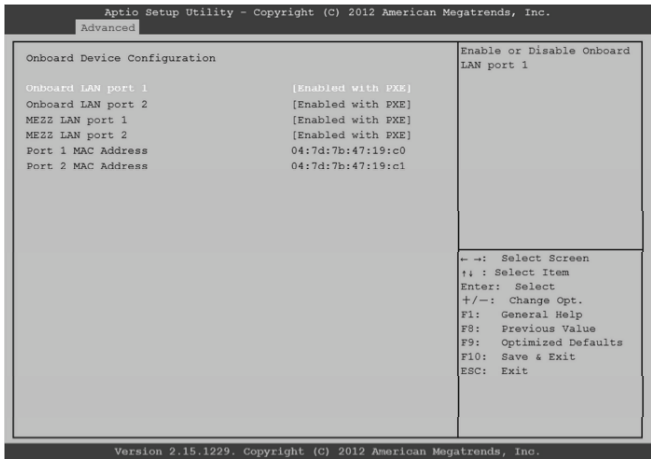

Figure 3-17. Onboard Device Configuration Screen

Table 3-19: Onboard Device Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Onboard LAN port 1	[Disabled] **[Enabled With PXE]** [Enabled Without PXE] [iSCSI Remote Boot]	Enable or Disable Onboard LAN port 1.	
Onboard LAN port 2	[Disabled] **[Enabled With PXE]** [Enabled Without PXE] [iSCSI Remote Boot]	Enable or Disable Onboard LAN port 2.	
MEZZ LAN port 1	**[Enabled With PXE]** [iSCSI Remote Boot]	Control Legacy PXE or iSCSI on MEZZ LAN port 1	The item is available only populate mezz i350 or x540.
MEZZ LAN port 2	**[Enabled With PXE]** [iSCSI Remote Boot]	Control Legacy PXE or iSCSI on MEZZ LAN port 2	The item is available only populate mezz i350 or x540.
Port 1 MAC Address			Information only. Display Port 1 MAC Address.
Port 2 MAC Address			Information only. Display Port 2 MAC Address.

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Console Redirection Screen

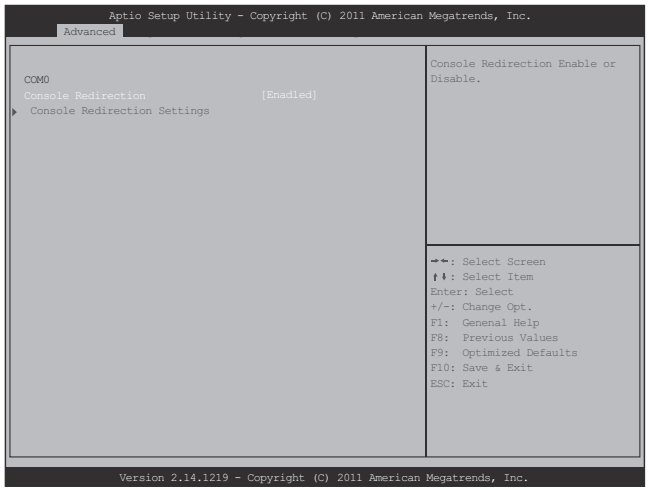

Figure 3-18. Console Redirection Screen

Table 3-20: Console Redirection Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Console Redirection	**[Disabled]** [Enabled]	Console Redirection Enable or Disable.	

Table 3-20: Console Redirection Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Console Redirection Settings		The settings specify how the host computer and the remote computer (which the user is using) will exchange data. Both computers should have the same or compatible settings.	Only accessible if Console Redirection is set to [Enabled].

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Console Redirection Settings

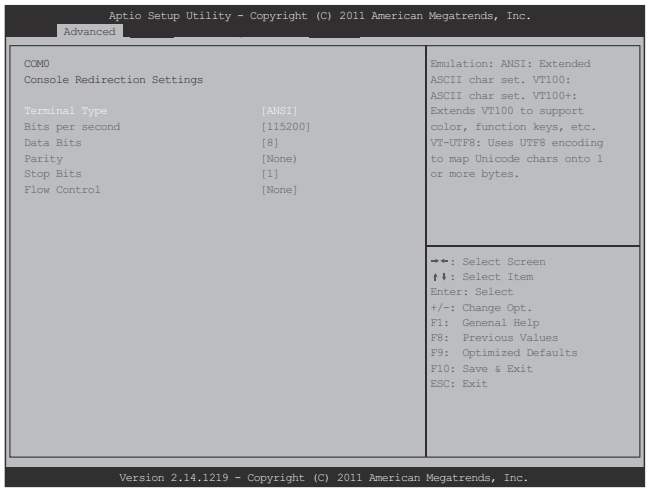

Figure 3-19. Console Redirection Settings Screen

Table 3-21: Console Redirection Settings Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Terminal Type	[VT100] [VT100+] [VT-UTF8] **[ANSI]**	Emulation: ANSI: Extended ASCII char set. VT100: ASCII char set. VT100+: Extends VT100 to support color, function keys, etc. VT-UTF8: Uses UTF8 encoding to map Unicode chars onto 1 or more bytes.	
Bits per second	[9600] [19200] [38400] [57600] **[115200]**	Selects serial port transmission speed. The speed must be matched on the other side. Long or noisy lines may require lower speeds.	
Data Bits	[7] **[8]**	Data Bits.	

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Table 3-21: Console Redirection Settings Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Parity	**[None]** [Even] [Odd] [Mark] [Space]	A parity bit can be sent with the data bits to detect some transmission errors. Even: parity bit is 0 if the num of 1's in the data bits is even. Odd: parity bit is 0 if num of 1's in the data bits is odd. Mark: parity bit is always 1. Space: Parity bit is always 0. Mark and Space Parity do not allow for error detection.	
Stop Bits	**[1]** [2]	Stop bits indicate the end of a serial data packet. (A start bit indicates the beginning). The standard setting is 1 stop bit. Communication with slow devices may require more than 1 stop bit.	

Table 3-21: Console Redirection Settings Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Flow Control	**[None]** [Hardware RTS/CTS]	Flow control can prevent data loss from buffer overflow. When sending data, if the receiving buffers are full, a 'stop' signal can be sent to stop the data flow. Once the buffers are empty, a 'start' signal can be sent to re-start the flow. Hardware flow control uses two wires to send start/stop signals.	

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iSCSI Configuration

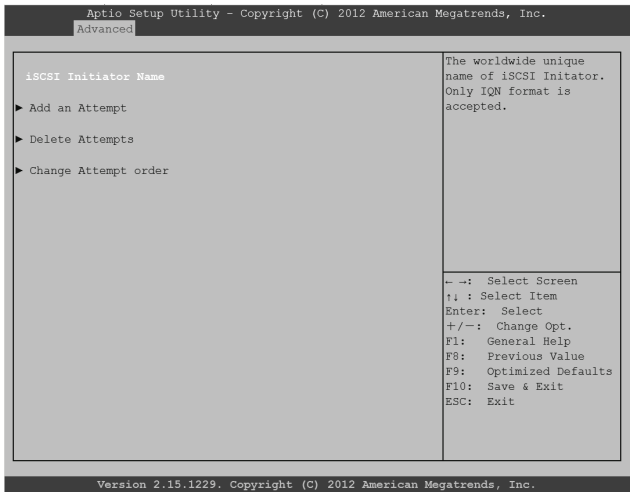

Figure 3-20. iSCSI Configuration Screen

Table 3-22: Console Redirection Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
iSCSI Initatiator Name		The worldwide unique name of iSCSI Initator. Only IQN format is accepted.	
Add an Attempt		Add an Attempt	

Table 3-22: Console Redirection Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Delete Attempts		Delete one or more attempts.	
Change Attempt Order		Change the order of Attempts using +/- keys, Use arrow keys to select the attempt then press +/- to move the attempt up/down in the attempt oder list.	

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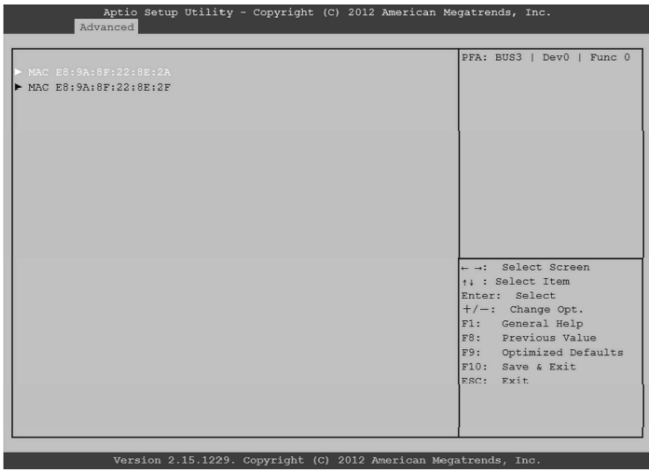

Figure 3-21.

Table 3-23:

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
MAC xx:xx:xx:xx:xx:xx		PFA: BUSx | Devx | Func x	MAC address and BUS/ Dev/Func are dependent on platform

Add an Attempt

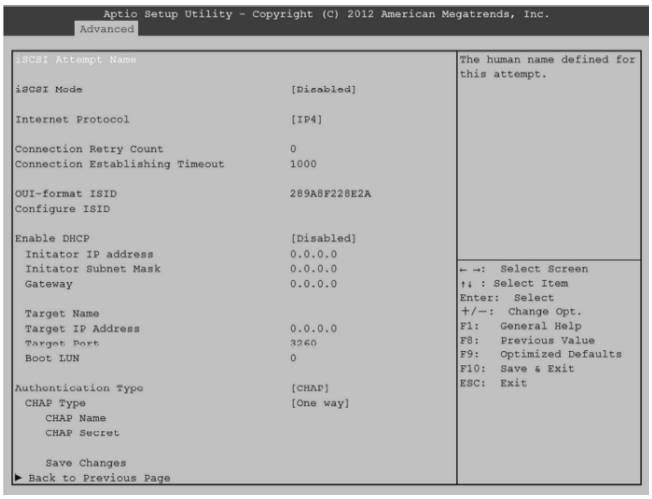

Figure 3-22. Add an Attempt Screen

Table 3-24: Console Redirection Settings Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
iSCSI Attempt Name		The human name defined for this attempt.	
iSCSI Mode	**[Disabled]** [Enabled] [Enabled for MPIO]	Disabled, Enabled, Enabled for MPIO	

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Table 3-24: Console Redirection Settings Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Internet Protocol	**IP4]** [IP6] [Autoconfigure]	Initiator IP address is system assigned in IP6 mode. In Autoconfigure mode, iSCSI driver will attempt to connect iSCSI target via IPv4 stack, if failed then attempt IPv6 stack	
Connect Retry Count		The minimum value is 0 and the maximum is 16. 0 means no retry	
Connection Establishing Time out		The timeout value in milliseconds. The minimum value is 100 milliseconds and the maximum is 20 seconds.	
OUI-format ISID			Information only. Display MAC Address.
Configure ISID		OUI-format ISID in 6bytes, default value are derived from MAC address. Only last 3 bytes are configurable. Example:update 0ABBCCDDEEFF to 0ABBCCF07900 by input F07901	

Table 3-24: Console Redirection Settings Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Enable DHCP	**[Disabled]** [Enabled]	Enable DHCP	
Initiator IP address		Enter IP address in dotted-decimal notation	
Initiator Subnet Mask		Enter IP address in dotted-decimal notation	
Gateway		Enter IP address in dotted-decimal notation	
Target Name		The worldwide unique name of iSCSI Initator. Only IQN format is accepted.	
Target IP address		Enter IP address in dotted-decimal notation	
Target IP Port		Target IP Port	
Boot LUN		Hexadecimal representation of the LU number. Examples are : 4752-3A4F-6b7e-2F99, 6734-9-156f-127, 4186-9	
Authentication Type	**[Chap]** [None]	Authentication method: CHAP, Kerberos, or None	
CHAP Type	**[One way]** [Mutual]	None, One way CHAP or Mutual CHAP	

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Table 3-24: Console Redirection Settings Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
CHAP Name		CHAP Name	
CHAP Secret		The minimum length is 12 bytes and the maximum length is 16 bytes.	
Save Changes		Must reboot system manually for changes to take place.	
Back to Previous Page		Back to Previous Page	

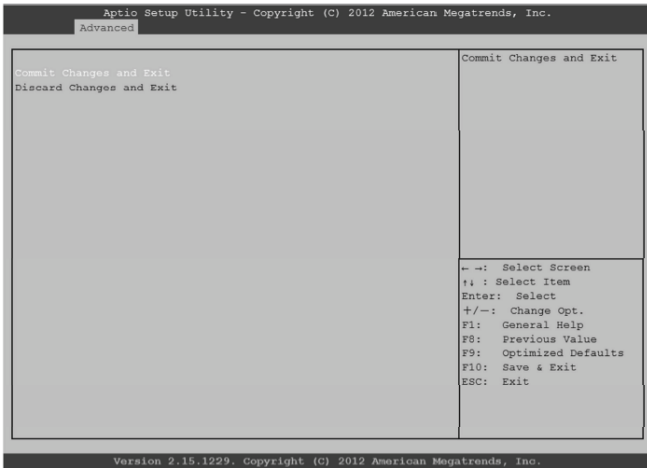

Figure 3-23.

Table 3-25:

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Commit Changes and Exit		Commit Changes and Exit	
Discard Changes and Exit		Discard Changes and Exit	

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Chipset Screen

The Chipset screen provides an access point to configure several options. On this screen, the user selects the option that is to be configured. Configurations are performed on the selected screen, not directly on the Chipset screen.

To access this screen from the Main screen, press the right arrow until the Chipset screen is chosen.

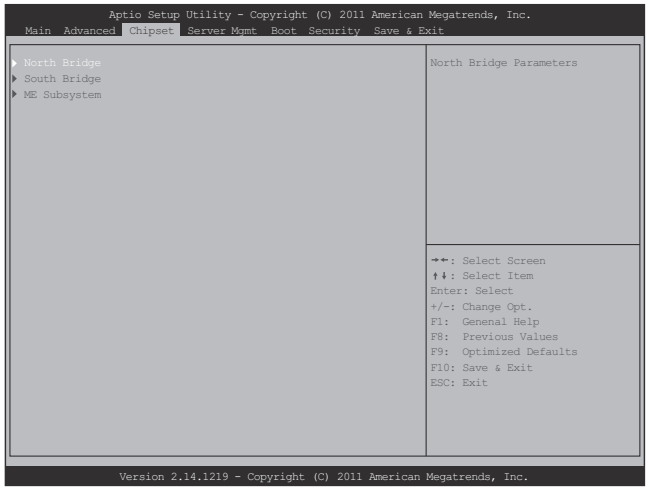

Figure 3-24. Chipset Screen

Table 3-26: Chipset Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
North Bridge		North Bridge Parameters.	
South Bridge		South Bridge Parameters.	
ME Subsystem		ME Subsystem Parameters.	

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North Bridge Screen

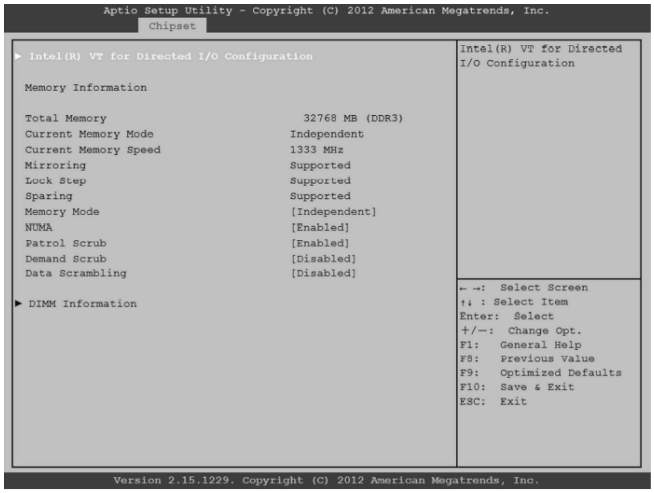

Figure 3-25. North Bridge Configuration Screen

Table 3-27: North Bridge Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Intel® VT for Directed I/O Configuration		Intel® VT for Directed I/O Configuration.	
Total Memory			Information only. Displays the Total Memory.

Table 3-27: North Bridge Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Current Memory Mode			Information only. Displays the Current Memory Mode.
Current Memory Speed			Information only. Displays the Current Memory Speed.
Mirroring			Information only. Displays the Miorring support state.
Lock Step			Information only. Displays the Lock Step support state.
Sparing			Information only. Displays the Sparing support state.
Memory Mode	[**Independent**] [Mirroring] [Lock Step] [Sparing]	Select the mode for memory initialization.	If unsupported memory mode is selected, BIOS will use “Independent” as current memory mode during next boot.

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Table 3-27: North Bridge Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
NUMA	[Disabled] [**Enabled**]	Enable or Disable Non uniform Memory Access (NUMA).	
Patrol Scrub	[Disabled] [**Enabled**]	Enable/Disable Patrol Scrub	
Demand Scrub	[**Disabled**] [Enabled]	Enable/Disable Demand Scrubing Feature	
Data Scrambling	[**Disabled**] [Enabled]	Enable/Disable Data Scrambling.	Only for S2T.
DIMM Information			Information only. Displays the DIMM information according to each platform.

Intel(R) VT-d Screen

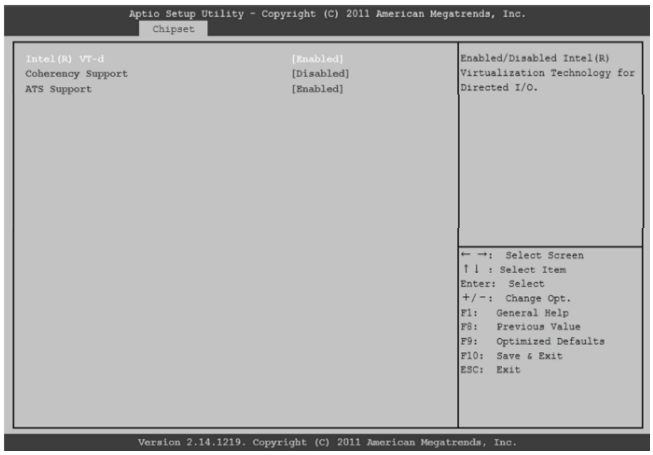

Figure 3-26. Intel(R) VT-d Screen

Table 3-28: Intel(R) VT-d Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Intel® VT-d	[**Disabled**] [Enabled]	Enable/Disable Intel® Virtualization Technology for Directed I/O.	

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Table 3-28: Intel(R) VT-d Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Coherency Support	[**Disabled**] [Enabled]	Enable/Disable VT-d Engine Coherency support	
ATS Support	[Disabled] [**Enabled**]	Enable/Disable VT-d Engine Address Translation Services (ATS) support.	

South Bridge Configuration Screen

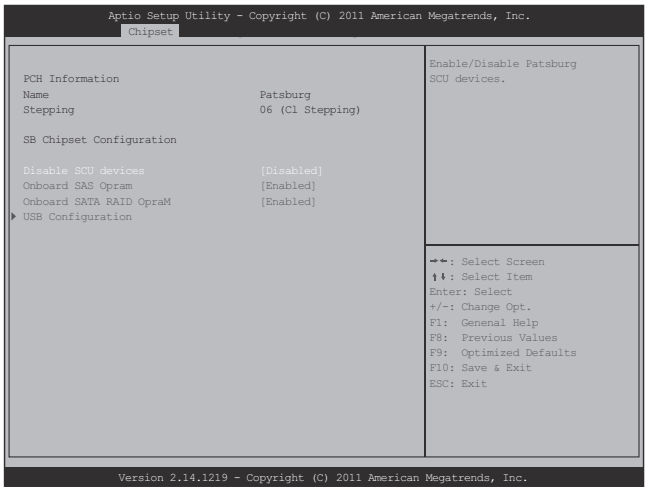

Figure 3-27. South Bridge Configuration Screen

Table 3-29: South Bridge Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Disable SCU devices	[**Disabled**] [Enabled]	Enable/Disable Patsburg SCU devices.	

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Table 3-29: South Bridge Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Onboard SAS Oprom	[Disabled] [**Enabled**]	Enabled/Disabled onboard SAS option rom if Launch Storage OpROM is enabled.	
Onboard SATA RAID Oprom	[Disabled] [**Enabled**]	Enabled/Disabled onboard SATA RAID option rom if Launch Storage OpROM is enabled.	
USB Configuration		USB Configuration.	

USB Configuration

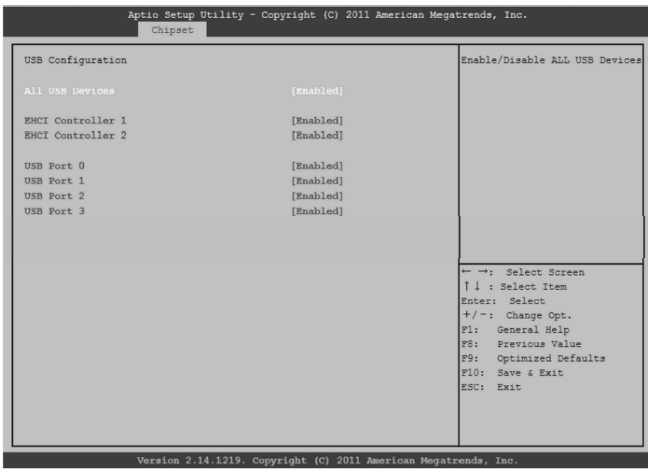

Figure 3-28. USB Configuration Screen

Table 3-30: USB Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
All USB Devices	[Disabled] **[Enabled]**	Enabled/Disabled ALL USB Devices.	

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Table 3-30: USB Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
EHCI Controller 1	[Disabled] **[Enabled]**	Enabled/Disabled USB EHCI Controller 1.	Disable the EHCI Controller would disable all USB ports from it.
EHCI Controller 2	[Disabled] **[Enabled]**	Enabled/Disabled USB EHCI Controller 2.	Disable the EHCI Controller would disable all USB ports from it. (Only for S2RS.)
USB Port 0	[Disabled] **[Enabled]**	Enabled/Disabled USB Port 0.	USB Port 0
USB Port 1	[Disabled] **[Enabled]**	Enabled/Disabled USB Port 1.	USB Port 1
USB Port 2	[Disabled] **[Enabled]**	Enabled/Disabled USB Port 2.	USB Port 2
USB Port 3	[Disabled] **[Enabled]**	Enabled/Disabled USB Port 3.	USB Port 3

ME Subsystem Screen

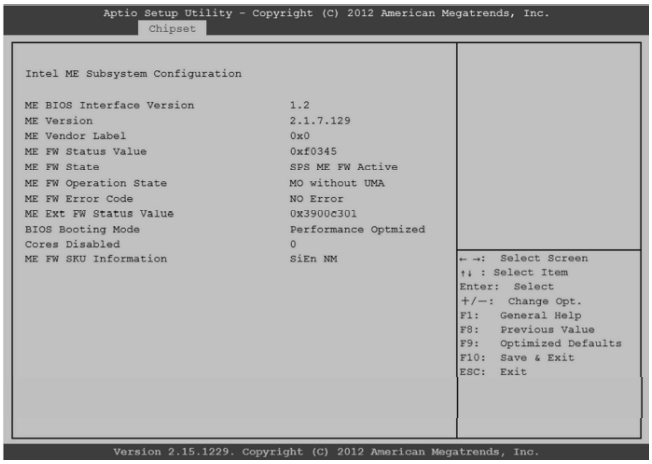

Figure 3-29. ME Subsystem Screen

Table 3-31: ME Subsystem Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
ME BIOS Interface Version			Information only. Displays the ME BIOS Interface Version.

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Table 3-31: ME Subsystem Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
ME Version			Information only. Displays the ME Version.
ME Vendor Label			Information only. Displays the ME Vendor Label.
ME FW Status Value			Information only. Displays the ME FW Status Value.
ME FW State			Information only. Displays the ME FW State.
ME FW Operation State			Information only. Displays the ME FW Operation State.
ME FW Error Code			Information only. Displays the ME FW Error Code.
ME Ext FW Status Value			Information only. Displays the ME Ext FW Status Value.
BIOS Booting Mode			Information only. Displays the BIOS Booting Mode.

Table 3-31: ME Subsystem Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Cores Disabled			Information only. Displays the Cores Disabled.
ME FW SKU Information			Information only. Displays the ME FW SKU Information.

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Server Management Screen

The Server Management screen displays information of the BMC, and allows the user to configure desired settings.

To access this screen from the Main screen, select Server Mgmt Options.

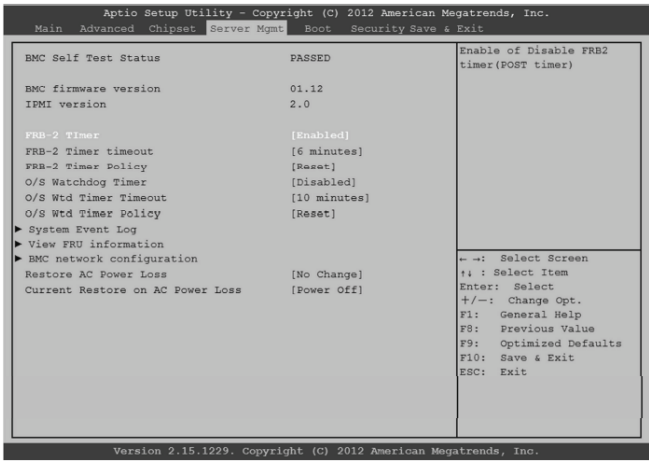

Figure 3-30. Server Management Configuration Screen

Table 3-32: Server Management Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
BMC firmware version			BMC firmware version.
IPMI version			IPMI version.
FRB-2 Timer	[**Enabled**] [Disabled]	Enable or Disable FRB2 timer (POST timer).	
FRB-2 Timer timeout	[3 minutes] [4 minutes] [5 minutes] [**6 minutes**]	Enter value Between 3 to 6 min for FRB2 Timer Expiration value.	Not available if FRB2 Timer is disabled.
FRB-2 Timer Policy	[Do Nothing] [**Reset**] [Power Down]	Configure how the system should respond if the FRB2 Timer expires. Not available if FRB2 Timer is disabled.	

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Table 3-32: Server Management Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
O/S Watchdog Timer	[Enabled] [**Disabled**]	If enabled, starts a BIOS timer which can only be shut off by Intel Management Software after the OS loads. Helps determine that the OS successfully loaded or follows the O/S Boot Watchdog Timer policy.	
O/S Wtd Timer Timeout	[5 minutes] [**10 minutes**] [15 minutes] [20 minutes]	Configure the length of the O/S Boot Watchdog Timer. Not available if O/S Boot Watchdog Timer is disabled.	
O/S Wtd Timer Policy	[Do Nothing] [**Reset**] [Power Down]	Configure how the system should respond if the O/S Boot Watchdog Timer expires. Not available if O/S Boot Watchdog Timer is disabled.	

Table 3-32: Server Management Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
System Event Log		Press <Enter> to change the SEL event log configuration.	
View FRU information		Press <Enter> to view FRU information.	
BMC network configuration		Configure BMC network parameters.	
Restore on AC Power Loss	[Power On] [Power Off] [Last State] [**No Change**]	System action to take on AC power loss.	
Current Restore on AC Power Loss			Information only. Displays Restore AC Power Loss version

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System Event Log Screen

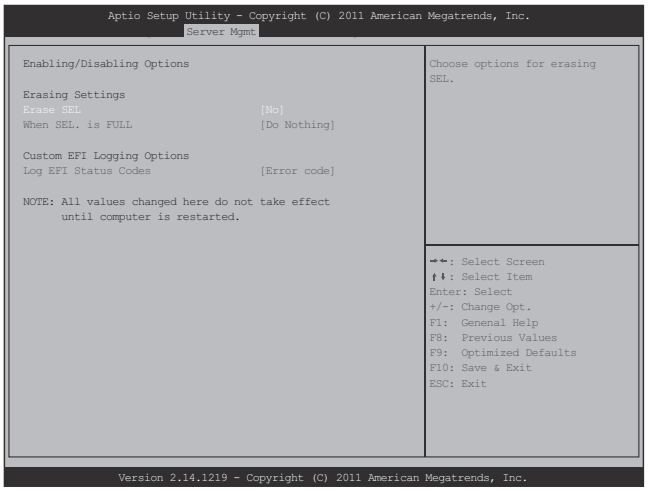

Figure 3-31. System Event Log Screen

Table 3-33: System Event Log Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Erase SEL	[**No**] [Yes, On next reset] [Yes, On every reset]	Choose options for erasing SEL.	

Table 3-33: System Event Log Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
When SEL is Full	[**Do Nothing**] [Erase Immediately]	Choose options for reactions to a full SEL.	
Log EFI Status Codes	[Disabled] [Both] [**Error code**] [Progress code]	Disable the logging of EFI Status Codes or log only error code or only progress code or both.	

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FRU Information

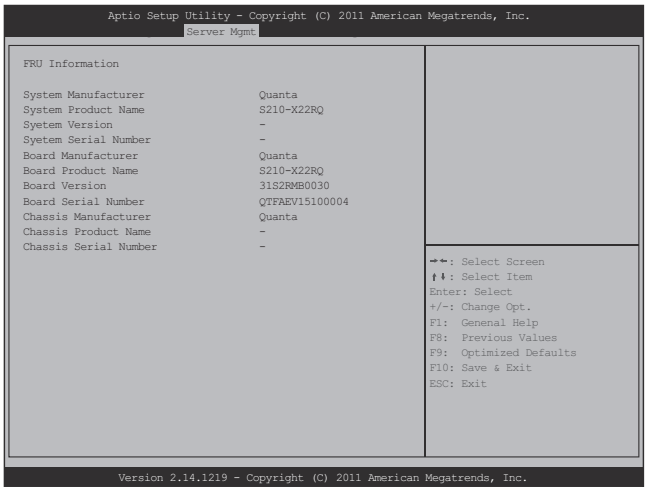

Figure 3-32. FRU Information Screen

Table 3-34: FRU Information Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
System Manufacturer			Information only. Displays the System Manufacturer.

Table 3-34: FRU Information Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
System Product Name			Information only. Displays the System Product Name.
System Version			Information only. Displays the System Version.
System Serial Number			Information only. Displays the System Serial Number.
Board Manufacturer			Information only. Displays the Board Manufacturer.
Board Product Name			Information only. Displays the Board Product Name.
Board Version			Information only. Displays the Board Version.
Board Serial Number			Information only. Displays the Board Serial Number.

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Table 3-34: FRU Information Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Chassis Manufacturer			Information only. Displays the Chassis Manufacturer.
Chassis Product Name			Information only. Displays the Chassis Product Name.
Chassis Serial Number			Information only. Displays the Chassis Serial Number.

BMC Network Configuration

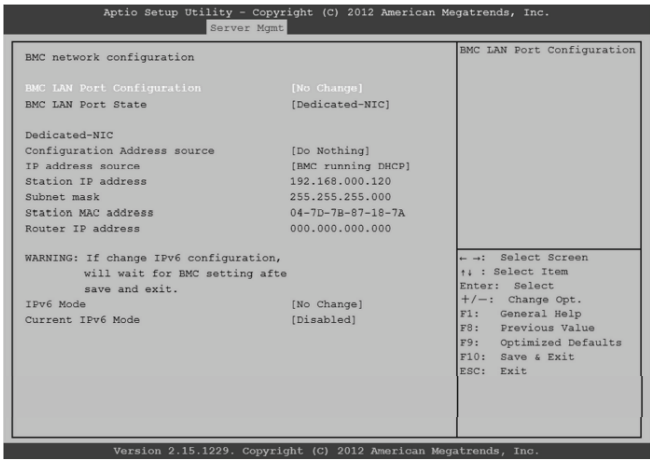

Figure 3-33. BMC Network Configuration Screen

Table 3-35: BMC Network Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
BMC LAN Port Configuration	[Dedicated-NIC] [Shared-NIC] **[No Change]**	BMC LAN Port Configuration.	Options for send to Dedicated NIC or Shared NIC

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Table 3-35: BMC Network Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
BMC LAN Port State			Information only. Displays the BMC LAN Port State.
Configuration Address source	[Static on next reset] [Dynamic-obtain by BMC] **[Do Nothing]**	Select to configure LAN channel parameters statically or dynamically (DHCP). Do nothing option will not modify any BMC network parameters during BIOS phase.	
IP address source			Information only. Displays the IP address source.
Station IP address		Enter station IP address	Available if Configuration Address is static on next reset.
Subnet mask		Enter subnet mask	Available if Configuration Address is static on next reset.

Table 3-35: BMC Network Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Station MAC address			Information only. Displays the Station MAC address.
Router IP address		Enter Router IP address	Available if Configuration Address is static on next reset.
IPv6 Mode	[Disabled] [Enabled] **[No Change]**	Disabled/Enabled IPv6 BMC LAN channel function. Disable option will not modify any BMC network during BIOS Phase.	
IPv6 AutoConfig	[Unspecified] **[Static]** [Dynamic-Obtained by BMC running DHCP]	Select to configure LAN channel parameter statically or dynamically (by BIOS or BMC)	Options only show when "IPv6 Mode" = [Enabled]
IPv6 IP Address		Enter IPv6 BMC LAN IP address	Options only show when "IPv6 Mode" = [Enabled] and "IPv6 AutoConfig" = [Disabled]

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Table 3-35: BMC Network Configuration Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
IPv6 Prefix Length		Enter IPv6 BMC LAN IP Prefix Length	Options only show when "IPv6 Mode" = [Enabled] and "IPv6 AutoConfig" = [Disabled]
IPv6 GateWay Address		Enter IPv6 BMC LAN Default Gateway	Options only show when "IPv6 Mode" = [Enabled] and "IPv6 AutoConfig" = [Disabled]

Boot Option Screen

The Boot Options screen displays any bootable media encountered during POST, and allows the user to configure desired boot device.

To access this screen from the Main screen, select Boot Options.

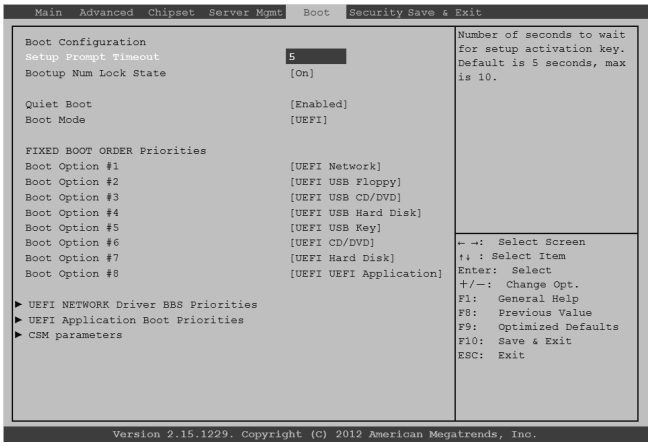

Figure 3-34. Boot Option Screen

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Table 3-36: Boot Option Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Setup Prompt Timeout	[<number>]	Number of seconds to wait for setup activation key. Max = 10 seconds.	
Boot up NumLock State	**[On]** [Off]	Select the keyboard NumLock state.	
Quiet Boot	**[Disabled]** [Enabled]	Enables or disables Quiet Boot option.	Default "Enabled" for S2RS
Boot Mode	[Legacy] **[UEFI]**	Enables/Disable CSM Support. If Auto is selected, based on OS, CSM will be enabled/ disabled automatically.	
1st Boot	[<Device String 1>] [<Device String 2>] … [Disabled]	Sets the system boot order.	
2nd Boot	[<Device String 1>] [<Device String 2>] … [Disabled]	Sets the system boot order.	

Table 3-36: Boot Option Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
3rd Boot	[<Device String 1>] [<Device String 2>] … [Disabled]	Sets the system boot order.	
4th Boot	[<Device String 1>] [<Device String 2>] … [Disabled]	Sets the system boot order.	
5th Boot	[<Device String 1>] [<Device String 2>] … [Disabled]	Sets the system boot order.	
6th Boot	[<Device String 1>] [<Device String 2>] … [Disabled]	Sets the system boot order.	

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Table 3-36: Boot Option Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
7th Boot	[<Device String 1>] [<Device String 2>] … [Disabled]	Sets the system boot order.	
CD/DVD ROM Drive BBS Priorities		Specifies the Boot Device Priority sequence from available CD/DVD Drives.	
Hard Disk Drive BBS Priorities		Specifies the Boot Device Priority sequence from available Hard Drives.	
USB Floppy Drive BBS Priorities		Specifies the Boot Device Priority sequence from available USB Floppy Drives.	
USB CD/DVD ROM Drive BBS Priorities		Specifies the Boot Device Priority sequence from available USB CD/DVD Drives.	

Table 3-36: Boot Option Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
USB Hard Disk Drive BBS Priorities		Specifies the Boot Device Priority sequence from available USB Hard Disk Drives.	
USB KEY Drive BBS Priorities		Specifies the Boot Device Priority sequence from available USB KEY Drives.	
NETWORK Device BBS Priorities		Specifies the Boot Device Priority sequence from available NETWORK Drives.	Only selectable when this type device exists
CSM parameters		OpROM execution, boot options filter, etc.	

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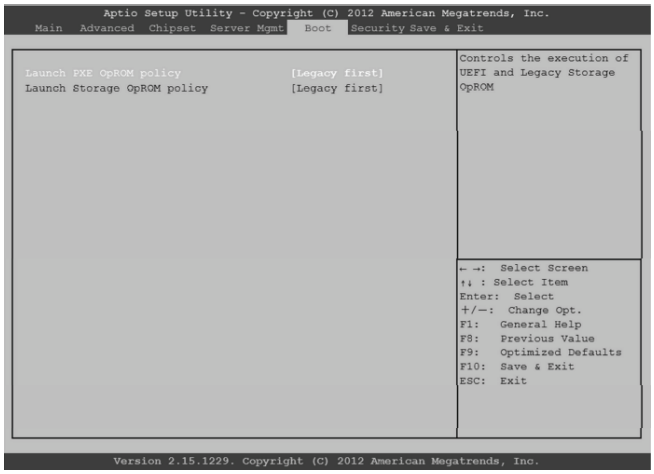

Figure 3-35.

Table 3-37:

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Launch PXE OpROM policy	[Do not launch] [UEFI only] [Legacy only] [**Legacy first**] [UEFI first]	Controls the execution of UEFI and Legacy PXE OpROM.	

Table 3-37: (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Launch Storage OpROM policy	[Do not launch] [UEFI only] [Legacy only] [**Legacy first**] [UEFI first]	Controls the execution of UEFI and Legacy Storage OpROM.	

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Security Screen

The Security screen provides fields to enable and set the user and administrative password and to lockout the front panel buttons so they cannot be used.

To access this screen from the Main screen, select the Security option.

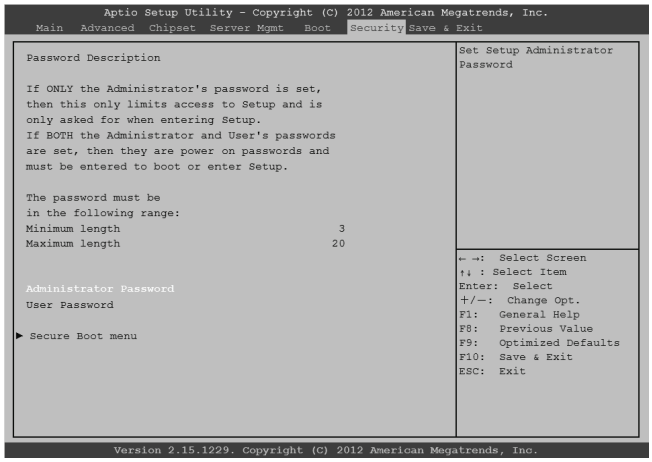

Figure 3-36. Security Configuration Screen

Table 3-38: Security Configuration Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Administrator Password		Set Setup Administrator Password.	
User Password		Set User Password.	
Secure Boot menu		Customizable Secure Boot Settings	

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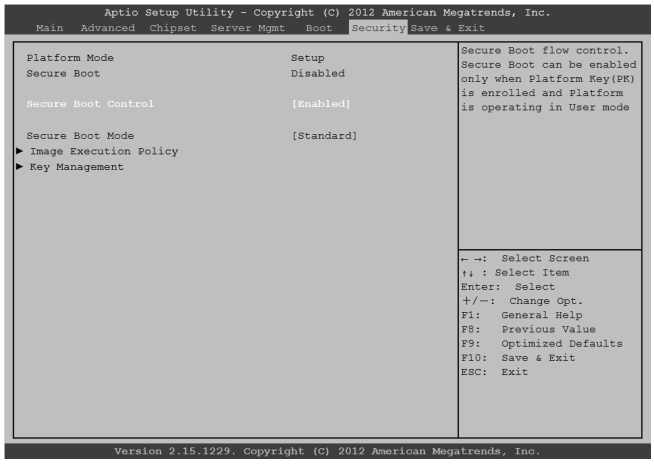

Figure 3-37.

Table 3-39:

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Platform Mode			Information only. Displays the Platform Mode state.

Table 3-39: (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Secure Boot			Information only. Displays the Secure Boot Mode state.
Secure Boot Control	[**Enabled**] [Disable]	Secure Boot flow control. Secure Boot can enabled only wehn 1. Platform Keys(PK) is enrolled and Platform is operation in User mode and 2. CSM function is disabled in Setup.	
Secure Boot state	[**Standard**] [Custom]	Secure Boot mode selector. “Custom” Mode allows for more flexibility changing Image Execution Policy and Secure Boot Key management.	
Image Execution Policy		Manage Image Execution Policy on Security Violation	
Key Management		Key Management page, enables user to modify Secure Boot variables.	

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Figure 3-38.

Table 3-40:

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Internal FV	[**Always Execute**]	Image Execution Policy on Security Violation. Image load device path	

Table 3-40: (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Option ROM	[Always Execute] [Always Deny] [Allow Execute] [Defer Execute] [**Deny Execute**] [Query User]	Image Execution Policy on Security Violation. Image load device path	
Removable Media	[Always Execute] [Always Deny] [Allow Execute] [Defer Execute] [**Deny Execute**] [Query User]	Image Execution Policy on Security Violation. Image load device path	

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Table 3-40: (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Fixed Media	[Always Execute] [Always Deny] [Allow Execute] [Defer Execute] [**Deny Execute**] [Query User]	Image Execution Policy on Security Violation. Image load device path	

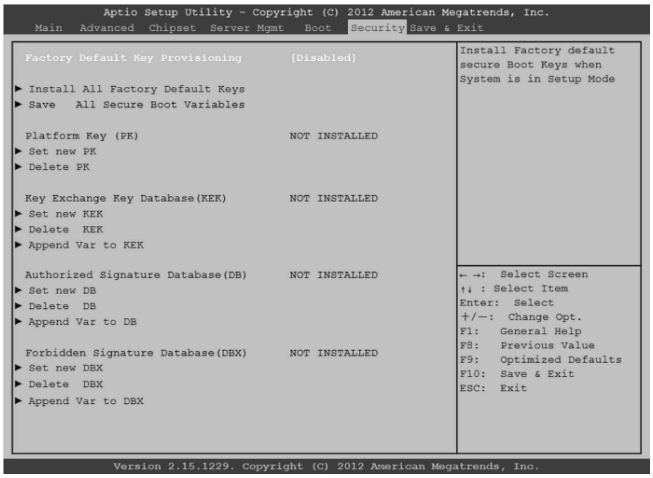

Figure 3-39.

Table 3-41:

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Factory Default Key Provisioning	[**Disabled**] [Enabled]	Install Factory default Secure Boot Keys when system is in Setup Mode.	

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Table 3-41: (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Install All Factory default keys		Force System to User Mode - install all Factory Default keys (PK, KEK, db, dbx.) Change takes effect after reboot.	
Set (PK, KEK, DB, DBX) from File		Launches the file browser to set Efi Variable from the file. The file data must be formatted as Efi Variable with TimeBased Authenticated Header.	
Get (PK, KEK, DB, DBX) to File		Dump content of the Variable to a file with a matching name in selected file system's root	
Delete the (PK, KEK, DB, DBX)		Delete the Variable	
Append an entry to (KEK, DB, DBX)		Launches the file browser to Append new Signature Database from the file. The file data must be formatted as Efi Variable with TimeBased Authenticated Header	

Exit Screen

The Exit screen allows the user to choose to save or discard the configuration changes made on the other screens. It also provides a method to restore the server to the factory defaults or to save or restore a set of user defined default values. If Restore Defaults is selected, the default settings, noted in bold in the tables in this chapter, will be applied. If Restore User Default Values is selected, the system is restored to the default values that the user saved earlier, instead of being restored to the factory defaults.

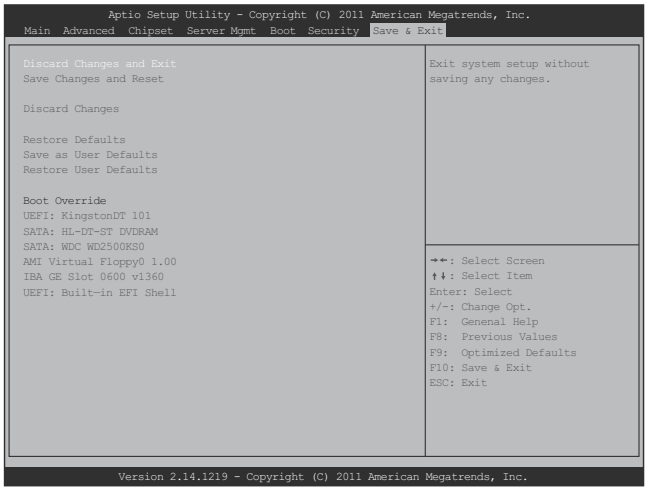

Figure 3-40. Exit Screen

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Table 3-42: Exit Fields

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
Discard Changes and Exit		Exit system setup without saving any changes.	
Save Changes and Reset		Reset the system after saving the changes.	
Discard Changes		Discards changes done so far to any of the setup questions.	
Restore Defaults		Restore/Load Default values for all the setup options.	
Save as User Defaults		Save the changes done so far as User Defaults.	
Restore User Defaults		Restore the User Defaults to all the setup options.	
[<Device String 1>]			Boot with Device <Device String 1>.
[<Device String 2>]			Boot with Device <Device String 2>.

Table 3-42: Exit Fields (Continued)

SETUP ITEM	OPTIONS	HELP TEXT	COMMENTS
[<Device String 3>]			Boot with Device <Device String 3>.
[<Device String 4>]			Boot with Device <Device String 4>.
[<Device String 5>]			Boot with Device <Device String 5>.
[<Device String 6>]			Boot with Device <Device String 6>.

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Loading BIOS Defaults

Different mechanisms exist for resetting the system configuration to the default values. When a request to reset the system configuration is detected, the BIOS loads the default system configuration values during the next POST. The request to reset the system to the defaults can be sent in the following way: A request to reset the system configuration can be generated by pressing <**F9**> from within the BIOS Setup utility.

A reset system configuration request can be generated by moving the clear system configuration jumper.

BIOS settings are stored on NVRAM. Only the D4 setting uses the CMOS. BIOS implements a mechanism to clear NVRAM and CMOS. BIOS does not show CMOS error messages after the CMOS has been cleared.

After clearing CMOS by battery or jumper, all variables in the BIOS setup Menu load with the default values. BMC related variables, like "Set BMC LAN Configuration" in BIOS setup menu, are able to synchronize from BMC.

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3.2 BIOS Update Utility

The flash ROM contains system initialization routines, the BIOS Setup Utility, and runtime support routines. The exact layout is subject to change, as determined by BIOS. The flash ROM also contains initialization code in compressed form for onboard peripherals, like SCSI, NIC and video controllers. The complete ROM is visible, starting at physical address 4 GB minus the size of the flash ROM device.

A 16-KB parameter block in the flash ROM is dedicated to storing configuration data that controls the system configuration (ESCD). Application software must use standard APIs to access these areas; application software cannot access the data directly.

BIOS Update Utility

Server platforms support the DOS-based, Windows-based, and Linux-based firmware update utility. This utility loads a fresh copy of the BIOS into the flash ROM.

The BIOS update may affect the following items:

- The system BIOS, including the recovery code, setup utility and strings.
- Onboard video BIOS, RAID BIOS, and other option ROMS for the devices embedded on the server board.
- Memory reference code.
- Microcode updates.

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Recovery Mode

Recovery process can be initiated by setting the recovery jumper. At startup BIOS detects the recovery jumper position and starts to execute the recovery code.

BIOS consists of the following three sections: Main BIOS, NVRAM, and Boot Block Recovery. The Main BIOS section will be updated during the recovery procedure. The NVRAM Section and Boot Block will be preserved.

A BIOS recovery can be accomplished from a USB mass storage device that includes the BIOS image file called *S2RQ_REC.ROM*. To recover BIOS, follow the steps in “Recovery Flow” section.

Recovery Flow

The BIOS recovery procedure is as follows:

1. Download the BIOS image file from the following web address: www.QuantaQCT.com.
2. Plug in a removable USB disk.
3. Save the BIOS image file into the USB mass storage disk.

   Example image file name: *S2RQ_REC.ROM*
4. Power off the system and all peripherials.
5. Short the recovery jumper.
6. Plug in the removable USB disk with the BIOS recovery file.
7. Power on the system.

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BIOS will automatically boot into Recovery page in BIOS setup menu.

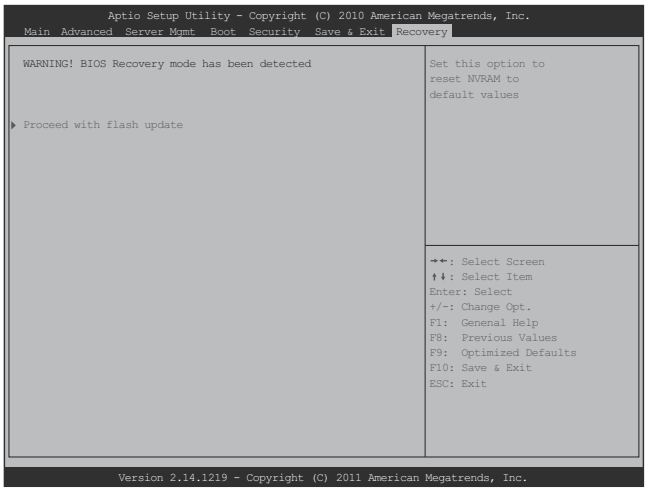

Figure 3-41. Recovery Page

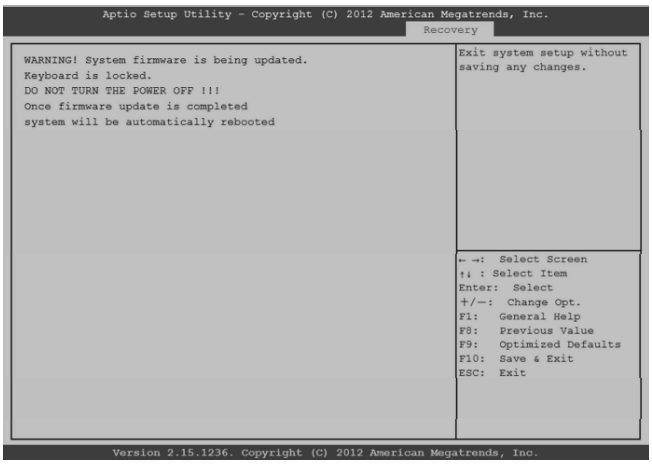

Figure 3-42. BIOS Recovery Completed

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3.3 Server Management

The BIOS supports many standard-based server management features and several proprietary features. The Intelligent Platform Management Interface (IPMI) is an industry standard and defines standardized, abstracted interfaces to platform management hardware. The BIOS implements many proprietary features that are allowed by the IPMI specification, but these features are outside the scope of the IPMI specification. This section describes the implementation of the standard and proprietary features.

Console Redirection

The BIOS supports redirection of both video and keyboard via a serial link (serial port). When console redirection is enabled, the local, or host server, keyboard input and video output are passed both to the local keyboard and video connections, and to the remote console through the serial link. Keyboard inputs from both sources are considered valid and video is displayed to both outputs.

As an option, the system can be operated without a host keyboard or monitor attached to the system and run entirely via the remote console. Utilities that can be executed remotely include BIOS Setup.

Serial Configuration Settings

For optimal configuration of Serial Over LAN or EMP see the BMC Specification.

The BIOS does not require that the splash logo be turned off for console redirection to function. The BIOS supports multiple consoles, some of which are in graphics mode and some in text mode. The graphics consoles can display the logo and the text consoles receive the redirected text.

Keystroke Mapping

During console redirection, the remote terminal sends keystrokes to the local server. The remote terminal can be a dumb terminal with a direct connection and running a communication program. The keystroke mapping follows VT-UTF8 format with the following extensions.

Table 3-43: Keystroke mappings

KEY	ANSI ESCAPE SEQUENCE	WINDOWS PLATFORM DESIGN NOTE
F1	<ESC><Shift>op	<ESC>1
F2	<ESC><Shift>oq	<ESC>2
F3	<ESC><Shift>or	<ESC>3
F4	<ESC><Shift>os	<ESC>4

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Table 3-43: Keystroke mappings (Continued)

KEY	ANSI ESCAPE SEQUENCE	WINDOWS PLATFORM DESIGN NOTE
F5		<ESC>5
F6		<ESC>6
F7		<ESC>7
F8		<ESC>8
F9		<ESC>9
F10		<ESC>0
F11		<ESC>!
F12		<ESC>@
Home	<ESC>[<Shift>h	<ESC>h
End	<ESC>[<Shift>k	<ESC>k
Ins		<ESC>+
Del		<ESC>-
Page Up		<ESC>?
Page Down		<ESC>/
System Reset		<ESC>R<ESC>r<ESC>R

Limitations

- BIOS Console redirection terminates after an operating system has being loaded. The operating system is responsible for continuing console redirection after that.
- BIOS console redirection is a text console. Graphical data, such as a logo, are not redirected.

Interface to Server Management

If the BIOS determines that console redirection is enabled, it will read the current baud rate and pass this value to the appropriate management controller via the Intelligent Platform Management Bus (IPMB).

PXE BIOS Support

The BIOS supports the EFI PXE implementation. To utilize this, the user must load EFI Simple Network Protocol driver and the UNDI driver specific for the network interface card being used. The UNDI driver should be included with the network interface card. The Simple Network Protocol driver can be obtained from http://developer.intel.com/technology/framework.

The BIOS supports legacy PXE option ROMs in legacy mode and includes the necessary PXE ROMs in the BIOS image for the onboard controllers. The legacy PXE ROM is required to boot a non-EFI operating system over the network.

Checkpoints

A checkpoint is either a byte or word value output to SuperIO printer port. The BIOS outputs checkpoints throughout boot-block and Power-On Self Test (POST) to indicate the task the system is currently executing. Checkpoints are very useful in

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aiding software developers or technicians in debugging problems that occur during the pre-boot process.

Checkpoint Ranges

Table 3-44: Checkpoint Ranges

STATUS CODE RANGE	DESCRIPTION
0x01 – 0x0B	SEC execution
0x0C – 0x0F	SEC errors
0x10 – 0x2F	PEI execution up to and including memory detection
0x30 – 0x4F	PEI execution after memory detection
0x50 – 0x5F	PEI errors
0x60 – 0x8F	DXE execution up to BDS
0x90 – 0xCF	BDS execution
0xD0 – 0xDF	DXE errors
0xE0 – 0xE8	S3 Resume (PEI)
0xE9 – 0xEF	S3 Resume errors (PEI)
0xF0 – 0xF8	Recovery (PEI)
0xF9 – 0xFF	Recovery errors (PEI)

Standard Checkpoints

SEC Phase

Table 3-45: SEC Phase

STATUS CODE	DESCRIPTION
0x00	Not used
Progress Codes	
0x01	Power on. Reset type detection (soft/hard).
0x02	AP initialization before microcode loading
0x03	North Bridge initialization before microcode loading
0x04	South Bridge initialization before microcode loading
0x05	OEM initialization before microcode loading
0x06	Microcode loading
0x07	AP initialization after microcode loading
0x08	North Bridge initialization after microcode loading
0x09	South Bridge initialization after microcode loading
0x0A	OEM initialization after microcode loading
0x0B	Cache initialization
SEC Error Codes	
0x0C – 0x0D	Reserved for future AMI SEC error codes
0x0E	Microcode not found
0x0F	Microcode not loaded

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PEI Phase

Table 3-46: PEI Phase

STATUS CODE	DESCRIPTION
Progress Codes	
0x10	PEI Core is started
0x11	Pre-memory CPU initialization is started
0x12	Pre-memory CPU initialization (CPU module specific)
0x13	Pre-memory CPU initialization (CPU module specific)
0x14	Pre-memory CPU initialization (CPU module specific)
0x15	Pre-memory North Bridge initialization is started
0x16	Pre-Memory North Bridge initialization (North Bridge module specific)
0x17	Pre-Memory North Bridge initialization (North Bridge module specific)
0x18	Pre-Memory North Bridge initialization (North Bridge module specific)
0x19	Pre-memory South Bridge initialization is started
0x1A	Pre-memory South Bridge initialization (South Bridge module specific)
0x1B	Pre-memory South Bridge initialization (South Bridge module specific)
0x1C	Pre-memory South Bridge initialization (South Bridge module specific)
0x1D – 0x2A	OEM pre-memory initialization codes

Table 3-46: PEI Phase (Continued)

STATUS CODE	DESCRIPTION
0x2B	Memory initialization. Serial Presence Detect (SPD) data reading
0x2C	Memory initialization. Memory presence detection
0x2D	Memory initialization. Programming memory timing information
0x2E	Memory initialization. Configuring memory
0x2F	Memory initialization (other).
0x30	Reserved for ASL (see ASL Status Codes section below)
0x31	Memory Installed
0x32	CPU post-memory initialization is started
0x33	CPU post-memory initialization. Cache initialization
0x34	CPU post-memory initialization. Application Processor(s) (AP) initialization
0x35	CPU post-memory initialization. Boot Strap Processor (BSP) selection
0x36	CPU post-memory initialization. System Management Mode (SMM) initialization
0x37	Post-Memory North Bridge initialization is started
0x38	Post-Memory North Bridge initialization (North Bridge module specific)
0x39	Post-Memory North Bridge initialization (North Bridge module specific)

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Table 3-46: PEI Phase (Continued)

STATUS CODE	DESCRIPTION
0x3A	Post-Memory North Bridge initialization (North Bridge module specific)
0x3B	Post-Memory South Bridge initialization is started
0x3C	Post-Memory South Bridge initialization (South Bridge module specific)
0x3D	Post-Memory South Bridge initialization (South Bridge module specific)
0x3E	Post-Memory South Bridge initialization (South Bridge module specific)
0x3F-0x4E	OEM post memory initialization codes
0x4F	DXE IPL is started
PEI Error Codes	
0x50	Memory initialization error. Invalid memory type or incompatible memory speed
0x51	Memory initialization error. SPD reading has failed
0x52	Memory initialization error. Invalid memory size or memory modules do not match.
0x53	Memory initialization error. No usable memory detected
0x54	Unspecified memory initialization error.
0x55	Memory not installed
0x56	Invalid CPU type or Speed
0x57	CPU mismatch

Table 3-46: PEI Phase (Continued)

STATUS CODE	DESCRIPTION
0x58	CPU self test failed or possible CPU cache error
0x59	CPU micro-code is not found or micro-code update is failed
0x5A	Internal CPU error
0x5B	reset PPI is not available
0x5C-0x5F	Reserved for future AMI error codes
S3 Resume Progress Codes	
0xE0	S3 Resume is stared (S3 Resume PPI is called by the DXE IPL)
0xE1	S3 Boot Script execution
0xE2	Video repost
0xE3	OS S3 wake vector call
0xE4-0xE7	Reserved for future AMI progress codes
S3 Resume Error Codes	
0xE8	S3 Resume Failed
0xE9	S3 Resume PPI not Found
0xEA	S3 Resume Boot Script Error
0xEB	S3 OS Wake Error
0xEC-0xEF	Reserved for future AMI error codes
Recovery Progress Codes	

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Table 3-46: PEI Phase (Continued)

STATUS CODE	DESCRIPTION
0xF0	Recovery condition triggered by firmware (Auto recovery)
0xF1	Recovery condition triggered by user (Forced recovery)
0xF2	Recovery process started
0xF3	Recovery firmware image is found
0xF4	Recovery firmware image is loaded
0xF5-0xF7	Reserved for future AMI progress codes
Recovery Error Codes	
0xF8	Recovery PPI is not available
0xF9	Recovery capsule is not found
0xFA	Invalid recovery capsule
0xFB – 0xFF	Reserved for future AMI error codes

DXE Phase

Table 3-47: DXE Phase

STATUS CODE	DESCRIPTION
0x60	DXE Core is started
0x61	NVRAM initialization
0x62	Installation of the South Bridge Runtime Services

Table 3-47: DXE Phase (Continued)

STATUS CODE	DESCRIPTION
0x63	CPU DXE initialization is started
0x64	CPU DXE initialization (CPU module specific)
0x65	CPU DXE initialization (CPU module specific)
0x66	CPU DXE initialization (CPU module specific)
0x67	CPU DXE initialization (CPU module specific)
0x68	PCI host bridge initialization
0x69	North Bridge DXE initialization is started
0x6A	North Bridge DXE SMM initialization is started
0x6B	North Bridge DXE initialization (North Bridge module specific)
0x6C	North Bridge DXE initialization (North Bridge module specific)
0x6D	North Bridge DXE initialization (North Bridge module specific)
0x6E	North Bridge DXE initialization (North Bridge module specific)
0x6F	North Bridge DXE initialization (North Bridge module specific)
0x70	South Bridge DXE initialization is started
0x71	South Bridge DXE SMM initialization is started
0x72	South Bridge devices initialization
0x73	South Bridge DXE Initialization (South Bridge module specific)
0x74	South Bridge DXE Initialization (South Bridge module specific)

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Table 3-47: DXE Phase (Continued)

STATUS CODE	DESCRIPTION
0x75	South Bridge DXE Initialization (South Bridge module specific)
0x76	South Bridge DXE Initialization (South Bridge module specific)
0x77	South Bridge DXE Initialization (South Bridge module specific)
0x78	ACPI module initialization
0x79	CSM initialization
0x7A – 0x7F	Reserved for future AMI DXE codes
0x80 – 0x8F	OEM DXE initialization codes
0x90	Boot Device Selection (BDS) phase is started
0x91	Driver connecting is started
0x92	PCI Bus initialization is started
0x93	PCI Bus Hot Plug Controller Initialization
0x94	PCI Bus Enumeration
0x95	PCI Bus Request Resources
0x96	PCI Bus Assign Resources
0x97	Console Output devices connect
0x98	Console input devices connect
0x99	Super IO Initialization

Table 3-47: DXE Phase (Continued)

STATUS CODE	DESCRIPTION
0x9A	USB initialization is started
0x9B	USB Reset
0x9C	USB Detect
0x9D	USB Enable
0x9E – 0x9F	Reserved for future AMI codes
0xA0	IDE initialization is started
0xA1	IDE Reset
0xA2	IDE Detect
0xA3	IDE Enable
0xA4	SCSI initialization is started
0xA5	SCSI Reset
0xA6	SCSI Detect
0xA7	SCSI Enable
0xA8	Setup Verifying Password
0xA9	Start of Setup
0xAA	Reserved for ASL (see ASL Status Codes section below)
0xAB	Setup Input Wait
0xAC	Reserved for ASL (see ASL Status Codes section below)
0xAD	Ready To Boot event

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Table 3-47: DXE Phase (Continued)

STATUS CODE	DESCRIPTION
0xAE	Legacy Boot event
0xAF	Exit Boot Services event
0xB0	Runtime Set Virtual Address MAP Begin
0xB1	Runtime Set Virtual Address MAP End
0xB2	Legacy Option ROM Initialization
0xB3	System Reset
0xB4	USB hot plug
0xB5	PCI bus hot plug
0xB6	Clean-up of NVRAM
0xB7	Configuration Reset (reset of NVRAM settings)
0xB8 – 0xBF	Reserved for future AMI codes
0xC0 – 0xCF	OEM BDS initialization codes
DXE Error Codes	
0xD0	CPU initialization error
0xD1	North Bridge initialization error
0xD2	South Bridge initialization error
0xD3	Some of the Architectural Protocols are not available
0xD4	PCI resource allocation error. Out of Resources

Table 3-47: DXE Phase (Continued)

STATUS CODE	DESCRIPTION
0xD5	No Space for Legacy Option ROM
0xD6	No Console Output Devices are found
0xD7	No Console Input Devices are found
0xD8	Invalid password
0xD9	Error loading Boot Option (LoadImage returned error)
0xDA	Boot Option is failed (StartImage returned error)
0xDB	Flash update is failed
0xDC	Reset protocol is not available

PEI Beep Codes

Table 3-48: PEI Beep Codes

# OF BEEPS	DESCRIPTION
1	Memory not Installed
1	Memory was installed twice (InstallPeiMemory routine in PEI Core called twice)
2	Recovery started
3	DXEIPL was not found
3	DXE Core Firmware Volume was not found
4	Recovery failed

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Table 3-48: PEI Beep Codes (Continued)

# OF BEEPS	DESCRIPTION
4	S3 Resume failed
7	Reset PPI is not available

DXE Beep Codes

Table 3-49: DXE Beep Codes

# OF BEEPS	DESCRIPTION
1	Invalid password
4	Some of the Architectural Protocols are not available
5	No Console Output Devices are found
5	No Console Input Devices are found
6	Flash update is failed
7	Reset protocol is not available
8	Platform PCI resource requirements cannot be met

ACPI/ASL Checkpoints

Table 3-50: ACPI/ASL Checkpoints

STATUS CODE	DESCRIPTION
0x01	System is entering S1 sleep state

Table 3-50: ACPI/ASL Checkpoints (Continued)

STATUS CODE	DESCRIPTION
0x02	System is entering S2 sleep state
0x03	System is entering S3 sleep state
0x04	System is entering S4 sleep state
0x05	System is entering S5 sleep state
0x10	System is waking up from the S1 sleep state
0x20	System is waking up from the S2 sleep state
0x30	System is waking up from the S3 sleep state
0x40	System is waking up from the S4 sleep state
0xAC	System has transitioned into ACPI mode. Interrupt controller is in APIC mode.
0xAA	System has transitioned into ACPI mode. Interrupt controller is in APIC mode.

Extra Checkpoint Ranges

Table 3-51: Extra Checkpoint Ranges

STATUS CODE	DESCRIPTION
0x05	SEC initialization before microcode loading
0x0A	SEC initialization after microcode loading
0x1D – 0x2A	Pre-memory initialization codes

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Table 3-51: Extra Checkpoint Ranges

STATUS CODE	DESCRIPTION
0x3F – 0x4E	PEI post memory initialization codes
0x80 – 0x8F	DXE initialization codes
0xC0 – 0xCF	BDS initialization codes

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BMC

Chapter 4

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4.1 Server Management Software

Server System Overview

In a server system, BMC, or the on-board to host server system, is an independent system of the host server system. This independent system has its own processor and memory; the host system can be managed by the BMC system even if the host hardware or OS hangs or is unable to function.

BMC Key Features and Functions

- Supports IPMI v1.5 and v2.0
- Out-of-band monitoring and control for sever management over LAN.
- Share NIC for remote management via network
- The FRU information report includes main board part number, product name, manufacturer, etc.)
- Health status/Hardware monitoring report.
- Events log, view, and clear.
- Event notification via chassis LED indicator and PET (Platform Event Trap).
- Platform Event Filtering (PEF) to take selected actions for selected events, including NMI.
- Chassis management includes power control and a status report, front panel buttons and LED control.
- Watchdog and auto server restart and recovery
- Supports multi-session users, and alert destination for LAN channel.
- Support IPMB connecter that advanced server management card can communicate with BMC.

Power System

BMC controls system power through GPIO pins and IPMI chassis commands.

Front Panel User Interface

The BMC provides control panel interface functionality including indicators (Fault/status and Identify LEDs) and buttons (Power/ID).

Power Button

The Power buttons allow to control the system status.

ID Button

The control panel Chassis Identify button toggles the state of the Chassis ID LED. If the ID LED is off, then a button press will turn the LED on (blinking). If the LED is on, a button press or IPMI Chassis Identify command will turn the LED off.

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LEDs

The following table contains information on Status, ID and Heartbeat LED's.

Table 4-1: Status LED, ID LED, and Heartbeat LED

LEDs	Color	Status	Description
Power / Status LED	Amber (Status LED)	Blinking	System Event (See following Status LED table.)
		Off	Normal status
	Green (Power LED controlled by BIOS)	On	Power On
		Off	Power Off
ID LED	Blue	Off	Normal status
		Blinking	Identify the system with interval
		Solid ON	Identify the system
Heartbeat LED	Green	Off	BMC is not Ready
		Blinking	BMC is Ready

The following table contains information on Status LED when amber blinking.

Table 4-2: Status LED Activity

No.	Status LED Activity	Description
1.	Temperature Sensor	Non-critical / critical event asserted
2	Fan Sensors	Non-critical / critical event asserted
3	Voltage Sensors	State asserted
4	Power Supply	State asserted
5.	Processor	Thermal trip
6	Event Logging Disable	• SEL almost full • SEL full
7	Post Error	System firmware error
8	Memory	• Correctable EEC error • Uncorrectable ECC error • Correctable ECC error logging limit reached
9.	PCIe Bus	• Bus correctable error • Bus uncorrectable error • Bus fatal error

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Table 4-2: Status LED Activity

NO.	STATUS LED ACTIVITY	DESCRIPTION
10	BMC Watch-dog	• Timer expired • Hard Reset • Power Down • Power cycle

LAN Interface

BMC LAN interface in AST2300 is assigned to its Shared NIC LAN (Default) and a dedicated NIC in this system. IPMI Specification v2.0 defines how IPMI messages, encapsulated in RMCP/RMCP+ packet format, can be sent to and from the BMC. This capability allows a remote console application to access the BMC and perform the following operations:

- Chassis control: obtain chassis status, reset and power-up the chassis
- Obtain system sensor status
- Obtain and Set system boot options
- Obtain Field Replaceable Unit (FRU) information
- Obtain System Event Log (SEL) entries
- Obtain Sensor Data Records (SDR)
- Set Platform Event Filtering (PEF)
- Set LAN configurations

In addition, the BMC supports LAN alerting in the form of SNMP traps that conform to the IPMI Platform Event Trap (PET) format.

Session and User

This BMC supports ten (10) user accounts. Each can have a different user name, password and privilege level. Four accounts can login simultaneously. The available user privilege levels are User, Operator, and Administrator.

Serial Over LAN

BMC supports 1 IPMI (Spec v2.0) specific SOL session. BMC supports redirect data from UART interface.

Time Sync

In BMC design, BMC does not have a local RTC to know what time it is. Each time the server powers on, BIOS will use Set SEL Time command to initialize BMC time. The remote console program interpret this time as pre-initial.

SEL

BMC supports IPMI 1.5/2.0 standard SEL operation. It can keep SEL log. Event happened in BIOS side will be logged by using Add SEL Entry command. BMC will store them in FLASH, the time stamp field will be filled by BMC. When SEL is full, the new

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SEL won't be logged but will go through PEF as usual. If AC powers off, all SELs will remain in NV.

Platform Event

Platform Event Filter

The BMC implements selectable action on an event or LAN alerting base on event. By default, no any PEF entries or actions exist, applications need to configure it to enable.

- Dedicated and Shared NIC
- The policy to match an event to Platform Event Filter Table entry is IPMI 1.5 standard.
- The action support Power Off, Power Reset, Power Cycle and NMI.
- All Platform Event Filter Table is default disabled.
- PEF Startup Delay and Last Processed Event tracking is not supported.
- PEF table lookup isn't correlated to log SEL to SEL Repository.
- Serial Alerting is no support.

BMC Firmware Update

The BMC will allow users to upgrade firmware image on following entities:

- BMC
- All other upgradable entities

The update capability is provided by local and remote interfaces.

DOS Recovery Utility

SOCFLASH Utility.

WebUI Update

Remote update can be performed through the remote Web console.

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4.2 BMC Recovery

This section provides guidelines on BMC recovery process in DOS and Linux systems.

Recovery Process in DOS System

To recover BMC on a DOS system, do as follows:

1. Boot into DOS.
2. Navigate to the *Upgrade Utilit*y folder.
3. Run *dos.bat*.

The BMC recovery is complete.

Recovery Process in Linux System

To recover BMC on a Linux system, do as follows:

1. Boot into Linux.
2. Navigate to the *Upgrade Utility* folder.
3. Run *linux.sh*.

The BMC recovery is complete.

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4.3. QUANTA SMASH

Quanta SMASH is a tool that allows you to use Secure Shell (SSH) to login in the embedded Linux of BMC from remote terminal and gather information as well as give you control over things like power resets, power off. The basic structure is shown as below:

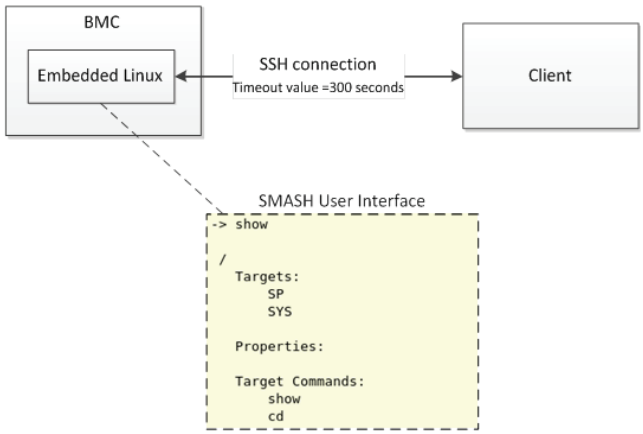

Figure 4-1. Using SSH to login in

Here presents an activity diagram, user could use SSH to login in embedded Linux of BMC from remote terminal. After login in successfully, SMASH would be executed automatically. In this time, SMASH is running and allowing user to input commands. The connection will be terminated if the terminal console is idle more than five minutes.

Default SSH UserName / Password (User Account in Linux): **sysadmin / superuser**

Input command in Linux: **ssh sysadmin@<Server IP>**

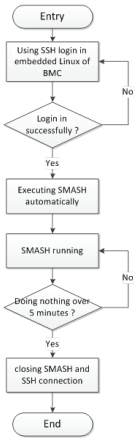

Figure 4-2. SMASH Activity Diagram

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Here provides you the commands about system level and BMC level.

System Level Commands

The system level commands provide you the information and power state control.

Table 4-3: Targets and Verbs

RELATED TARGETS	SUPPORTED VERBS										
	cd	exit	help	create	delete	set	show	reset	start	stop	version
/	v	v	v				v				v
/SYS	v	v	v				v	v	v	v	v
/SYS/voltage	v	v	v				v				v
/SYS/fan	v	v	v				v				v
/SYS/temperature	v	v	v				v				v
/SYS/powerSupply	v	v	v				v				v

Displays information for the board

show /SYS

Power-on system

start /SYS

Power-off system

stop /SYS

Power-reset system

reset /SYS

Display all system voltage

show /SYS/voltage

Display all system fan

show /SYS/fan

Display all system temperature

show /SYS/temperature

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Display all system power supply

show /SYS/powerSupply

/SYS

This command provides you the hig-level status of the system chassis and main power subsystem.

Table 4-4: /SYS

PROPERTY NAME	VALID VALUE	ACCESS	DESCRIPTION
SystemMAC0		R	Display system MAC0 address
SystemMAC1		R	Display system MAC1 address
ChassisStatus	powerIsOFF powerIsON	R	**PowerIsOFF** indicates the system power is off **PowerIsON** indicates the system power is on.

Q&A

Q: I tried to turn system power off by IPMI command "**power off**" when there is no response from operating system and system could not be shutdown. What is the Chassis Status?

A: The status of ChassisStatus is "**powerIsON.**"

/SYS/voltage

This command returns a high level version of the system voltages health status.

Table 4-5: /SYS/voltage

PROPERTY NAME	VALID VALUE	ACCESS	DESCRIPTION
Sensor name list of vlotage	na ok nonCritical critical	R	**na** indicates the status not available /unknown (typically because system power is OFF) **ok** indicates the monitored parameters within normal operating ranges **nonCritical** indicates the hardware outside normal operating range **critical** indicates the hardware exceeding specified ratings

*The sensor name list depends on the Server Hardware.

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/SYS/fan

This command returns a high level version of the system fan health status.

Table 4-6: /SYS/fan

PROPERTY NAME	VALID VALUE	ACCESS	DESCRIPTION
Sensor name list of fan	na ok nonCritical critical	R	**na** indicates the status not available /unknown (typically because system power is OFF) **ok** indicates the monitored parameters within normal operating ranges **nonCritical** indicates the hardware outside normal operating range **critical** indicates the hardware exceeding specified ratings

*The sensor name list depends on the Server Hardware.

/SYS/temperature

This command returns a high level version of the system temperature health status.

Table 4-7: /SYS/temperature

PROPERTY NAME	VALID VALUE	ACCESS	DESCRIPTION
Sensor name list of temperature	na ok nonCritical critical	R	**na** indicates the status not available /unknown (typically because system power is OFF) **ok** indicates the monitored parameters within normal operating ranges **nonCritical** indicates the hardware outside normal operating range **critical** indicates the hardware exceeding specified ratings

*The sensor name list depends on the Server Hardware.

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/SYS/powerSupply

This command provides the specification of the Sensor Type sensor-specific event.

Table 4-8: /SYS/powerSupply

PROPERTY NAME	VALID VALUE	ACCESS	DESCRIPTION
Sensor name list of power supply	Presence* FailDetected* PredictiveFail* InputLost(AC/DC)* AllDeasserted (*Note: Only for certain models.)	R	**Presence** indicates the Power Supply Presence detected **FailDetected** indicates the Powser Supply Failure detected **PredictiveFail** indicates the Power Supply Predictive Failure, the status supported or not depends on project **InputLost(AC/DC)** indicates the Power Supply input lost, such as power cord not inserted **AllDeasserted** indicates the power supply is not inserted

Table 4-8: /SYS/powerSupply (Continued)

PROPERTY NAME	VALID VALUE	ACCESS	DESCRIPTION
Redundancy	FullyRedundant RedundancyLost	R	The property is provided depend on project. **FullyRedundant** Indicates the power redundancy is OK. **RedundancyLost** Indicates the power redundancy is fail. One PSU is removed or AC lost.

*The sensor name list depends on the Server Hardware.

Q&A:

Q1: My system supports two power supply slots and only one power supply unit connected. What is the other power supply status?

A1: The other power supply status is " AllDeasserted ".

Q2: My system supports two power supply slots and two power supply units connected. But only one power cord plugged. What is the other power supply status?

A2: The other power supply status shows "Presence, PredictiveFail, InputLost(AC/DC) ".

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BMC Information

The BMC level commands provide several options to configure and display parameters of the management agent.

Table 4-9: Targets and Verbs

RELATED TARGETS	SUPPORTED VERBS										
	cd	exit	help	create	delete	set	show	reset	start	stop	version
/	v	v	v				v				v
/SP	v	v	v			v	v	v			v

Displays information for the board

show /SP

Reset BMC

reset /SP

Set server identify LED to be off

set /SP ServerIdentify=off

Set server identify LED to be on

set /SP ServerIdentify=on

Set server identify LED to be blinking

set /SP ServerIdentify=blinking

/SP

Table 4-10: /SYS/fan

PROPERTY NAME	VALID VALUE	ACCESS	DESCRIPTION
BMCVersion		R	Display BMC firmware revision
BMCGUID		R	Display BMC GUID
ServerIdentify	off on blinking	R/W	Configuring server identify LED
BMCMAC		R	Display the NIC physical address used by server management agent

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4.4 Web Graphical User Interface (GUI) for ESMS

Using the Web GUI

The BMC firmware features an embedded web server enabling users to connect to the BMC using a Web browser (e.g. Microsoft Internet Explorer). The Web GUI shows system information, system events, system status of managed servers, and other system-related information.

The Web-based GUI is supported on the following browsers:

- Internet Explorer 7 and above
- Firefox 2.0 and above
- Google Chrome 2.0 and above
- Safari 3.0 and above
- Opera 9.64 and above

Login

Enter the IP address or URL (default DHCP\static IP address) into the address bar of the web browser.

When connecting to the BMC the Login screen prompts for the username and password. This authentication with SSL protection prevents unauthorized intruders from gaining access to the BMC web server.

When a user is authenticated they can manage the server according to the privilege of their role.

The OEM Proprietary, Administrator and Operator privilege levels are authorized to login to the web interface. The User and No Access privilege levels do not allow access through the BMC web GUI.

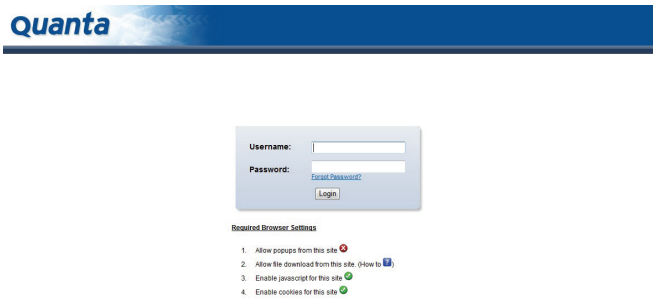

Figure 4-3. Login Web Page

Table 4-11: Default Username and Password

FIELD	DEFAULT
Username	admin
Password	admin

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After passing authentication, the following web page appears.

Note:
The default username and password are in lowercase characters. It is advised to change the admin password once you have logged in.

Click the **Help** button on the right corner of the page for assistance, the **Refresh** button to refresh the page, or the **Logout** button to exit.

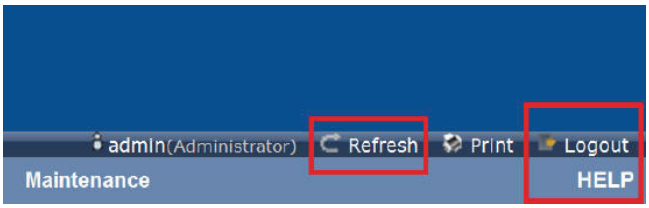

Figure 4-4. Main Web Page

Table 4-12: Main Web Page

MENU ITEM	DESCRIPTION
System Information	Shows system information.
Server Health	Monitoring status of the server.
Configuration	Configuration of the IPMI settings.
Remote Control	Launch KVM console and perform power control.
Maintenance	Allows the user to do firmware update.

Dashboard

In MegaRAC GUI, the Dashboard page displays the overall information on status of the device.

To open the **Dashboard** page, click Dashboard from the main menu. A sample screenshot of the Dashboard page is as follows:

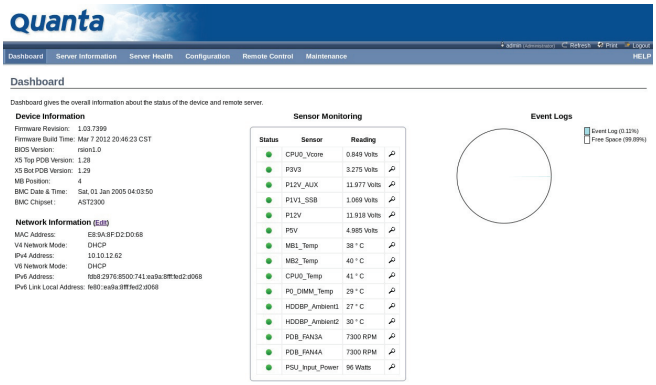

Figure 4-5. Dashboard

A brief description of the Dashboard page is given in the next section.

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Device Information

The Device Information displays the following information:

Table 4-13: Device Information Page

ITEM	DESCRIPTION
Firmware Revision	The revision number of the firmware.
Firmware Build Time	Firmware date and time.
BMC Chipset	This field shows BMC chipset type.

Note:

BMC Chipset type support list is as follows:

- AST2300: supports virtual KVM function and related setting item.
- AST2300 without RKVM: does not support virtual KVM function and related setting item.

If BMC Chipset type is AST2300 without RKVM, the Console Redirection, Mouse Mode, Remote Session, and Virtual Media menu items are not visible.

Network Information

The Network Information of the device with the following fields is shown in the following table. To edit the network Information, click **Edit**.

Table 4-14: Network Information

ITEM	DESCRIPTION
MAC Address	Read only field showing the IP address of the device.
V4 Network Mode	The v4 network mode options are the following disable, static, or DHCP.
IPv4 Address	The IPv4 address of the device (could be static or DHCP).
V6 Network Mode	The v6 network mode options are disable, static, or DHCP.
IPv6 Address:	The IPv6 address of the device.

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Sensor Monitoring

Lists all the available sensors on the device.

The status column displays the state of the device as follows:

STATUS (ICON)	DESCRIPTION
	Normal state
	Warning state
	Critical state

If you click on , the sensor page for that particular sensor will be displayed.

Event Logs

A graphical representation of all events incurred by various sensors as well as occupied/available space in logs. Clicking on the color-coded rectangle in the Legend for the chart, allows to view a list of specific events only.

Server Information

The Server Information Group consists of the following three items:

- FRU Information
- Server Component
- Server Identify

The following screenshot displays the Server Information menu items:

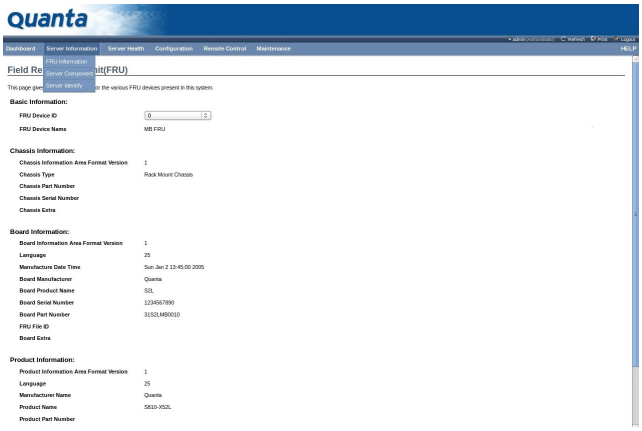

Figure 4-6. Server Information – Menu

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FRU Information

In the MegaRAC GUI, the FRU Information Page displays the BMC FRU file information. The information displayed in this page is Basic Information, Common Header Information, Chassis Information, Board Information and Product Information of the FRU device.

To open the FRU Information Page, click on **FRU Information** on top menu. Select a FRU Device ID from the Basic Information section to view the details of the selected device. A screenshot of FRU Information page is shown as follows:

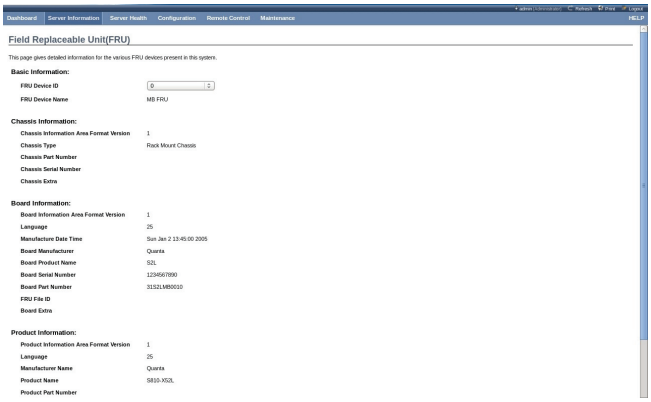

Figure 4-7. FRU Information Page

A brief description of the fields is given in the following sections.

Basic Information

Table 4-15: Basic Information

ITEM	DESCRIPTION
FRU device ID	The ID of the device.
FRU Device Name	The device name of the selected FRU device.

Chassis Information

- Chassis Information Area Format Version
- Chassis Type
- Chassis Part Number
- Chassis Serial Number
- Chassis Extra

Board Information

- Board Information Area Format Version
- Language
- Manufacture Date Time
- Board Manufacturer
- Board Product Name
- Board Serial Number
- Board Part Number
- FRU File ID

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- Board Extra

Product Information

- Product Information Area Format Version
- Language
- Manufacturer Name
- Product Name
- Product Part Number
- Product Version
- Product Serial Number
- Asset Tag

Server Component

The Component Information page displays the CPU and memory information.

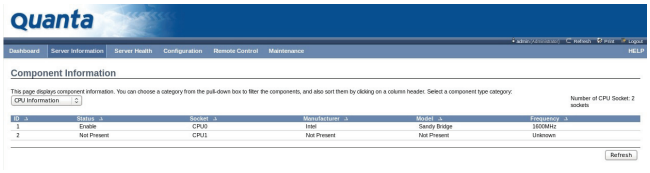

Figure 4-8. Component Information Page

Table 4-16: Component Information Page

ITEM	DESCRIPTION
CPU Information	Displays the following information: • CPU ID, • Status, • Socket, • Manufacturer, • Model, • Frequency
Memory Information	Displays the following information: • Memory ID, • Status, • Socket, • Module Size, • Model, • Frequency, and • Memory type*.

Note:
DDR3 ECC or non-ECCUDIMM, RDIMM and LRDIMM memory types support both normal voltage (1.5V) and low voltage (1.35V).

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Server identify

The Server Identify page displays the indicator LED status. You can select a Server Identify Operation to control the indicator LED.

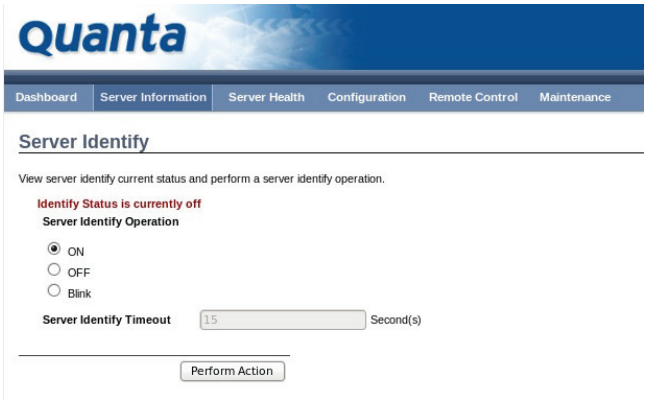

Figure 4-9. Server Identify Page

Table 4-17: Server Identify Page

ITEM	DESCRIPTION
Current Server Identify Status	The server status: On or Off.

Table 4-17: Server Identify Page (Continued)

ITEM	DESCRIPTION
Server Identify Operation	Server identify LED operation with the following options: • ON • OFF • Blink
Server Identify Timeout	Server timeout value when a Blink Identify Operation is selected. For Blink Operation the time period must be from 1 to 255 seconds. When 255 seconds is selected, the blinking is continuous.
Perform Action	Executes the selected Server Identify Operation.

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Server Health Group

The Server Health Group consists of the following two items:

- Sensor Readings
- Event Log

The Server Health screenshot allows to select Sensor Readings or Event Log as shown in the following image:

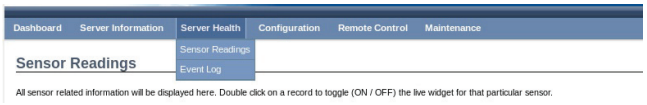

Figure 4-10. Server Health – Menu

Sensor Readings

In MegaRAC GUI, the Sensor Readings page displays all the sensor related information.

To open the Sensor readings page, click **Server Health > Sensor Readings** from the top menu. Click on a record to display more information on a particular sensor, including thresholds and a graphical representation of all associated events. A screenshot of Sensor Readings page is shown in the following image:

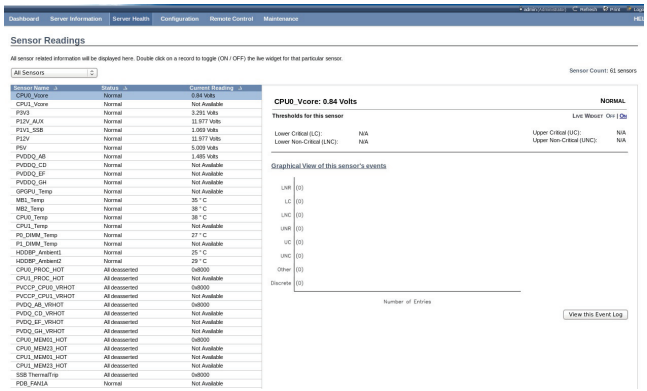

Figure 4-11. Sensor Readings Page

A brief description of the Sensor Readings page fields is given in the following sections.

Sensor Type

You can select a specific type of sensor from a drop-down menu. The list of sensors include the Sensor Name, Status, and Current Reading. The All Sensors option allows to view all the available sensor details, and select a specific type of sensor.

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When a specific type of sensor is selected, on the right hand side of the screen will be displayed the Thresholds for the sensor.

The total of six thresholds are available as follows:

- Lower Non-Recoverable (LNR)
- Lower Critical (LC)
- Lower Non-Critical (LNC)
- Upper Non-Recoverable (UNR)
- Upper Critical (UC)
- Upper Non-Critical (UNC)

The threshold states can be Lower Non-critical - going low, Lower Non-critical - going high, Lower Critical - going low, Lower Critical - going high, Lower Non-recoverable - going low, Lower Non-recoverable - going high, Upper Non-critical - going low, Upper Non-critical - going high, Upper Critical - going low, Upper Critical - going high, Upper Non-recoverable - going low, Upper Non-recoverable - going high.

Live Widget

The widget window can be turned On and Off for a selected sensor. Widget provides a dynamic representation of the readings for the sensor. The following image shows and example widget:

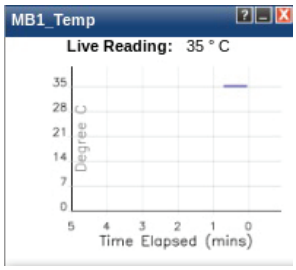

Figure 4-12. Widget Window

Note:
Widgets provide real time information on a particular sensor. User can track a sensor's behavior over a specific amount of time at specific intervals. The result will be displayed as a line graph on the widget. The data on widgets are updating in real time until the widget is closed.

View this Event Log

View the Event Log page for the selected sensor.

Event Log

In MegaRAC GUI, this page displays the list of event logs occurred by the different sensors on this device. Double click on a record to see the details of that entry. You can use the sensor

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type or sensor name filter options to view those specific events or you can also sort the list of entries by clicking on any of the column headers.

To open the Event Log page, click **Server Health > Event Log** from the top menu. A sample screenshot of Event Log page is shown below.

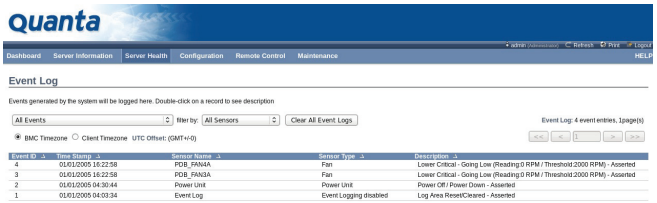

Figure 4-13. Event Log Page

A brief description of the Even Log page fields is given in the following sections.

Event Log Category

The category could be either a sensor-specific event, BIOS generated event or system management software event.

Table 4-18: Event Log Category

ITEM	DESCRIPTION
Filter Type	The type of filter listed. **Note:** Once the Event Log category and Filter type are selected, the list of events will be displayed with the Event ID, Time Stamp, Sensor Type, Sensor Name and Description.
BMC Timezone	BMC UTC offset timestamp value of the events.
Client Timezone	Events of client UTC offset timestamp.
Clear All Event Logs	Deletes all the existing records for all the sensors.

Procedure:

1. From the **Event Log Category** drop down menu select the event categories.
2. From the **Filter Type** drop down list select the sensor name filter to view the event for the selected filter.
3. Select either **BMC Timezone** or **Client Timezone**.
4. To clear all events from the list, click **Clear All Event Logs** button.

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Configuration Group

Configuration Group page allows access to various configuration settings. A screenshot of the Configuration Group menu is shown in the following figure:

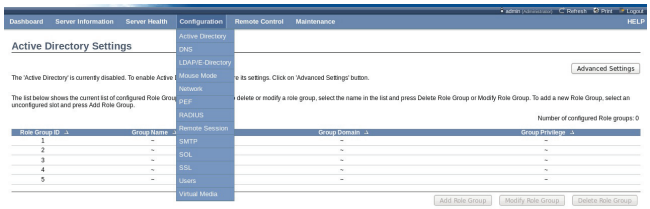

Figure 4-14. Configuration Group Menu

A detailed description of the Configuration menu is given in the following sections.

Active Directory

An Active Directory (AD) is a directory structure used in Microsoft Windows based computers and servers, to store information and data on networks and domains. AD provides a variety of functions including the ability to provide information on objects, organizes these objects for easy retrieval and access, allows the access of end users and administrators, and allows the administrator to set up security for the directory.

This Active Directory Settings page in MegaRAC SP-X as shown on the following figure, allows to Configure Active Directory Server Settings.

To open Active Directory Settings page, click **Configuration > Active Directory** from the main menu. A sample screenshot of Active Directory Settings Page is shown in the following screenshot:

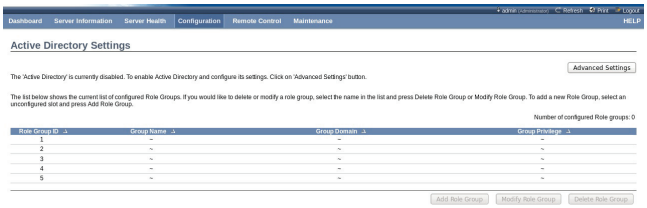

Figure 4-15. Active Directory Settings Page

Table 4-19: Active Directory Settings Page

ITEM	DESCRIPTION
Advanced Settings	Active Directory advanced settings configuration options are as follows: • Enable Active Directory Authentication, • User Domain name, • Time Out, and • Up to three Domain Controller Server Addresses.

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Table 4-19: Active Directory Settings Page (Continued)

ITEM	DESCRIPTION
Role Group ID	The name that identifies the role group in the Active Directory. • Role Group Name is a string of 255 alpha-numeric characters. • Special symbols hyphen and underscore are allowed.
Group Name	This name identifies the role group in Active Directory. • Role Group Name is a string of 255 alpha-numeric characters. • Special symbols hyphen and underscore are allowed.
Group Domain	The domain where the role group is located. **Note:** • Domain Name is a string of 255 alpha-numeric characters. • Special symbols hyphen, underscore and dot are allowed.
Group Privilege	The level of privilege to assign to this role group.
Add Role Group	To add a new role group to the device.
Modify role Group	To modify that role group. Alternatively, double click on the configured slot.
Delete Role Group	To delete an existing Role Group.

Procedure:

Entering the details in Advanced Active Directory Settings Page

1. Click on **Advanced Settings** to open the Advanced Active Directory Settings Page.

Figure 4-16. Advanced Active Directory Settings Page

2. In the Active Directory Settings Page, enter the following details.
3. **Active Directory Authentication:** To enable/disable Active Directory, check or uncheck the **Enable** checkbox respectively.

Note:
If you have enabled Active Directory Authentication, enter the required information to access the Active Directory server.

4. Specify the Domain Name for the user in the **User Domain Name** field. e.g. MyDomain.com
5. Specify the time (in seconds) to wait for Active Directory queries to complete in the **Time Out** field.

Note:
- Default Time out value: 120 seconds.
- Range from 15 to 300 allowed.

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6. Configure IP addresses in **Domain Controller Server Address1, Domain Controller Server Address2 & Domain Controller Server Address3**.

Note:
IP address of Active Directory server: At least one Domain Controller Server Address must be configured.

- IP Address made of 4 numbers separated by dots as in "xxx.xxx.xxx.xxx".
- Each number ranges from 0 to 255.
- First number must not be 0.

Domain Controller Server Addresses will supports IPv4 Address format and IPv6 Address format.

7. Click **Save** to save the entered settings and return to Active Directory Settings Page.
8. Click **Cancel** to cancel the entry and return to Active Directory Settings Page.

To add a Role Group

9. In the Active Directory Settings Page, select a blank row and click **Add Role Group** to open the Add Role group Page as shown in the screenshot below.

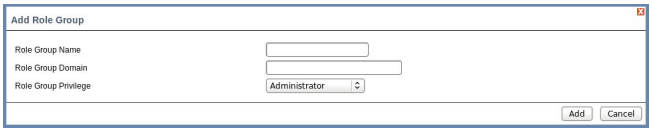

Figure 4-17. Add Role group Page

10. In the **Role Group Name** field, enter the name that identifies the role group in the Active Directory.

Note:
- Role Group Name is a string of 255 alpha-numeric characters.
- Special symbols hyphen and underscore are allowed.

11. In the **Role Group Domain** field, enter the domain where the role group is located.

Note:
- Domain Name is a string of 255 alpha-numeric characters.
- Special symbols hyphen, underscore and dot are allowed.

12. In the **Role Group Privilege** field, enter the level of privilege to assign to this role group.
13. Click **Add** to save the new role group and return to the Role Group List.
14. Click **Cancel** to cancel the settings and return to the Role Group List.

To modify a Role Group

15. In the Advanced Directory Settings Page, select the row that you wish to modify and click **Modify Role Group**.
16. Make the necessary changes and click **Save**.

To delete a Role Group

17. In the Advanced Directory Settings Page, select the row that you wish to delete and click **Delete Role Group**.

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DNS

The **Domain Name System (DNS)** is a distributed hierarchical naming system for computers, services, or any resource connected to the Internet or a private network. It associates the information with domain names assigned to each of the participants. Most importantly, it translates domain names meaningful to humans into the numerical (binary) identifiers associated with networking equipment for the purpose of locating and addressing these devices worldwide. In MegaRAC GUI, the DNS Server settings page is used to manage the DNS settings of a device.

To open DNS Server Settings page, click **Configuration > DNS** from the main menu. A sample screenshot of DNS Server Settings Page is shown in the screenshot below.

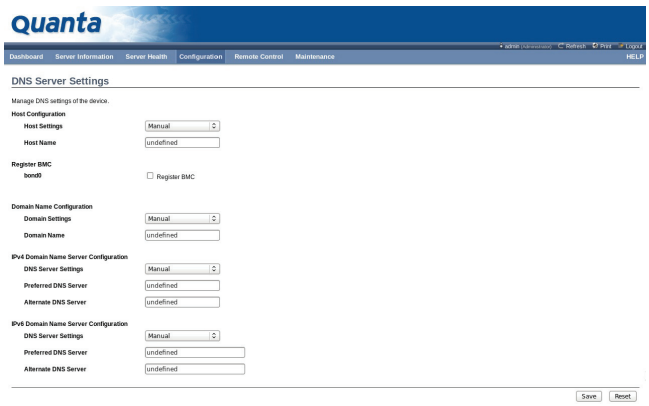

Figure 4-18. DNS Server Settings Page

The fields of DNS Server Settings page are explained below.

Table 4-20: DNS Server Settings Page

ITEM	DESCRIPTION
Host configuration	
Host Settings	Choose either Automatic or Manual settings.
Host Name	It displays hostname of the device. If the Host setting is chosen as Manual, then specify the hostname of the device.

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Table 4-20: DNS Server Settings Page (Continued)

ITEM	DESCRIPTION
Domain Name Configuration	
Domain Settings	It lists the option for domain interface as Manual, v4 or v6 for multiLAN channels. **Note:** If you choose DHCP, then select v4 or v6 for DHCP servers.
Domain Name	It displays the domain name of the device. If the Domain setting is chosen as Manual, then specify the domain name of the device. If you chose Automatic, the Domain Name cannot be configured as it will be done automatically. The field will be disabled.
IPv4 Domain Name Server Configuration	
DNS Server Settings:	It lists the option for v4 DNS settings for the device, Manual and available LAN interfaces.
Preferred DNS Server	The DNS (Domain Name System) server v4 address to be configured to the device. • IP Address made of 4 numbers separated by dots as in "xxx.xxx.xxx.xxx". • Each number ranges from 0 to 255. • First number must not be 0.
Alternate DNS Server	
IPv6 Domain Name Server Configuration	

Table 4-20: DNS Server Settings Page (Continued)

ITEM	DESCRIPTION
DNS Server Settings	It lists the option for v6 DNS settings for the device, Manual and available LAN interfaces. If you choose Manual setting, you have to configure the DNS Server IP addresses. If you have chosen DHCP, then you have to select the interface from which the IP address is to be received. Example of IPv6 address - 2001:db8:0:101
Preferred DNS Server, Alternate DNS Server	Specify the DNS (Domain Name System) server v6 address to be configured to the device.
Save	To save the entered changes.
Reset	To reset the entered changes.

Procedure:

1. Choose the **Host Configuration** as either Automatic or Manual.

Note:
Under Automatic, a Host Name is not necessary but under Manual, a Host Name is not.

2. Enter the **Host Name** in the given field if you have chosen Manual Configuration.
3. Under **Register BMC**,
   - Check the option **Register BMC** to register with this DNS settings.

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- Choose the option Direct Dynamic DNS to register with direct dynamic DNS or choose **DHCP Client FQDN** to register through DHCP server.

4. In the **Domain name Configuration Settings**,
   - Select the domain settings from the dropdown list.
   - Enter the **Domain Name** in the given field
5. In **IPv4 Domain Name Server Configuration**,
   - Select the **DNS Server Settings**, from the dropdown list.
   - In the **Preferred DNS Server** field, enter the preferred IP address.
   - In the **Alternate DNS Server** field, enter the alternate address.
6. In **IPv6 Domain Name Server Configuration**,
   - Select the **DNS Server Settings**, from the dropdown list.
   - In the **Preferred DNS Server** field, enter the preferred IP address.
   - In the **Alternate DNS Server** field, enter the alternate address.
7. Click **Save** to save the entries.
8. Click **Reset** to reset the entries.

LDAP/E-Directory

The **Lightweight Directory Access Protocol (LDAP)** is an application protocol for querying and modifying data of directory services implemented in Internet Protocol (IP) networks.

In MegaRAC GUI, LDAP is an Internet protocol that MegaRAC® card can use to authenticate users. If you have an LDAP server configured on your network, you can use it as an easy way to add, manage and authenticate MegaRAC® card users. This is done by passing login requests to your LDAP Server. This means that there is no need to define an additional authentication mechanism, when using the MegaRAC card. Since your existing LDAP Server keeps an authentication centralized, you will always know who is accessing the network resources and can easily define the user or group-based policies to control access.

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To open LDAP Settings page, click **Configuration > LDAP** from the main menu. A sample screenshot of LDAP Settings Page is shown in the screenshot below.

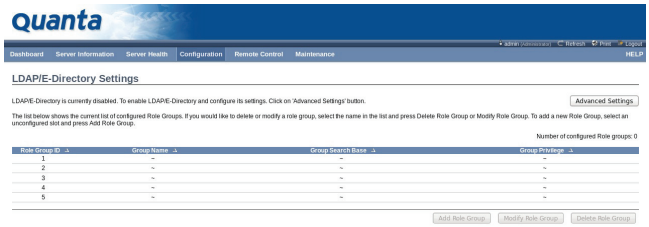

Figure 4-19. LDAP Settings Page

The fields of LDAP Settings Page are explained below.

Table 4-21: LDAP Settings Page

ITEM	DESCRIPTION
Advanced Settings	To configure LDAP Advanced Settings. Options are Enable LDAP Authentication, IP Address, Port and Search base.
Add Role Group	To add a new role group to the device. Alternatively, double click on a free slot to add a role group.
Modify Role Group	To modify the particular role group.
Delete Role Group	To be delete a role group from the list.

Procedure:

Entering the details in Advanced LDAP Settings Page

1. In the LDAP Settings Page, click Advanced Settings. A sample screenshot of LDAP Settings page is given below.

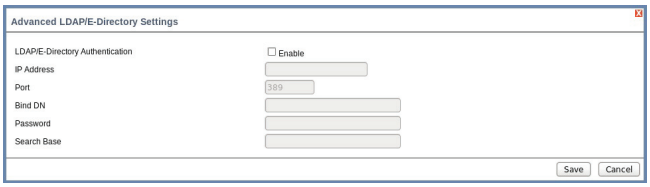

Figure 4-20. Advanced LDAP Settings

2. To enable/disable LDAP Authentication, check or uncheck the **Enable** checkbox respectively.

Note:
At login prompt, enter username to login as an LDAP group member.

3. Enter the IP address of LDAP server in the **IP Address** field.

Note:
- IP Address made of 4 numbers separated by dots as in 'xxx.xxx.xxx.xxx'.
- Each Number ranges from 0 to 255.
- First Number must not be 0.
- Supports IPv4 Address format and IPv6 Address format.

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4. Specify the LDAP Port in the **Port** field.

Note:
Default Port is 389. For Secure connection, default port is 636.

5. Enter the Search Base. The Search base tells the LDAP server which part of the external directory tree to search. The search base may be something equivalent to the organization, group of external directory.

Note:
- Searchbase is a string of 4 to 63 alpha-numeric characters.
- It must start with an alphabetical character.
- Special Symbols like dot(.), comma(,), hyphen(-), underscore(_), equal-to(=) are allowed.
- Example: ou=login,dc=domain,dc=com

6. Click **Save** to save the settings.
7. Click **Cancel** to cancel the modified changes.

To add a new Role Group

8. In the LDAP Settings Page, select a blank row and click **Add Role Group** to open the Add Role group Page as shown in the screenshot below.

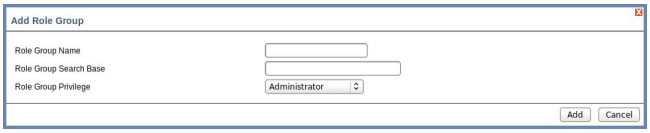

Figure 4-21. Add Role group Page

9. In the **Role Group Name** field, enter the name that identifies the role group.

Note:
- Role Group Name is a string of 255 alpha-numeric characters.
- Special symbols hyphen and underscore are allowed.

10. In the Role Group Search Base field, enter the path from where the role group is located to Base DN.

Note:
- Search Base is a string of 255 alpha-numeric characters.
- Special symbols hyphen, underscore and dot are allowed.

11. In the **Role Group Privilege** field, enter the level of privilege to assign to this role group.
12. Click **Add** to save the new role group and return to the Role Group List.

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13. Click **Cancel** to cancel the settings and return to the Role Group List.

To Modify Role Group

14. In the LDAP Settings Page, select the row that you wish to modify and click Modify Role Group.
15. Make the necessary changes and click Save.

To Delete a Role Group

16. In the LDAP Settings Page, select the row that you wish to delete and click Delete Role Group.

Mouse Mode

In MegaRAC GUI, Redirection Console handles mouse emulation from local window to remote screen in either of two methods. User has to be an Administrator to configure this option.

To open Mouse Mode page, click **Configuration > Mouse Mode** from the main menu. A sample screenshot of Mouse Mode Settings Page is shown in the screenshot below.

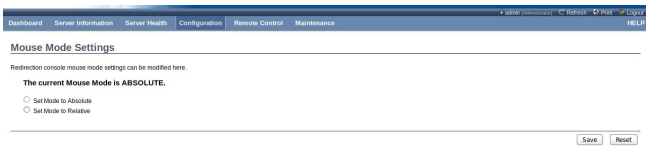

Figure 4-22. Mouse Mode Settings Page

The fields of Mouse Mode Settings page are explained below.

Table 4-22: Mouse Mode Settings Page

ITEM	DESCRIPTION
Absolute Mode	The absolute position of the local mouse is sent to the server.
Relative Mode	Relative mode sends the calculated relative mouse position displacement to the server.
Save	To save any changes made.
Reset	To Reset the modified changes.

Procedure:

1. Choose either of the following as your requirement:
   - Set Mode to Absolute

Note:
Applicable for all Widnows versions, versions above RHEL6, and versions above FC14.

   - Set Mode to Relative

Note:
Applicable for all Linux versions, versions above RHEL6, and versions less than FC14.

2. Click **Save** button to save the changes made.
3. Click **Reset** to reset the modified changes.

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Network

In MegaRAC GUI, the Network Settings Page is used to configure the network settings for the available LAN channels.

To open Network Settings page, click **Configuration > Network** from the main menu. A sample screenshot of Network Settings Page is shown in the screenshot below.

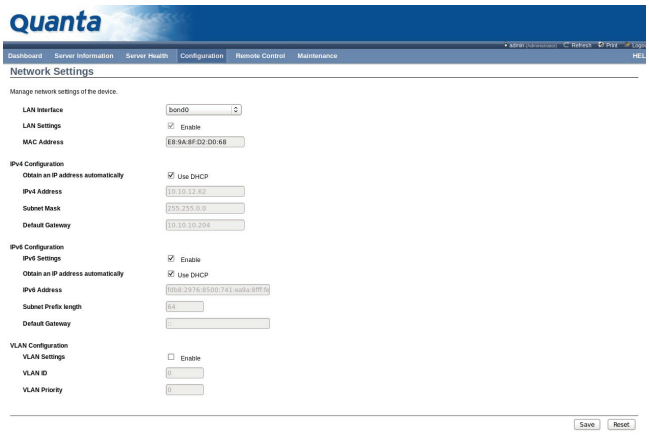

Figure 4-23. Network Settings Page

The fields of Network Settings page are explained below.

Table 4-23: Network Settings Page

ITEM	DESCRIPTION
LAN Interface	Lists the LAN interfaces.
LAN Settings	To enable or disable the LAN Settings.
MAC Address	This field displays the MAC Address of the device. This is a read only field.
IPv4 Settings	This option lists the IPv4 configuration settings. • Obtain IP Address automatically: This option is to dynamically configure IPv4 address using DHCP (Dynamic Host Configuration Protocol). • IPv4 Address, Subnet Mask, and Default Gateway: These fields are for specifying the static IPv4 address, Subnet Mask and Default Gateway to be configured to the device. **Note:** • IP Address made of 4 numbers separated by dots as in "xxx.xxx.xxx.xxx". • Each Number ranges from 0 to 255. • First Number must not be 0.

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Table 4-23: Network Settings Page (Continued)

ITEM	DESCRIPTION
IPv6 Configuration	This option lists the following IPv6 configuration settings. ● IPv6 Settings: This option is to enable the IPv6 settings in the device. ● Obtain an IPv6 address automatically: This option is to dynamically configure IPv6 address using DHCP (Dynamic Host Configuration Protocol). ● IPv6 Address: To specify a static IPv6 address to be configured to the device. Eg: 2004:2010 ● Subnet Prefix length: To specify the subnet prefix length for the IPv6 settings. **Note:** Value ranges from 0 to 128. ● Default Gateway: Specify v6 default gateway for the IPv6 settings.
VLAN Configuration	It lists the VLAN configuration settings. ● VLAN Settings: To enable/disable the VLAN support for selected interface. ● VLAN ID: The Identification for VLAN configuration. ■ Value ranges from 1 to 4095. ● VLAN Priority: The priority for VLAN configuration. ■ Value ranges from 1 to 7. ■ 7 is the highest priority for VLAN.
Save	To save the entries.

Table 4-23: Network Settings Page (Continued)

ITEM	DESCRIPTION
Reset	To Reset the modified changes.

Procedure:

1. Select the **LAN Interface** from the drop down list.
2. Check **Enable** to enable the LAN Settings.
3. In IPv4 Configuration, enable **Use DHCP to Obtain an IP address automatically** to dynamically configure IPv4 address using DHCP.
4. If the field is disabled, enter the **IPv4 Address**, **Subnet Mask** and **Default Gateway** in the respective fields.
5. In IPv6 Configuration, if you wish to enable the IPv6 settings, check **Enable**.
6. If the IPv6 setting is enabled, enable or disable the option **Use DHCP for obtaining the IP address automatically**.
7. If the field is disabled, enter the **IPv6 Address, Subnet Prefix length** and **Default Gateway** in the given field.
8. In VLAN Configuration, if you wish to enable the VLAN settings, check **Enable**.
9. Enter the **VLAN ID** in the specified field.
10. Enter the **VLAN Priority** in the specified field.
11. Click **Save** to save the entries.
12. Click **Reset** if you want to reset the modified changes.

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PEF

Platform Event Filtering (PEF) provides a mechanism for configuring the BMC to take selected actions on event messages that it receives or has internally generated. These actions include operations such as system power-off, system reset, as well as triggering the generation of an alert. In MegaRAC GUI, the PEF Management is used to configure the following features:

- Event Filter
- Alert Policy
- LAN Destination

To open PEF Management Settings page, click **Configurations > PEF** from the main menu. A sample screenshot of PEF Management Settings Page is shown in the screen shot below along with an explanation of each of the tabs.

Event Filter Tab

A PEF implementation is recommended to provide at least 16 entries in the event filter table. A subset of these entries should be pre-configured for common system failure events, such as over-temperature, power system failure, fan failure events, etc. Remaining entries can be made available for 'OEM' or System Management Software configured events. Note that individual entries can be tagged as being reserved for system use - so this ratio of pre-configured entries to run-time configurable entries can be reallocated if necessary.

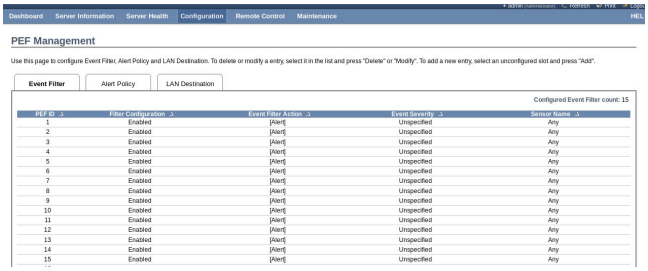

Figure 4-24. PEF Management – Event Filter

The fields of PEF Management – Event Filter Tab are explained below.

This page contains the list of configured PEF's.

Table 4-24: PET Management - Event Filter

ITEM	DESCRIPTION
PEF ID	This field displays the ID for the newly configured PEF entry (read-only).
Filter configuration	Check box to enable the PEF settings.
Event Filter Action	Check box to enable PEF Alert action. This is a mandatory field.
Event Severity	To choose any one of the Event severity from the list.

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Table 4-24: PET Management - Event Filter (Continued)

ITEM	DESCRIPTION
Sensor Name	To choose the particular sensor from the sensor list.
Add	To add the new event filter entry and return to Event filter list.
Modify	To modify the existing entries.
Cancel	To cancel the modification and return to Event filter list.

Procedure:

1. Click the **Event Filter** Tab to configure the event filters in the available slots
2. To Add an Event Filter entry, select a free slot and click **Add** to open the Add event Filter entry Page. A sample screenshot of Add Event Filter Page is in seen the screenshot below.

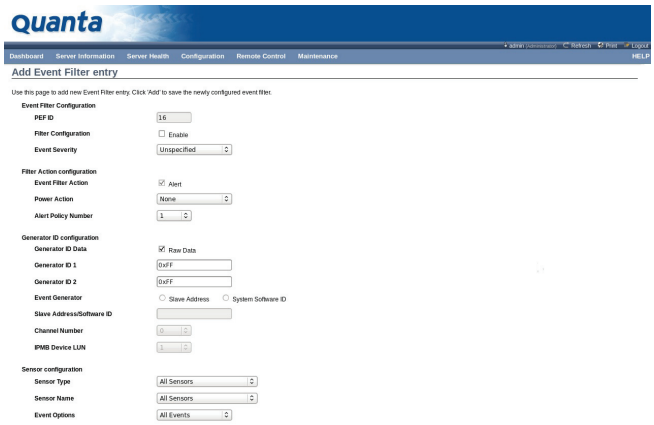

Figure 4-25. Add Event Filter Entry Page

3. In the Event Filter Configuration section,
   - PEF ID displays the ID for configured PEF entry (read-only).
   - In filter configuration, check the box to enable the PEF settings.
   - In Event Severity, select any one of the Event severity from the list.
4. In the Filter Action configuration section,

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- Event Filter Action is a mandatory field and checked by default, which enable PEF Alert action (read-only).
- Select any one of the Power action either Power down, Power reset or Power cycle from the drop down list
- Choose any one of the configured alert policy number from the drop down list.

Note:
Alert Policy has to be configured - under **Configuration > PEF > Alert Policy**.

5. In the Generator ID configuration section,
   - Check Generator ID Data option to fill the Generator ID with raw data.
   - Generator ID 1 field is used to give raw generator ID1 data value.
   - Generator ID 2 field is used to give raw generator ID2 data value.

Note:
In RAW data field, to specify hexadecimal value prefix with '0x'.

   - In the Event Generator section, choose the event generator as Slave Address - if event was generated from IPMB.Otherwise as System Software ID - if event was generated from system software.
   - In the Slave Address/Software ID field, specify corresponding I2C Slave Address or System Software ID.
   - Choose the particular channel number that event message was received over. Or choose '0' if the event message was received via the system interface, primary IPMB, or internally generated by the BMC.
   - Choose the corresponding IPMB device LUN if event generated by IPMB.
6. In the Sensor configuration section,
   - Select the s type of sensor that will trigger the event filter action.
   - In the sensor name field, choose the particular sensor from the sensor list.
   - Choose event option to be either All Events or Sensor Specific Events.
7. In the Event Data configuration section,
   - Event Trigger field is used to give Event/Reading type value.

Note:
Value ranges from 1 to 255.

   - Event Data 1 AND Mask field is used to indicate wildcarded or compared bits.

Note:
Value ranges from 0 to 255.

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- Event Data 1 Compare 1 & Event Data 1 Compare 2 field is used to indicate whether each bit position's comparison is an exact comparison or not.

Note:
Value ranges from 0 to 255.

8. In the Event Data 2 configuration section,
   - Event Data 2 AND Mask field is similar to Event Data 1 AND Mask.
   - Event Data 2 Compare 1 & Event Data 2 Compare 2 fields are similar to Event Data 1 Compare 1 and Event Data 1 Compare 2 respectively.
9. In the Event Data 3 configuration section,
   - Event Data 3 AND Mask field is similar to Event Data 1 AND Mask.
   - Event Data 3 Compare 1 & Event Data 3 Compare 2 fields are similar to Event Data 1 Compare 1 and Event Data 1 Compare 2 respectively.
10. Click **Modify** to accept the modification and return to Event filter list.
11. Click **Reset** to reset the modification done.
12. Click on Cancel to cancel the modification and return to Event filter list.
13. In the Event filter list, click **Modify** to modify the existing filter.
14. In the Event filter list, click **Delete** to delete the existing filter.

Alert Policy Tab

This page is used to configure the Alert Policy and LAN destination. You can add, delete or modify an entry in this page.

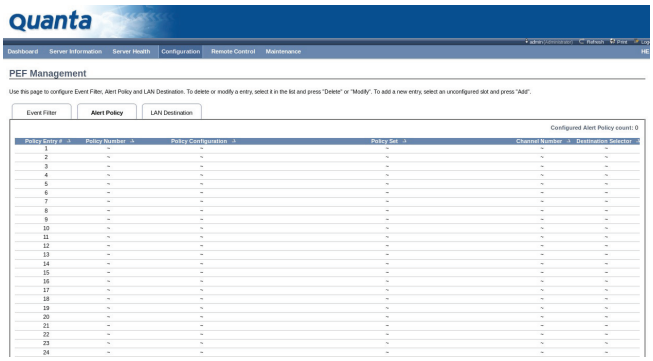

Figure 4-26. PEF Management – Alert Policy

The fields of the PEF Management – Alert Policy Tab are explained below.

Table 4-25: PEF Management - Alert Policy

ITEM	DESCRIPTION
Policy Entry #	Displays Policy entry number for the newly configured entry (read-only).

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Table 4-25: PEF Management - Alert Policy (Continued)

ITEM	DESCRIPTION
Policy Number	Displays the Policy number of the configuration.
Policy Configuration	To enable or disable the policy settings.
Policy Set	To choose any one of the Policy set values from the list. ● 0: Always send alert to this destination. ● 1: If alert to previous destination was successful, do not send alert to this destination. Proceed to ESMS Functional Specification For SI v0.11 next entry in this policy set. ● 2: If alert to previous destination was successful, do not send alert to this destination. Do not process any more entries in this policy set. ● 3: If alert to previous destination was successful, do not send alert to this destination. Proceed to next entry in this policy set that is to a different channel. ● 4: If alert to previous destination was successful, do not send alert to this destination. Proceed to next entry in this policy set that is to a different destination type.
Channel Number	To choose a particular channel from the available channel list.
Destination Selector	To choose a particular destination from the configured destination list. **Note:** LAN Destination has to be configured - under **Configuration > PEF > LAN Destination**.

Table 4-25: PEF Management - Alert Policy (Continued)

ITEM	DESCRIPTION
Add	To save the new alert policy and return to Alert Policy list.
Modify	To modify the existing entries.
Cancel	To cancel the modification and return to Alert Policy list.

Procedure:

1. In the Alert Policy Tab, select the slot for which you have to configure the Alert policy. That is, In the **Event Filter Entry Page**, if you have chosen Alert Policy number as 4, you have to configure the 4th slot (the slot with Policy Number 4) in the Alert Policy Tab.
2. Select the slot and click **Add** to open the **Add Alert Policy Entry Page** as shown in the screenshot below.

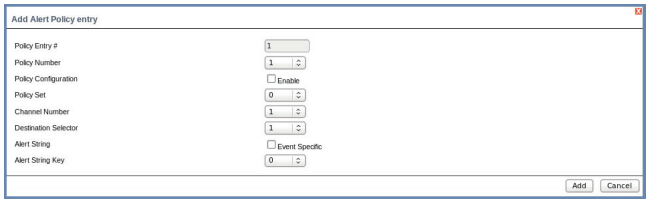

Figure 4-27. Add Alert Policy Entry Page

3. **Policy Entry #** is a read only field.
4. Select the **Policy Number** from the list.

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5. In the **Policy Configuration** field, check **Enable** if you wish to enable the policy settings.
6. In the **Policy Set** field, choose any of the Policy set from the list.
7. In the **Channel Number** field, choose particular channel from the available channel list.
8. In the **Destination Selector** field, choose particular destination from the configured destination list.

Note:
LAN Destination has to be configured under **Configuration > PEF > LAN Destination**. That is if you select the number 4 for destination selector in Alert Policy Entry page, then you have to configure the 4th slot (LAN Destination Number 4) in the LAN Destination tab.

9. In the **Alert String** field, enable the check box if the Alert policy entry is Event Specific.
10. In the **Alert String Key** field, choose any one value that is used to look up the Alert String to send for this Alert Policy entry.
11. Click **Add** to save the new alert policy and return to Alert Policy list.
12. Click **Cancel** to cancel the modification and return to Alert Policy list.
13. In the Alert Policy list, to modify a configuration, select the slot to be modified and click **Modify**.
14. In the Modify Alert Policy Entry Page, make the necessary changes and click **Modify**.
15. In the Alert Policy list, to delete a configuration, select the slot and click **Delete**.

PEF Management LAN Destination Page

This page is used to configure the Event filter, Alert Policy and LAN destination. A sample screenshot of PEF Management LAN Destination Page is given below.

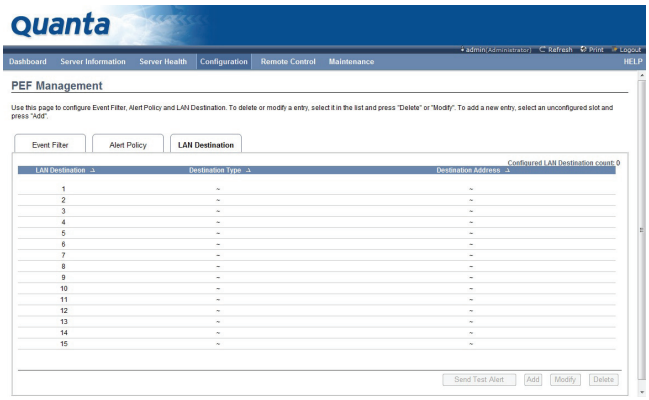

Figure 4-28. PEF Management LAN Destination

The fields of PEF Management – LAN Destination Tab are explained below.

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Table 4-26: PEF Management - LAN Destination

ITEM	DESCRIPTION
LAN Destination	Displays Destination number for the newly configured entry (read-only).
Destination Type	Destination type can be either an SNMP Trap or an Email alert. For Email alerts, the 3 fields - destination Email address, subject and body of the message needs to be filled. The SMTP server information also has to be added - under **Configuration > SMTP**. For SNMP Trap, only the destination IP address has to be filled.
Destination Address	If Destination type is SNMP Trap, then enter the IP address of the system that will receive the alert. Destination address will support the following: ● IPv4 address format. ● IPv6 address format. If Destination type is Email Alert, then give the email address that will receive the email.
Subject & Message	These fields must be configured if email alert is chosen as destination type. An email will be sent to the configured email address in case of any severity events with a subject specified in subject field and will contain the message field's content as the email body.
Add	To save the new LAN destination and return to LAN destination list.
Cancel	To cancel the modification and return to LAN destination list.

Procedure:

1. In the **LAN Destination Tab**, choose the slot to be configured. This should be the same slot that you have selected in the Alert Policy Entry- Destination Selector field. That is if you have chosen the Destination Selector as 4 in the Alert Policy Entry page of Alert Policy Tab, then you have to configure the 4th slot of LAN Destination Page.
2. Select the slot and click **Add**. This opens the **Add LAN Destination entry**.

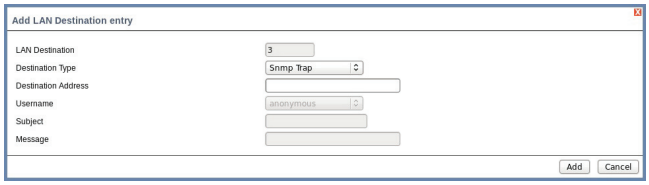

Figure 4-29. Add LAN Destination entry Page

3. In the **LAN Destination** field, the destination for the newly configured entry is displayed and this is a read only field.
4. In the **Destination Type** field, select the one of the types.
5. In the **Destination Address** field, enter the destination address.

Note:
If Destination type is Email Alert, then give the email address that will receive the email.

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6. Select the **User Name** from the list of users.
7. In the **Subject** field, enter the subject.
8. In the **Message** field, enter the message.
9. Click **Add** to save the new LAN destination and return to LAN destination list.
10. Click **Cancel** to cancel the modification and return to LAN destination list.
11. In the LAN Destination Tab, to modify a configuration, select the row to be modified and click **Modify**.
12. In the Modify LAN Destination Entry page, make the necessary changes and click **Modify**.
13. In the LAN Destination Tab, to delete a configuration, select the slot and click **Delete**.

Configuring the SNMP:

1. Navigate to **PEF Management**.
2. Select **LAN Destination** tab in **Configuration** section.
3. Select from **LAN Destination** menu **Snmp Trap**.

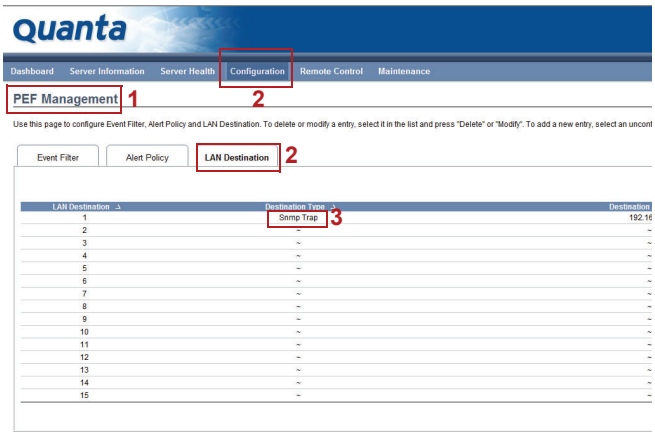

Figure 4-30. Selecting Snmp Trap

A **Modify LAN Destination entry** menu opens.

4. Key in an IP address to the **Destination Address** field.

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5. To complete the SNMP configuration procedure, click **Modify**.

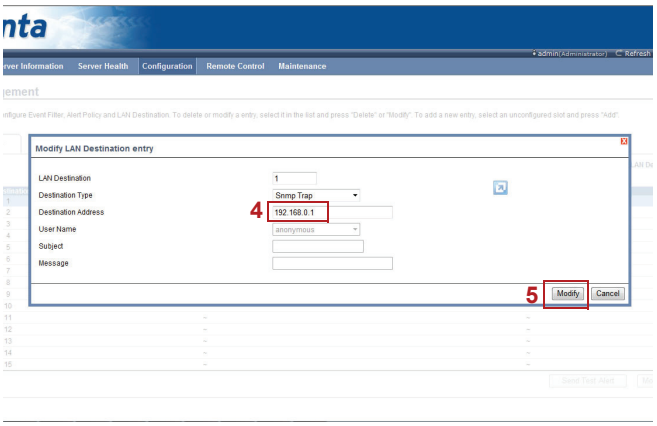

Figure 4-31. Inserting the IP Address

RADIUS

RADIUS is a modular, high performance and feature-rich RADIUS suite including server, clients, development libraries and numerous additional RADIUS related utilities.

In MegaRAC GUI, this page is used to set the RADIUS Authentication.

To open RADIUS Settings page, click Configuration > RADIUS from the main menu. A sample screenshot of RADIUS Settings Page is shown in the screenshot below.

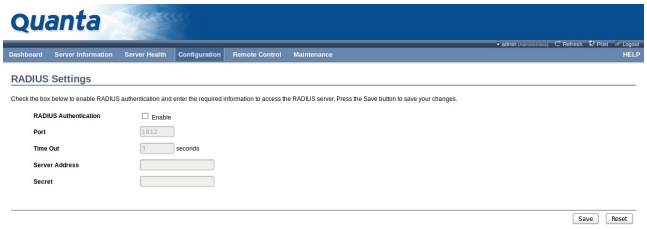

Figure 4-32. RADIUS Settings Page

The fields of RADIUS Settings Page are explained below.

Table 4-27: RADIUS Settings Page

ITEM	DESCRIPTION
RADIUS Authentication	Option to enable RADIUS authentication.
Port	The RADIUS Port number. **Note:** Default Port is 1812.
Time Out	The Time out value in seconds. **Note:** • Default Timeout value is 3seconds. • Timeout value ranges from 3 to 300.

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Table 4-27: RADIUS Settings Page (Continued)

ITEM	DESCRIPTION
Server Address	The IP address of RADIUS server. **Note:** • IP Address made of 4 numbers separated by dots as in "xxx.xxx.xxx.xxx". • Each Number ranges from 0 to 255. • First Number must not be 0.
Secret	The Authentication Secret for RADIUS server. **Note:** • This field will not allow more than 31 characters. • Secret must be at least 4 characters long. • White space is not allowed.
Save	To save the settings.
Reset	To reset the modified changes.

Procedure:

1. Enable the **RADIUS Authentication** checkbox to authenticate the RADIUS.
2. Enter the port number in the **Port Number** field.
3. Enter the time out value in seconds in the **Time out** field.
4. Enter the address of the server in the **Server Address** field.
5. Enter the authentication secret for RADIUS Server in the **Secret** field.
6. Click **Save** to save the entered details.
7. Click **Reset** to reset the entered details.

Remote Session

In MegaRAC SP, use this page to configure virtual media configuration settings for the next redirection session. Encryption is disabled by default.

To open Remote Session page, click **Configuration > Remote Session** from the main menu. A sample screenshot of Remote Session Page is shown in the screenshot below.

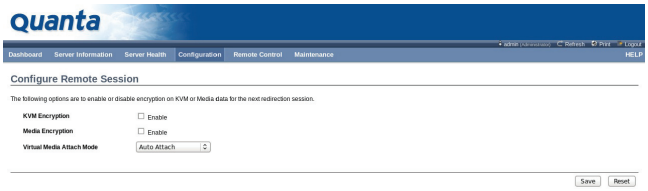

Figure 4-33. Remote Session

The fields of Remote Session Settings Page are explained below.

Table 4-28: Remote Session Settings Page

ITEM	DESCRIPTION
KVM Encryption	Enable/Disable encryption on KVM data for the next redirection session.

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Table 4-28: Remote Session Settings Page (Continued)

ITEM	DESCRIPTION
Media Encryption	Enable/Disable encryption on Media data for the next redirection session.
Virtual Media Attach Mode	Two types of VM attach mode are available: ● Attach: Immediately attaches Virtual Media to the server upon bootup. ● Auto Attach: Attaches Virtual Media to the server only when a virtual media session is started.
Save	To save the current changes. **Note:** It will automatically close the existing remote redirection either KVM or Virtual media sessions, if any.
Reset	To reset the modified changes.

Procedure:

1. In **KVM encryption**, check or uncheck the option **Enable**.
2. In **Media Encryption**, check or uncheck the option **Enable**.
3. In **Virtual media Attach mode**, select **Auto Attach** or **Attach** from the dropdown list as required.
4. Click **Save** to save the entries.
5. Click **Reset** to reset the entries

Note:

- If we choose more than one virtual CDROMs, then the RHEL5 host displays only one CDROM in the "Computer" window. When we redirect second CDROM, the second CDROM device will appear in "Computer" window.
- If we choose more than 2 virtual Hard disks, then the RHEL5 host displays only two hard disks in "Computer" window. When we redirect third hard disk, the third hard disk will appear in "Computer" window

SMTP

Simple Mail Transfer Protocol (SMTP) is an Internet standard for electronic mail (e-mail) transmission across Internet Protocol (IP) networks.

Using MegaRAC GUI, you can configure the SMTP settings of the device.

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To open SMTP Settings page, click **Configuration > SMTP** from the main menu. A sample screenshot of SMTP Settings Page is shown in the screenshot below.

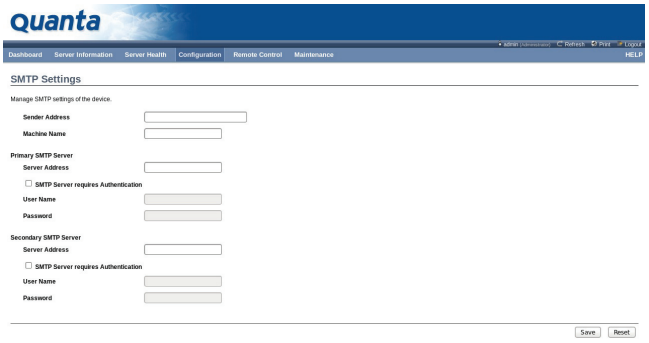

Figure 4-34. SMTP Settings Page

The fields of SMTP Settings Page are explained below.

Table 4-29: SMTP Settings Page

ITEM	DESCRIPTION
LAN Channel Number	Displays the list of LAN channels available.
Sender Address	The 'Sender Address' valid on the SMTP Server.
Machine Name	The 'Machine Name' of the SMTP Server. ● Machine Name is a string of maximum 15 alpha-numeric characters. ● Space, special characters are not allowed.

Table 4-29: SMTP Settings Page (Continued)

ITEM	DESCRIPTION
Primary SMTP Server	Lists the Primary SMTP Server configuration.
Server Address	The 'IP address' of the SMTP Server. It is a mandatory field. **Note:** ● IP Address made of 4 numbers separated by dots as in "xxx.xxx. xxx.xxx". ● Each Number ranges from 0 to 255. ● First Number must not be 0. ● Supports IPv4 Address format and IPv6 Address format.
SMTP Server requires Authentication	To enable/disable SMTP Authentication. **Note:** SMTP Server Authentication Types supported are: ● CRAM-MD5 ● LOGIN ● PLAIN If the SMTP server does not support any one of the above authentication types, the user will get an error message stating, "Authentication type is not supported by SMTP Server"

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Table 4-29: SMTP Settings Page (Continued)

ITEM	DESCRIPTION
Username	The username to access SMTP Accounts. **Note:** ● User Name can be of length 4 to 64 alpha-numeric characters. ● It must start with an alphabet. ● Special characters ','(comma), ':'(colon), ';'(semicolon), ' '(space) and '\'(backslash) are not allowed.
Password	The password for the SMTP User Account. **Note:** ● Password must be at least 4 characters long. ● White space is not allowed. ● This field will not allow more than 64 characters.
Secondary SMTP Server	It lists the Secondary SMTP Server configuration. It is an optional field. If the Primary SMTP server is not working fine, then it tries with Secondary SMTP Server configuration.
Save	To save the new SMTP server configuration.
Reset	To reset the modified changes.

Procedure:

1. Select the **LAN Channel Number** from the dropdown list.
2. Enter the **Sender Address** in the specified field.
3. Enter the **Machine Name** in the specified field.
4. In Primary SMTP Server, enter the **Server Address** in the specified field.
5. Enable the check box **SMTP Server requires Authentication** if you want to authenticate SMTP Server.
6. Enter your **User name** and **Password** in the respective fields.
7. In Secondary SMTP Server, enter the **Server Address** in the specific field.
8. Enable the check box **SMTP Server requires Authentication** if you want to authenticate SMTP Server.
9. Enter your **User name** and **Password** in the respective fields.
10. Click **Save** to save the entered details.
11. Click **Reset** to update the entered details.

SNMP

Simple Network Management Protocol (SNMP) is an "Internet-standard protocol for managing devices on IP networks."

Using MegaRAC GUI, you can configure the SNMP settings of the device.

BMC alert could issue by SNMP trap.

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To open SNMP Settings page, click **Configuration > SNMP** from the main menu. A sample screenshot of SNMP Settings Page is shown in the screenshot below.

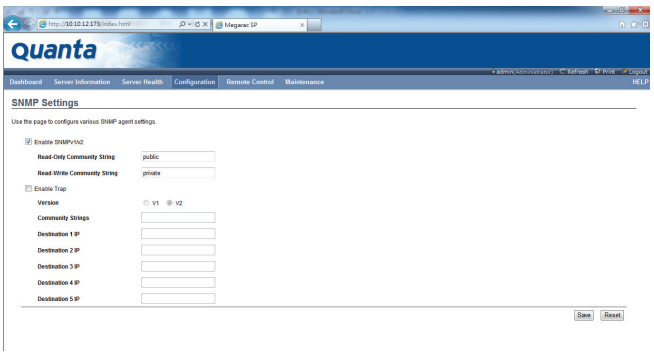

Figure 4-35. SNMP Settings Page

The fields of SNMP Settings Page are explained below.

Table 4-30: SMTP Settings Page

ITEM	DESCRIPTION
Enable SNMPv1/v2	Enable or disable the SNMP feature
Read-Only/Read-Write Community String	Set SNMP community string for read-only and read-write.
Enable Trap	Select it to make BMC send SNMP trap
Version	Select which trap version to send.

Table 4-30: SMTP Settings Page (Continued)

ITEM	DESCRIPTION
Community String	SNMP trap community
Destination IP	The destination IP for trap to send. BMC provides the 5 destinations.

SOL

Here, you can configure the Serial over LAN settings, select or change values for each attribute and click the Save button to save any changes.

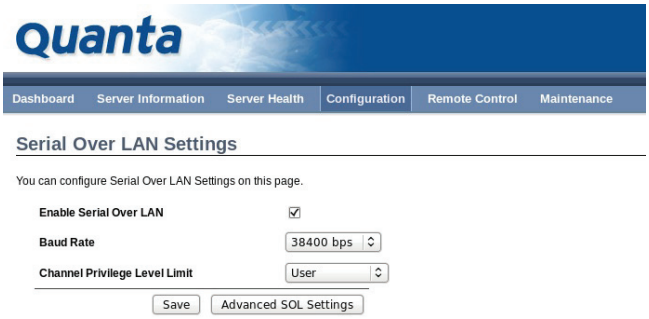

Figure 4-36. SOL Settings Page

The fields of SOL Settings Page are explained below.

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Table 4-31: SOL Settings Page

ITEM	DESCRIPTION
Enable Serial over LAN	Checked=Enabled; Unchecked=Disabled.
Channel Privilege Level Limit	Select the IPMI Serial over LAN minimum user privilege: • Administrator • Operator • User
Save	Use this button to save your settings.
Advanced SOL Settings	Use this button to go to advanced SOL page.

SSL

The Secure Socket Layer protocol was created by Netscape to ensure secure transactions between web servers and browsers. The protocol uses a third party, a Certificate Authority (CA), to identify one end or both end of the transactions.

Using MegaRAC GUI, configure SSL certificate into the BMC. Using this, the device can be accessed in a secured mode.

To open SSL Certificate Configuration page, click **Configuration > SSL** from the main menu. There are three tabs on this page.

- Upload SSL option is used to upload the certificate and private key file into the BMC.
- Generate SSL option is used to generate the SSL certificate based on configuration details.
- View SSL option is used to view the uploaded SSL certificate in readable format.

A sample screenshot of SSL Management Page is shown in the screenshot below.

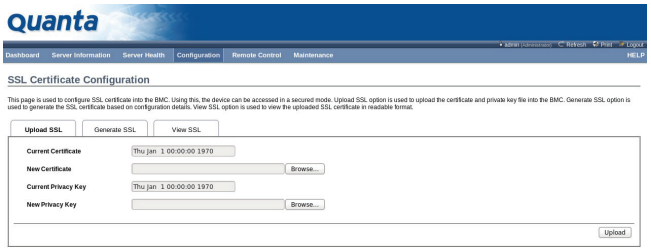

Figure 4-37. SSL Certificate Configuration – Upload SSL

The fields of SSL Certificate Configuration – Upload SSL tab are explained below.

Table 4-32: SSL Certificate Configuration - Upload SSL

ITEM	DESCRIPTION
Current Certificate	Current certificate information will be displayed (read-only).
New Certificate	Certificate file should be of pem type
Current Privacy Key	Current privacy key information will be displayed (read-only).

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Table 4-32: SSL Certificate Configuration - Upload SSL (Continued)

ITEM	DESCRIPTION
New Privacy Key	Privacy key file should be of pem type.
Upload	To upload the SSL certificate and privacy key into the BMC.

Note:
Upon successful upload, HTTPs service will get restarted to use the newly uploaded SSL certificate.

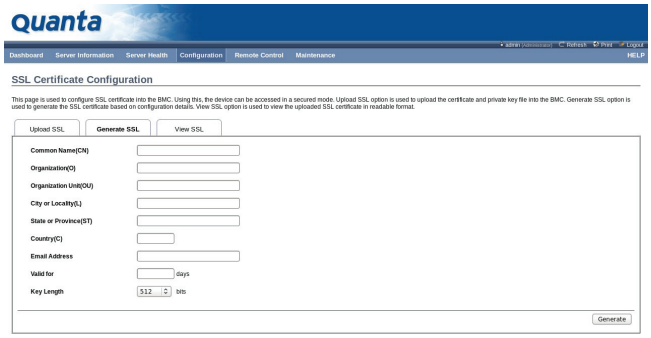

Figure 4-38. SSL Certificate Configuration – Generate SSL

The fields of SSL Certificate Configuration – Generate SSL tab are explained below.

Table 4-33: SSL Certificate Configuration - Generate SSL

ITEM	DESCRIPTION
Common Name (CN)	Common name for which certificate is to be generated. ● Maximum length of 64 characters. ● Special characters '#' and '$' are not allowed.
Organization (O)	Organization name for which the certificate is to be generated. ● Maximum length of 64 characters. ● Special characters '#' and '$' are not allowed.
Organization Unit (OU)	Over all organization section unit name for which certificate is to be generated. ● Maximum length of 64 characters. ● Special characters '#' and '$' are not allowed.
City or Locality (L)	City or Locality of the organization (mandatory). ● Maximum length of 64 characters. ● Special characters '#' and '$' are not allowed.
State or Province (ST)	State or Province of the organization (mandatory). ● Maximum length of 64 characters. ● Special characters '#' and '$' are not allowed.
Country (C)	Country code of the organization (mandatory). ● Only two characters are allowed. ● Special characters are not allowed.
Email Address	Email Address of the organization (mandatory).
Valid for	Validity of the certificate. ● Value ranges from 1 to 3650 days.
Key Length	The key length bit value of the certificate.

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Table 4-33: SSL Certificate Configuration - Generate SSL (Continued)

ITEM	DESCRIPTION
Generate	To generate the new SSL certificate.

Note:
HTTPs service will get restarted, to use the newly generated SSL certificate.

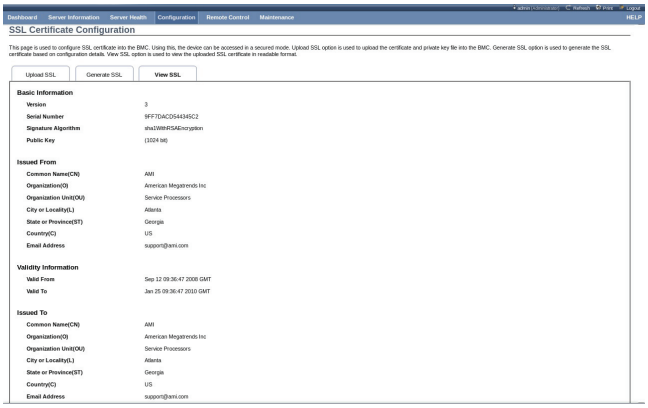

Figure 4-39. SSL Certificate Configuration – View SSL

The fields of SSL Certificate Configuration – View SSL tab are explained below.

Table 4-34: SSL Certificate Configuration – View SSL

ITEM	DESCRIPTION
Basic Information	This section displays the basic information about the uploaded SSL certificate. It displays the following fields. • Version • Serial Number • Signature Algorithm • Public Key
Issued From	This section describes the following Certificate Issuer information • Common Name (CN) • Organization (O) • Organization Unit(OU) • City or Locality (L) • State or Province (ST) • Country (C) • Email Address
Validity Information	This section displays the validity period of the uploaded certificate. • Valid From • Valid To

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Table 4-34: SSL Certificate Configuration – View SSL (Continued)

ITEM	DESCRIPTION
Issued To	This section display the information about the certificate issuer. • Common Name (CN • Organization (O • Organization Unit (OU • City or Locality (L • State or Province (ST • Country (C • Email Address

Procedure:

1. Click the Upload SSL Tab, **Browse** the **New Certificate** and **New Privacy** key.
2. Click **Upload** to upload the new certificate and privacy key.
3. In **Generate SSL** tab, enter the following details in the respective fields
   - The **Common Name** for which the certificate is to be generated.
   - The **Name of the Organization** for which the certificate is to be generated.
   - The **Overall Organization Section Unit** name for which certificate to be generated.
   - The **City or Locality** of the organization
   - The **State or Province** of the organization
   - The **Country** of the organization
   - The **email address** of the organization.
   - The number of days the certificate will be valid in the **Valid For** field.
4. Choose the **Key Length** bit value of the certificate
5. Click **Generate** to generate the certificate.
6. Click **View SSL** tab to view the uploaded SSL certificate in user readable format.

Note:

- Once you Upload/Generate the certificates, only HTTPs service will get restarted.
- You can now access your Generic MegaRAC® SP securely using the following format in your IP Address field from your Internet browser: https://<your MegaRAC® SP's IP address here>
- For example, if your MegaRAC® SP's IP address is 192.168.0.30, enter the following: https://192.168.0.30
- Please note the <s> after <http>.You must accept the certificate before you are able to access your Generic MegaRAC® SP.

User Management

In MegaRAC GUI, the User Management page allows you to view the current list of user slots for the server. You can add a new user and modify or delete the existing users.

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To open User Management page, click **Configuration > Users** from the main menu. A sample screenshot of User Management Page is shown in the screenshot below.

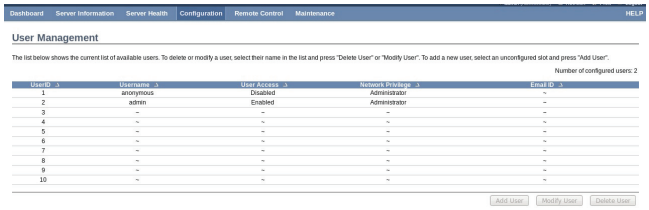

Figure 4-40. User Management

The fields of User Management Page are explained below.

Table 4-35: User Management Page

ITEM	DESCRIPTION
User ID	Displays the ID number of the user. **Note:** The list contains a maximum of ten users only.
User Name	Displays the name of the user.
User Access	To enable or disable the access privilege of the user.
Network Privilege	Displays the network access privilege of the user.
SNMP Status	Displays if the SNMP status for the user is enabled or Disabled.

Table 4-35: User Management Page (Continued)

ITEM	DESCRIPTION
Email ID	Displays email address of the user.
Add User	To add a new user. Modify.
User	To modify an existing user.
Delete User	To delete an existing user.

Procedure:

Note:
The Free slots are denoted by "~" in all columns for the slot.

Add a new user:

1. To add a new user, select a free slot and click **Add User**. This opens the Add User screen as shown in the screenshot below.

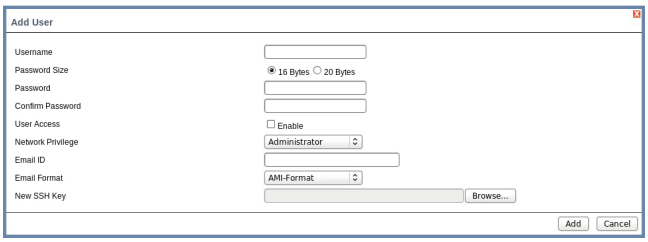

Figure 4-41. Add User Page

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2. Enter the name of the user in the **User Name** field.

Note:
- User Name is a string of 4 to 16 alpha-numeric characters.
- It must start with an alphabetical character.
- It is case-sensitive.
- Special characters ','(comma), '.'(period), ':'(colon), ';'(semicolon), ' '(space), '/'(slash), '\'(backslash), '('(left bracket) and ')'(right bracket) are not allowed.

3. In the **Password and Confirm Password** fields, enter and confirm your new password.

Note:
- Password must be at least 8 characters long.
- White space is not allowed.
- This field will not allow more than 20 characters.

4. Enable or Disable the User Access Privilege.
5. In the **Network Privilege** field, enter the network privilege assigned to the user which could be Administrator, Operator, User or No Access.
6. Check the **SNMP Status** check box to enable SNMP access for the user.

Note:
Password field is mandatory, if SNMP Status is enabled.

7. Choose the SNMP Access level option for user from the **SNMP Access** dropdown list. Either it can be Read Only or Read Write.
8. Choose the **Authentication Protocol** to use for SNMP settings from the drop down list.

Note:
Password field is mandatory, if Authentication protocol is changed.

9. Choose the Encryption algorithm to use for SNMP settings from the **Privacy protocol** dropdown list.
10. In the **Email ID** field, enter the email ID of the user. If the user forgets the password, the new password will be mailed to the configured email address.

Note:
SMTP Server must be configured to send emails.

- Email Format: Two types of formats are available:
  - AMI-Format: The subject of this mail format is 'Alert from (your Hostname)'. The mail content shows sensor information, ex: Sensor type and Description.
  - Fixed-Subject Format: This format displays the message according to user's setting. You must set the subject and message for email alert.

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11. In the **New SSK Key** field, click Browse and select the SSH key file.

Note:
SSH key file should be of pub type.

12. Click **Add** to save the new user and return to the users list.
13. Click **Cancel** to cancel the modification and return to the users list.

Modify an existing user:

14. Select an existing user from the list and click **Modify User**. This opens the Add User screen as shown in the screenshot below.

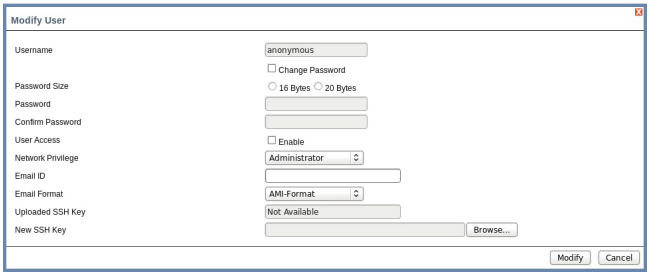

Figure 4-42. Modify User Page

15. Edit the required fields.
16. To change the password, enable the **Change Password** option.
17. After editing the changes, click **Modify** to return to the users list page.

Delete an existing User

18. To delete an existing user, select the user from the list and click **Delete User**.

Note:
There is a list of reserved users which cannot be added / modified as BMC users. Please Refer “MEGARAC SP-X Platform Porting Guide” section “Changing the Configurations in PMC File-> User Configurations in PMC File” for the list of reserved users.

Virtual Media

In MegaRAC GUI, this page to configure Virtual Media device settings. If you change the configuration of the virtual media in this page, it shows the appropriate device in the JViewer Vmedia dialog. For example, if you select two floppy devices in Configure Virtual Media page, then in JViewer > Vmedia, you can view two floppy device panel.

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To open Virtual Media page, click **Configuration > Virtual Media** from the main menu. A sample screenshot of User Management Page is shown in the screenshot below.

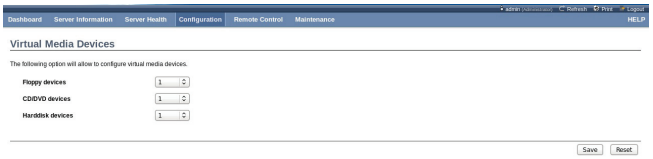

Figure 4-43. Configure Virtual Media Devices

The following fields are displayed in this page.

Table 4-36: Configure Virtual Media Devices

ITEM	DESCRIPTION
Floppy devices	The number of floppy devices that support for Virtual Media redirection.
CD/DVD devices	The number of CD/DVD devices that support for Virtual Media redirection.
Hard disk devices	The number of hard disk devices that support for Virtual Media redirection.
Local Media Support	To enable or disable the local media support for Virtual Media redirection.
Save	To save the configured settings.
Reset	To reset the previously-saved values.

Procedure:

1. Select the number of Floppy devices, CD/DVD devices and Hard disk devices from the dropdown list.

Note:
Maximum of two devices can be added in Floppy, CD/DVD and Hard disk drives.

2. Enable the **Local Media Support** if needed.
3. Click **Save** to save the changes made else click Reset to reset the previously saved values.

Note:
If there are two device panels for each device, and when you click the Connect button, then the redirected device panel will be disabled.

Remote Control

The Remote Control consists of the following menu items.

- Console Redirection
- Server Power Control

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A sample screenshot of the Remote Control menu is given below.

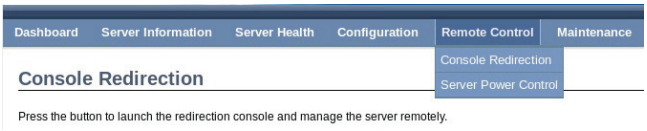

Figure 4-44. Remote Control Menu

A detailed description of the menu items are given ahead

Console Redirection

The remote console application, which is started using the WebGUI, allows you to control your server's operating system remotely, using the screen, mouse, and keyboard, and to redirect local CD/DVD, Floppy diskette and Hard disk/USB thumb drives as if they were connected directly to the server.

List of Supported Client Operating Systems

- WinXP
- W2K3 - 32 bit
- W2K3 - 64 bit
- RHEL 4 - 32 bit
- RHEL 4 - 64 bit
- RHEL 5.4 - 32 bit
- RHEL 5.4 - 64 bit
- RHEL 6.0 - 64 bit
- RHEL 6.0 - 32 bit
- Ubuntu 9.10 LTS - 32
- Ubuntu 9.10 LTS - 64
- Ubuntu 8.10 -32
- Ubuntu 8.10 -64
- OpenSuse 11.2 -32
- OpenSuse 11.2 -64
- FC 9 - 32
- FC 9 - 64
- FC 10 - 32
- FC 10 - 64
- FC 12 - 32
- FC 12 - 64
- FC 13 - 32
- FC 13 - 64
- FC 14 - 32
- FC 14 - 64
- MAC -32
- MAC-64

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List of Supported Host OS

- RHEL 5
- RHEL 6
- W2K3
- W2K8
- RHEL 4
- OpenSuse 11.2
- OpenSuse 10.x
- Ubuntu 8.10
- Ubuntu 9.10
- Ubuntu 11.04

Browser Settings

For Launching the KVM, pop-up block should be disabled. For Internet explorer, enable the download file options from the settings.

Java Console

This is an OS independent plug-in which can be used in Windows as well as Linux with the help of JRE. JRE should be installed in the client's system. You can install JRE from the following link.

http://www.java.com/en/download/manual.jsp

Procedure:

In MegaRAC GUI, the Java Console can be launched in two ways:

1. Open the Dashboard Page and in Remote control section, click Launch for Java Console.
2. Open **Remote Control > Console Redirection** Page and click **Java Console**.

This will download the **.jnlp** file from the BMC.

To open the .jnlp file, use the appropriate JRE version (Javaws)

The Console Redirection window opens when the downloading is done.

The Console Redirection main menu consists of the following menu items.

- Video
- Keyboard
- Mouse
- Options
- Media
- Keyboard Layout
- Video Record
- Power
- Active Users

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- Help

A detailed explanation of these menu items are given below.

Video

This menu contains the following sub menu items.

Table 4-37: Video

ITEM	DESCRIPTION
Pause redirection	This option is used for pausing Console Redirection.
Resume Redirection	This option is used to resume the Console Redirection when the session is paused.
Refresh Video	This option can be used to update the display shown in the Console Redirection window.
Turn Off Host display	If you enable this option, the server display will be blank but you can view the screen in Console Redirection. If you disable this option, the display will be back in the server screen.
Full Screen	This option is used to view the Console Redirection in full screen mode (Maximize). This menu is enabled only when both the client and host resolution are same.
Exit	This option is used to exit the console redirection screen.

Keyboard

This menu contains the following sub menu items.

Table 4-38: Keyboard

ITEM	DESCRIPTION
Hold Right Ctrl Key	This menu item can be used to act as the right-side <CTRL> key when in Console Redirection.
Hold Right Alt Key	This menu item can be used to act as the right-side <ALT> key when in Console Redirection.
Hold Left Ctrl Key	This menu item can be used to act as the left-side <CTRL> key when in Console Redirection.
Hold Left Alt Key	This menu item can be used to act as the left-side <ALT> key when in Console Redirection.
Left Windows Key	This menu item can be used to act as the left-side <WIN> key when in Console Redirection. You can also decide how the key should be pressed: Hold Down or Press and Release.
Right Windows Key	This menu item can be used to act as the right-side <WIN> key when in Console Redirection. You can also decide how the key should be pressed: Hold Down or Press and Release.
Alt+Ctrl+Del	This menu item can be used to act as if you depressed the <CTRL>, <ALT> and <DEL> keys down simultaneously on the server that you are redirecting.

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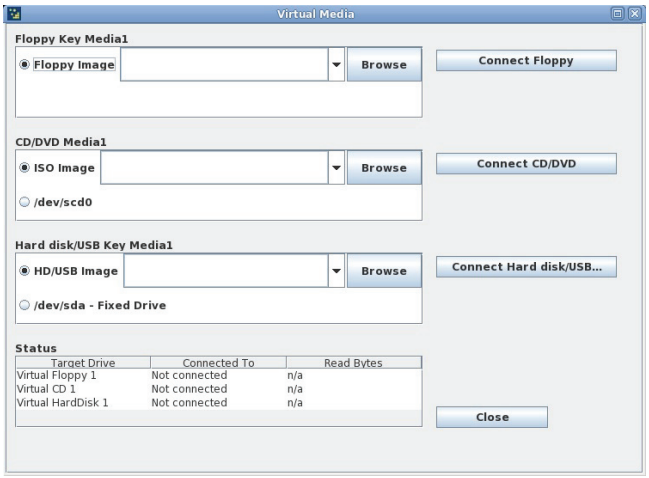

Figure 4-45. Virtual Media

Table 4-39: Virtual Media

ITEM	DESCRIPTION
Floppy Key Media	This menu item can be used to start or stop the redirection of a physical floppy drive and floppy image types such as **.img*. **Note:** Floppy Redirection is not an available feature on all versions of the MegaRACR SPs.

Table 4-39: Virtual Media (Continued)

ITEM	DESCRIPTION
CD/DVD Media	This menu item can be used to start or stop the redirection of a physical DVD/CD-ROM drive and cd image types such as iso.
Hard disc/USB Key Media	This menu item can be used to start or stop the redirection of a Hard Disk/USB key image and USB key image such as **.img*. **Note:** • For windows client, if the logical drive of the physical drive is dismount then the logical device is redirected with Read/Write Permission else it is redirected with Read permission only. • For MAC client, External USB Hard disk redirection is only supported. • For Linux client, fixed hard drive is redirected only as Read Mode. It is not Write mode supported. • For USB key image redirection, support FAT 16, FAT 32 and NTFS.

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Keyboard Layout

Table 4-40: Keyboard Layout

ITEM	DESCRIPTION
Auto Detect	This option is used to detect keyboard layout automatically. The languages supported automatically are English – US, French – France, Spanish – Spain, German- Germany, Japanese- Japan. If the client and host languages are same, then for all the languages other than English mentioned above, you must select this option to avoid typo errors.
Soft Keyboard	This option allows you to select the keyboard layout. It will show the dialog as similar to onscreen keyboard. If the client and host languages are different, then for all the languages other than English mentioned above, you must select the appropriate language in the list shown in JViewer and use the soft keyboard to avoid typo errors. **Note:** Soft keyboard is applicable only for JViewer Application not for other application in the client system.

Video Record

Note:
This option is available only when you launch the Java Console.

Table 4-41: Video Record

ITEM	DESCRIPTION
Important	To view this menu option you must download the Java Media FrameWork (JMF). It can be downloaded from the link http://www.oracle.com/technetwork/java/javase/download-142937.html
Start Record	This option is to start recording the screen.
Stop Record	This option is used to stop the recording.
Settings	To set the settings for video recording.

Procedure:

Note:
Before you start recording, you have to enter the settings.

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1. Click **Video Record > Settings** to open the settings page as shown in the screenshot below.

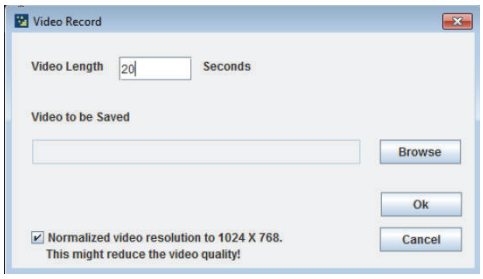

Figure 4-46. Video Record Settings Page

2. Enter the **Video Length** in seconds.
3. **Browse** and enter the location where you want the video to be saved.
4. Enable the option **Normalized video resolution to 1024X768**.
5. Click **OK** to save the entries and return to the Console Redirection screen.
6. Click **Cancel** if you don't wish to save the entries.
7. In the Console Redirection window, click **Video Record > Start Record**.
8. Record the process.
9. To stop the recording, click **Video Record > Stop Record**.

Power

The power option is to perform any power cycle operation. Click on the required option to perform the **following operation. Reset Server**: To reboot the system without powering off (warm boot).

Table 4-42: Power

ITEM	DESCRIPTION
Power Off Server - Immediate	To immediately power off the server.
Power Off Server - Orderly Shutdown	To initiate operating system shutdown prior to the shutdown.
Power On Server	To power on the server.
Power Cycle Server	To first power off, and then reboot the system (cold boot).

Active Users

Click this option to displays the active users and their system IP address.

Help

Jviewer: Displays the copyright and version information

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Quick Buttons

The lower right of Console Redirection windows displays all the quick buttons. These quick buttons helps you to perform these functions by just clicking them.

Note:
This option is available only when you launch the Java Console.
Server Power Control.

Server Power Control

This page allows you to view and control the power of your server.

To open Power Control and Status page, click **Remote Control > Server Power Control** from the main menu. A sample screenshot of Power Control and Status page is shown in the screenshot below.

Figure 4-47. Power Control and Status Page

The various options of Power Control are given below.

Table 4-43: Server Power Control

ITEM	DESCRIPTION
Reset Server	This option will reboot the system without powering off (warm boot).
Power Off Server – Immediate	This option will immediately power off the server.
Power Off Server – Orderly Shutdown	This option will initiate operating system shutdown prior to the shutdown.
Power On Server	This option will power on the server.
Power Cycle Server	This option will first power off, and then reboot the system (cold boot).
Perform Action	Click this option to perform the selected operation.

Procedure:

Select an action and click Perform Action to proceed with the selected action.

Note:
You will be asked to confirm your choice. Upon confirmation, the command will be executed and you will be informed of the status.

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Maintenance Group

This group of pages allows you to do maintenance tasks on the device. The menu contains the following items:

- Firmware Update
- Preserve Configuration
- Restore Factory Defaults
- System Administrator

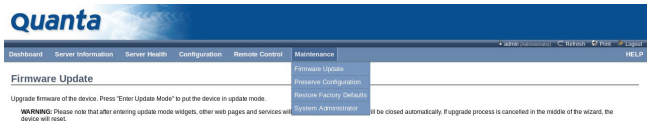

Figure 4-48. Maintenance Menu

Firmware Update

In MegaRAC GUI, this wizard takes you through the process of firmware up gradation. A reset of the box will automatically follow if the upgrade is completed or cancelled. An option to preserve configuration will be presented. Enable it, if you wish to preserve configured settings through the upgrade.

WARNING!

Please note that after entering update mode widgets, other web pages and services will not work. All open widgets will be closed automatically. If upgrade process is cancelled in the middle of the wizard, the device will be reset.

Note:

The firmware upgrade process is a crucial operation. Make sure that the chances of a power or connectivity loss are minimal when performing this operation.

Once you enter into Update Mode and choose to cancel the firmware flash operation, the MegaRAC® card must be reset. This means that you must close the Internet browser and log back onto the MegaRAC® card before you can perform any other types of operations.

To open Firmware Update page, click **Maintenance > Firmware Update** from the main menu. A sample screenshot of Firmware Update Page is shown in the screenshot below.

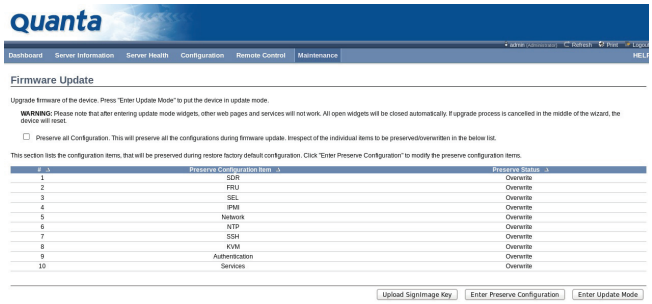

Figure 4-49. Firmware Update Page

Procedure:

Click **Enter Update Mode** to upgrade the current device firmware. As below step by step:

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1. Closing all active client requests.
2. Preparing device for firmware upgrade.
3. Uploading firmware image.
4. Verifying firmware image.
5. Flashing firmware image.
6. Resetting Device.

Note:
You can now follow the instructions presented in the subsequent pages to successfully update the card's firmware. The device will reset if update is canceled. The device will also reset upon successful completion of firmware update.

Preserve Configuration

This page allows the user to configure the preserve configuration items, which will be used by the Restore factory defaults to preserve the existing configuration without overwriting with default configuration.

Restore Factory Defaults

In MegaRAC GUI, this option is used to restore the factory defaults of the device firmware.

WARNING!
Please note that after entering restore factory widgets, other web pages and services will not work. All open widgets will be closed automatically. The device will reset and reboot within a few minutes.

To open Restore Factory Defaults page, click **Maintenance > Restore Factory Defaults** from the main menu. A sample screenshot of Restore Factory Defaults Page is shown in the screenshot below.

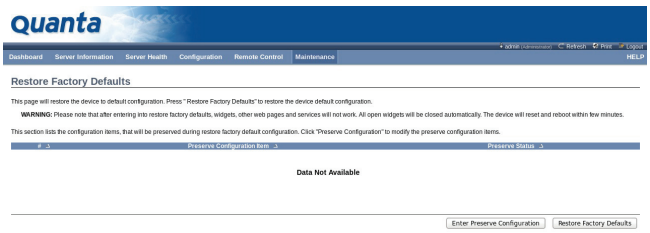

Figure 4-50. Restore Factory Defaults Page

Procedure:

Click Restore Factory to restore the factory defaults of the device firmware.

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System Administrator

This page is used to configure the System Administrator settings.

To open System Administrator page, click **Maintenance > System Administrator** from the main menu. A sample screenshot of System Administrator page is shown in the screenshot below.

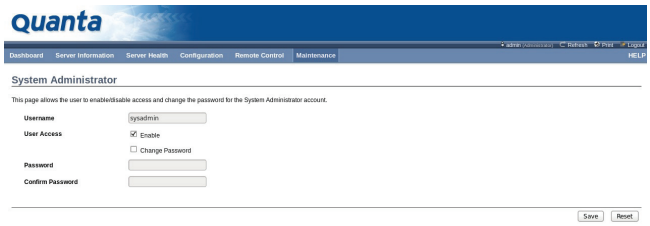

Figure 4-51. System Administrator

The various fields of System Administrator page are given below. Username: Username of System

Administrator is a read only field.

Table 4-44: System Administrator Page

ITEM	DESCRIPTION
User Access	To enable user access for system administrator.

Table 4-44: System Administrator Page (Continued)

ITEM	DESCRIPTION
Change Password	To change the user's password. **Note:** • Password, Confirm Password: ▪ Password must be at least 8 characters long. ▪ White space is not allowed. • This field will not allow more than 64 characters.
Save	This field will not allow more than 64 characters.
Reset	To reset the modified changes.

Procedure:

1. To enable User Access, check the Enable option.
2. Enable Change Password option to change the user password. This action enables the password fields.
3. Enter the new password in the Password field.
4. Re-enter the password in the Confirm Password field.
5. Click Save to save the changes.
6. Click Reset to reset the changes.

Log Out

To log out of the MegaRAC GUI, click the logout link on the top right corner of the screen.

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User Privilege

Table 4-45: User Privilege

WEB GUI PRIVILEGE LIST	PRIVILEGE ASSOCIATION BETWEEN IPMI AND WEB GUI			
	ADMINISTRATOR	OPERATOR	USER	OEM
login BMC from Web GUI, SSH	O	O	X	O
configure BMC from Web GUI	O	X	X	X
configure users from Web GUI	O	X	X	X
clear logs from Web GUI	O	X	X	X
execute server power control from Web GUI	O	X	X	X
virtual KVM redirection	O	X	X	X
virtual media	O	X	X	X
View Users	O	O	X	X
View DNS	O	O	X	X
View Network	O	O	X	X
View PEF	O	O	X	X

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Mainboard Jumpers and Connectors

Chapter 5

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5.1 Mainboard Jumpers and Connectors

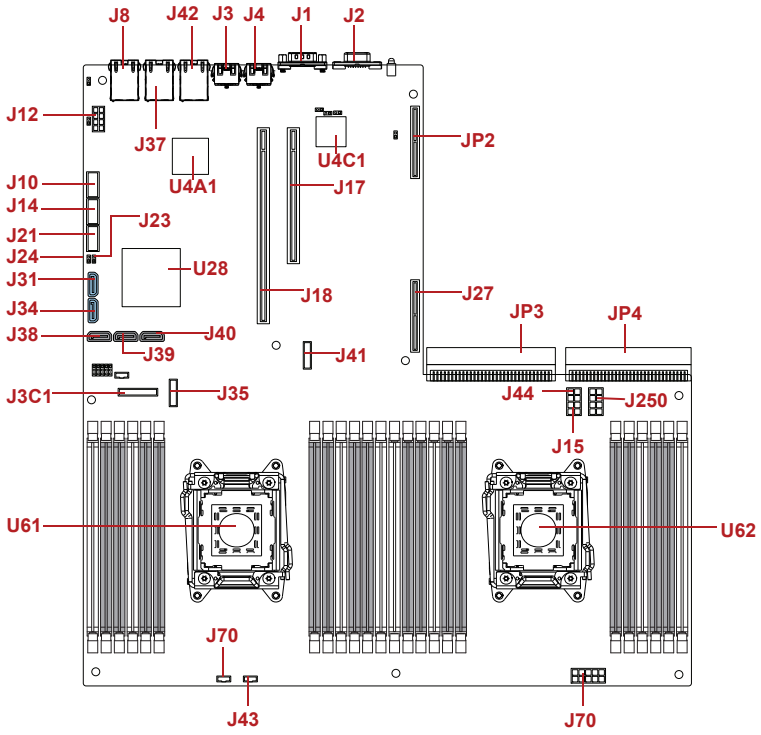

Figure 5-1. Mainboard Connectors

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MAINBOARD JUMPERS AND CONNECTORS

Mainboard Connectors

Table 5-1: Mainboard Connectors

CONNECTOR LOCATION	DESCRIPTION
JP3, JP4	Power Supply connector x2
U61	CPU0 Socket
U62	CPU1 Socket
U28	PCH Chip
U4A1	LAN Chip
U4C1	BMC Chip
J41	XDP CPU connector
J35	XDP PCH connector
J250, J70	Mother Board to back plane Power Connectors
J12	TPM module connector
J38, J39	Mini-SAS (SAS/SATA 6G x4) connector x 2
J40	Mini-SAS (SATA 3G x4) connector
J31, J34	SATA (6G) Connector x 2
J27	SAS2 MEZZ connector

Table 5-1: Mainboard Connectors (Continued)

CONNECTOR LOCATION	DESCRIPTION
JP2	10Gb NIC MEZZ connector
J1	COM (Rear) connector
J10	COM (Internal) connector
J14	CPLD JTAG connector
J2	VGA (Rear) connector - DSUB15
J3C1	Front Panel connector
J21	USB (Front) connector (2-port)
J3	USB (Rear) connector USB_CONN0
J4	USB (Rear) connector USB_CONN1
J17	PCIe x20 Slot6
J18	PCIe x16 Slot7
J71	IPMB connector
J5	Intrusion Detect Jumper
J7	Mfg. mode Jumper
J44, J15	VGA card power connector
J23, J24	SAS SES0/1

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MAINBOARD JUMPERS AND CONNECTORS

Table 5-1: Mainboard Connectors (Continued)

CONNECTOR LOCATION	DESCRIPTION
BT1	Battery connector

Mainboard Jumpers

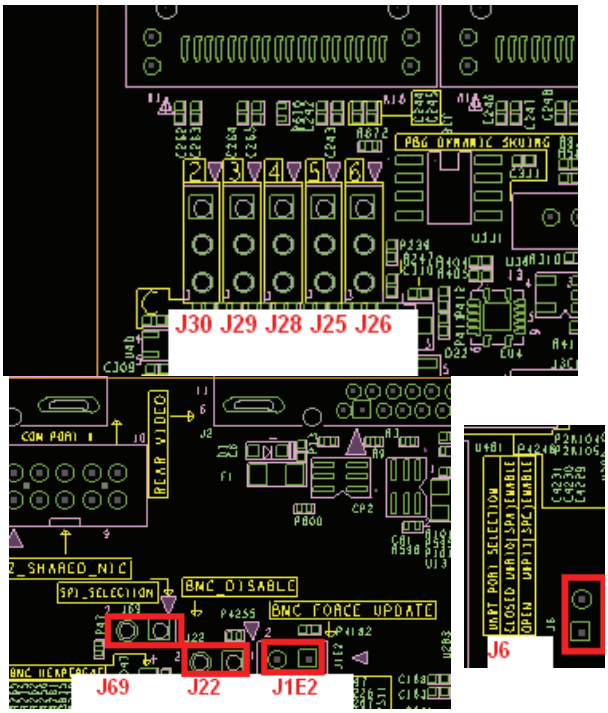

Figure 5-2. Mainboard Jumpers

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MAINBOARD JUMPERS AND CONNECTORS

Table 5-2: User-Accessible Jumpers

JUMPER LOCATION	DEFAULT SETTING	FUNCTION
J30	1-2	PASSWORLD CLEAR • 1-2 : NORMAL • 2-3 : PASSWORD CLEAR
J16	1-2	PLD DEBUG MODE • 1-2 : DISABLE DEBUG MODE • 2-3 : ENABLE DEBUG MODE
J28	1-2	CMOS CLEAR • 1-2 NORMAL RTC 2-3 CLR • RTC REGISTERS
J26	1-2	BIOS RECOVERY MODE • 1-2 : NORMAL OPERATION • 2-3 : RECOVER BIOS
J29	1-2	ME FIRMWARE UPDATE • 1-2: NORMAL MODE • 2-3: ME IN FORCE UPDATE MODE
J25	1-2	QPI SLOW MODE SELECTION • 1-2 : QPI FAST MODE • 2-3 : QPI SLOW MODE (66MT/s)

Table 5-2: User-Accessible Jumpers (Continued)

JUMPER LOCATION	DEFAULT SETTING	FUNCTION
J1E2	Non-Pop	FORCE IBMC UPDATE • Non-pop: NORMAL MODE • 1-2 : FORCE UPDATE MODE
J22	Non-Pop	IBMC DISABLE • Non-pop: NORMAL MODE • 1-2 : DISABLE BMC
J69	Non-Pop	SPI SELECTION • Non-pop : System BIOS used • 1-2 : BMC used
J6	Non-Pop	RS232 SELECTION • Non-pop : System used • 1-2 : BMC debug used
J7	OPEN	MFG MODE (Reserved)
J5	OPEN	PBG INTRUDER DETECT (Reserved)

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Rail Kit Assembly

Chapter 6

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RAIL KIT ASSEMBLY

6.1 Rail Kit Assembly

Installing the Rack Brackets

CAUTION!
BEFORE REMOVING THE SERVER COVER, DISCONNECT ALL POWER. UNPLUG THE AC POWER CORD; DISCONNECT ALL PERIPHERIALS, AND ALL LAN LINES.

When the server is to be installed into a rack, it is necessary to install two rack mount brackets onto the server chassis prior to installing the server into the rack. Follow the procedures below to install the rack mount brackets onto the server.

Preparing for the Installation

- Review all the tasks required to install and configure the system
- Review the electrical requirements and power supply input voltage information
- Examine the system's shipping kit contents and assemble the required tools for installation
- Review ESD requirements and take safety precautions
- Before installing the system into a rack, install any optional components into the system
- Stabilize the rack before installing the system, if necessary

WARNING!
Use caution when removing the inner rail from the rail kit assembly. The rails can slide out and cause personal injury or damage to the server.

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Installing the Inner Rail

Follow these instructions to install inner rail:

1. Slide the release tab to the open position and remove the inner rail from the rail kit assembly.

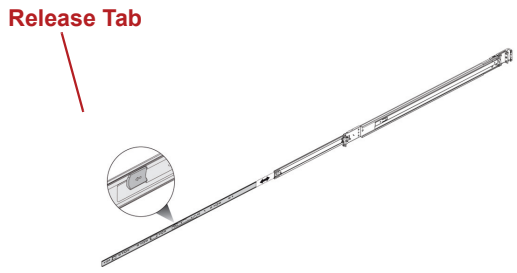

Figure 6-1. Disassembling the Rails

2. Align and insert the holes on the inner rail into the pins on the side of the chassis.

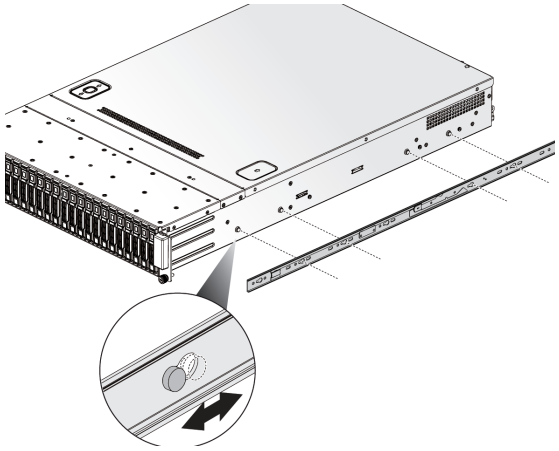

Figure 6-2. Installing the Inner Rail

3. Slide the inner rail until the latch key on the rail locks and is secure to the chassis.
4. Secure the rail with the single screw.

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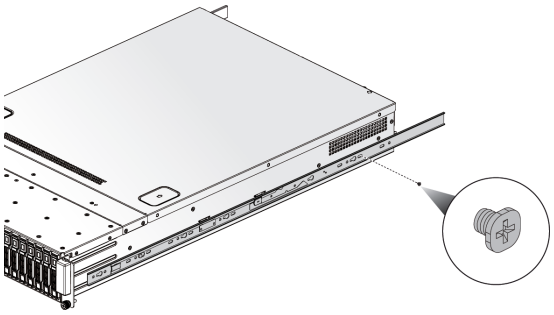

Figure 6-3. Securing the Inner Rail

5. Repeat steps 1 through 3 for the remaining side.

Installing the Outer Rail

1. Align the screws on the outer rail with the screw holes on the front cabinet column.
2. Guide the screws on outer rail through the screw holes on the front cabinet column until the rail clicks in place.
3. Extend the rear side of the outer rail, to reach the rear cabinet column.
4. Guide the screws on outer rail through the screw holes on the rear cabinet column until the rail clicks in place.

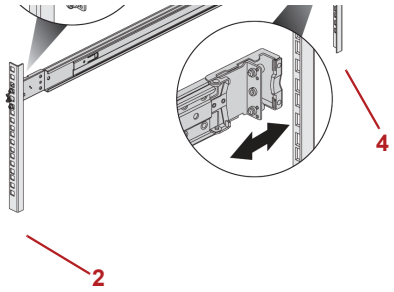

Figure 6-4. Installing the Outer Rail

5. Repeat steps 1 to 4 for the second rail.

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WARNING!
Heavy object. Two people are required to safely handle the server. Always ask for assistance to install or remove the server.

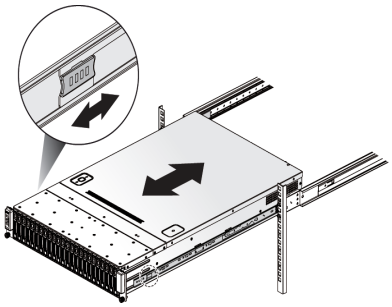

Figure 6-5. Aligning the Rails

6. Align the inner rail with the outer rail.
7. Horizontally install the system half way into the slide rail.
8. Slide the release tab into locked position.
9. Slide the server into the rails till flush with the rack.

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Troubleshooting

Chapter 7

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TROUBLESHOOTING

7.1 Troubleshooting

Server Boot Issue Topics

System does not Boot after initial installation:

- “Power Cord Not Plugged In”
- Mainboard Module Configuration Issues
- “Processor Issues”
- “Memory Issues”
- “Monitor Issues”
- “Power Supply, Chassis and Fan Issues”
- “Cable Issues”
- “Electrical Short or Overload”
- “Defective Components”

System does not boot after configuration changes:

- “Hardware Changes”
- “Software Changes”
- “BIOS Changes”
- “Installation Problems”
- “Troubleshooting External Connections”

System does not Boot after Initial Installation

Power Cord Not Plugged In

If the power supply cable is not plugged into the chassis power connector, the system cannot boot up, even though chassis front panel LEDs and the fan may be operational. Verify that the power connections are good.

Processor Issues

Boot failure situations are also caused by the following:

Incompatible processor - ensure the selected processor model is correct for your server board. If the processor is compatible, try removing and reinstalling the processor to ensure it is installed correctly.

Processor overheat-the system does not boot or shuts down shortly after booting.

- Ensure that the cooling fans are correctly installed and running.
- Ensure that the correct thermal interface material or the thermal grease is applied to the processor.
- Ensure that the power supply fan is running.

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- Ensure that the air intakes for the fans are unobstructed.

Memory Issues

If you have installed incompatible memory modules, the system may not boot. Verify the memory you've installed has been tested with your board (Please refer to www.QuantaQCT.com for details on valid memory). If the installed memory is compatible, remove and reinstall the memory modules. Defective memory modules may cause boot errors. To isolate a specific memory module as defective, boot the system with just one memory module installed at a time.

Monitor Issues

Monitor configurations can cause boot failure. Run through the following checklist to verify monitor operation:

- Ensure the monitor is plugged in and turned on.
- Ensure all cables are connected properly between the monitor and the computer.
- Check the brightness and contrast controls on the monitor are not too low.

Most monitors employ indicator LEDs showing status. Refer to the monitor's documentation to confirm operation. If the problem still persists, try replacing the monitor or test the monitor on a different AC outlet/different system.

Power Supply, Chassis and Fan Issues

- Ensure that the chassis and power supply is appropriate for system requirement.
- Ensure all power cables and connectors are firmly connected to the power supply and the AC outlet.
- If the power supply or the AC outlet has an on/off switch, make sure that it is on and verify that the outlet is supplying current.
- Check for foreign objects inside the chassis such as screws that can short circuit connections.
  - To isolate a specific PSU as defective, boot the system with just one PSU installed at a time.
  - Check fan speed in WEBUI & event log to find out if there are any defective fans. If failure happens, please contact your dealer for assistance.

Cable Issues

Ensure that all cable connections, both internal and external, are attached correctly and securely.

Electrical Short or Overload

Remove non-essential items such as extra controller cards (e.g SAS 6G Mezz/B, 10G Mezz/B) or HDD devices to check for shorts and overloads.

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If the system boots correctly, there may be a short or overload associated with one of the components.

Replace each of non-essential items one at a time to isolate which one is causing the problem.

If the problem occurs even after removing the non-essential components, the problem has to be with the server board, power supply, memory, or processor.

Defective Components

Defective components, especially processor and memory, can cause system boot issues.

- Swap the memory modules with known good memory. Verify correct operation of the suspected memory in a known working system.
- Swap the processor with a known good processor. Verify correct operation of the suspected processor in a known working system.

System does not boot after Configuration Changes

Hardware Changes

If the system does not boot after making changes to hardware or adding new components, verify that the component installed is compatible with the server.

Software Changes

If you recently installed new software or new device drivers:

- Try booting into Safe Mode and uninstall the new software or driver. If you can now boot normally, there may be a compatibility issue between the new software or driver and some component in your system. Contact the software manufacturer for assistance.

BIOS Changes

Changes to some advanced BIOS settings can cause boot issues. Changes to Advanced BIOS settings should only be made by experienced users.

If the BIOS Setup Utility is accessible by pressing **F2** during boot, reset the BIOS to factory defaults by pressing **F9**. Save and exit the BIOS Setup

If you cannot access the BIOS Setup Utility, clear the CMOS by performing the following steps:

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1. Power down the server. Do not unplug the power cord.
2. Open the server chassis
3. Move the CMOS CLEAR jumper from the default operation position, covering pins 1 and 2, to the reset / clear CMOS, covering pins 2 and 3.
4. Remove AC power.
5. Wait 5 seconds.
6. Move the jumper back to default position, covering pins 1 and 2.
7. Close the server chassis and power up the server.

The CMOS is now cleared and can be reset by going into BIOS setup.

Please refer to www.QuantaQCT.com for the BIOS update.

Installation Problems

Perform the following checks if you are troubleshooting an installation problem:

Check all cable and power connections (including all rack cable connections). Unplug the power cord, and wait one minute. Then reconnect the power cord and try again. If the network is reporting an error, see if the server has enough memory and disk space available. Remove all added options, one at a time, and try to power up the system. If after removing an option the server works, you may find that it is a problem with the option or a configuration problem between the option and the server. Contact the option vendor for assistance.

- If the system doesn't power on, check the LED display. If the power LED is not on, you may not be receiving AC power. Check the AC power cord to make sure that it is securely connected.

Troubleshooting External Connections

Loose or improperly connected cables are the most likely source of problems for the system, monitor, and other peripherals (such as a keyboard, mouse, or other external device). Ensure that all external cables are securely attached to the external connectors on your system.

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Installation and Assembly Safety Instructions

Chapter 8

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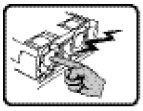
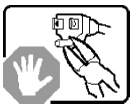
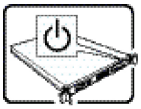
INSTALLATION AND ASSEMBLY SAFETY INSTRUCTIONS

8.1 Installation and Assembly Safety Instructions

Guidelines

	The power supply in this product contains no user-serviceable parts. Refer servicing only to qualified personnel.
	Do not attempt to modify or use the supplied AC power cord if it is not the exact type required. A product with more than one power supply will have a separate AC power cord for each supply.
	The power button on the system does not turn off system AC power. To remove AC power from the system, you must unplug each AC power cord from the wall outlet or power supply. The power cord(s) is considered the disconnect device to the main (AC) power. The socket outlet that the system plugs into shall be installed near the equipment and shall be easily accessible.

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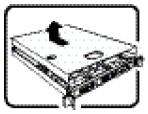
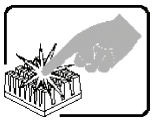
	SAFETY STEPS: Whenever you remove the chassis covers to access the inside of the system, follow these steps: 1. Turn off all peripheral devices connected to the system. 2. Turn off the system by pressing the power button. 3. Unplug all AC power cords from the system or from wall outlets. 4. Label and disconnect all cables connected to I/O connectors or ports on the back of the system. 5. Provide some electrostatic discharge (ESD) protection by wearing an antistatic wrist strap attached to chassis ground of the system-any unpainted metal surface-when handling components. 6. Do not operate the system with the chassis covers removed.
	After you have completed the six SAFETY steps above, you can remove the system covers. To do this: 1. Unlock and remove the padlock from the back of the system if a padlock has been installed. 2. Remove and save all screws from the covers. 3. Remove the cover(s).
	A microprocessor and heat sink may be hot if the system has been running. Also, there may be sharp pins and edges on some board and chassis parts. Contact should be made with care. Consider wearing protective gloves.

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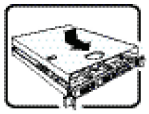
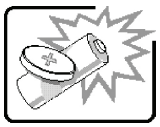
	For proper cooling and airflow, always reinstall the chassis covers before turning on the system. Operating the system without the covers in place can damage system parts. To install the covers: 1. Check first to make sure you have not left loose tools or parts inside the system. 2. Check that cables, add-in cards, and other components are properly installed. 3. Attach the covers to the chassis with the screws removed earlier, and tighten them firmly. 4. Insert and lock the padlock to the system to prevent unauthorized access inside the system. 5. Connect all external cables and the AC power cord(s) to the system.
	Danger of explosion if the battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the equipment manufacturer. Dispose of used batteries according to manufacturer's instructions.

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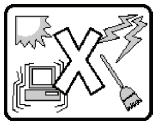	The system is designed to operate in a typical office environment. Choose a site that is: • Clean and free of airborne particles (other than normal room dust). • Well ventilated and away from sources of heat including direct sunlight. • Away from sources of vibration or physical shock. • Isolated from strong electromagnetic fields produced by electrical devices. • In regions that are susceptible to electrical storms, we recommend you plug your system into a surge suppressor and disconnect telecommunication lines to your modem during an electrical storm. • Provided with a properly grounded wall outlet. • Provided with sufficient space to access the power supply cord(s), because they serve as the product's main power disconnect.
	The server system is safety certified as rack-mounted equipment for use in a server room or computer room, using the customer rack kit. The rail racks are designed to carry only the weight of the server system. Do not place additional load onto any rail-mounted equipment. System rack kits are intended to be installed in a rack by trained service technicians.
	Heavy object. Indicates two people are required to safely handle the system.

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Safety Information

Chapter 9

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9.1 Server Safety Information

To reduce the risk of bodily injury, electrical shock, fire, and equipment damage, read this document and observe all warnings and precautions in this guide before installing or maintaining your server product.

In the event of a conflict between the information in this document and information provided with the product or on the website for a particular product, the product documentation takes precedence.

Your server should be integrated and serviced only by technically qualified persons.

You must adhere to the guidelines in this guide and the assembly instructions in your server manuals to ensure and maintain compliance with existing product certifications and approvals. Use only the described, regulated components specified in this guide. Use of other products / components will void the UL Listing and other regulatory approvals of the product, and may result in noncompliance with product regulations in the region(s) in which the product is sold.

Safety Warnings and Cautions

To avoid personal injury or property damage, before you begin installing the product, read, observe, and adhere to all of the following safety instructions and information. The following safety symbols may be used throughout the documentation and may be marked on the product and / or the product packaging.

Table 9-1: Safety Warning and Cautions

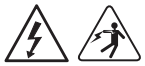
CAUTION	Indicates the presence of a hazard that may cause minor personal injury or property damage if the CAUTION is ignored.
WARNING	Indicates the presence of a hazard that may result in serious personal injury if the WARNING is ignored.
	Indicates potential hazard if indicated information is ignored.
	Indicates shock hazards that result in serious injury or death if safety instructions are not followed.
	Indicates hot components or surfaces.
	Indicates do not touch fan blades, may result in injury.

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Table 9-1: Safety Warning and Cautions (Continued)

	Indicates to unplug all AC power cord(s) to disconnect AC power.
	Please recycle battery.
	The rail racks are designed to carry only the weight of the server system. Do not use rail-mounted equipment as a workspace. Do not place additional load onto any rail-mounted equipment.
	Indicates two people are required to safely handle the system.
	Restricted Access Location: The server is intended for installation only in a Server Room or Computer Room where both these conditions apply: ■ access can only be gained by SERVICE PERSONS or by USERS who have been instructed about the reasons for the restrictions applied to the location and about any precautions that shall be taken; and ■ access is through the use of a TOOL or lock and key, or other means of security, and is controlled by the authority responsible for the location.

Intended Application Uses

This product was evaluated as Information Technology Equipment (ITE), which may be installed in offices, schools, computer rooms, and similar commercial type locations. The suitability of this product for other product categories and environments (such as medical, industrial, residential, alarm systems, and test equipment), other than an ITE application, may require further evaluation.

Site Selection

The system is designed to operate in a typical office environment. Choose a site that is:

- Clean, dry, and free of airborne particles (other than normal room dust).
- Well-ventilated and away from sources of heat including direct sunlight and radiators.
- Away from sources of vibration or physical shock.
- Isolated from strong electromagnetic fields produced by electrical devices.
- In regions that are susceptible to electrical storms, we recommend you plug your system into a surge suppressor and disconnect telecommunication lines to your modem during an electrical storm.
- Provided with a properly grounded wall outlet.
- Provided with sufficient space to access the power supply cord(s), because they serve as the product's main power disconnect.
- Provided with either two independent AC power sources or two independent phases from a s single source.

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Equipment Handling Practices

Reduce the risk of personal injury or equipment damage:

- Conform to local occupational health and safety requirements when moving and lifting equipment.
- Use mechanical assistance or other suitable assistance when moving and lifting equipment.
- To reduce the weight for easier handling, remove any easily detachable components.

Power and Electrical Warnings

CAUTION!

THE POWER BUTTON, INDICATED BY THE STAND-BY POWER MARKING, DOES NOT COMPLETELY TURN OFF THE SYSTEM AC POWER, 5V STANDBY POWER IS ACTIVE WHENEVER THE SYSTEM IS PLUGGED IN. TO REMOVE POWER FROM SYSTEM, YOU MUST UNPLUG THE AC POWER CORD FROM THE WALL OUTLET. YOUR SYSTEM MAY USE MORE THAN ONE AC POWER CORD. MAKE SURE ALL AC POWER CORDS ARE UNPLUGGED. MAKE SURE THE AC POWER CORD(S) IS / ARE UNPLUGGED BEFORE YOU OPEN THE CHASSIS, OR ADD OR REMOVE ANY NON HOT-PLUG COMPONENTS.

CAUTION!

DO NOT ATTEMPT TO MODIFY OR USE AN AC POWER CORD IF IT IS NOT THE EXACT TYPE REQUIRED. A SEPARATE AC CORD IS REQUIRED FOR EACH SYSTEM POWER SUPPLY.

CAUTION!

SOME POWER SUPPLIES IN SERVERS USE NEUTRAL POLE FUSING. TO AVOID RISK OF SHOCK USE CAUTION WHEN WORKING WITH POWER SUPPLIES THAT USE NEUTRAL POLE FUSING.

CAUTION!

SOME POWER SUPPLIES IN SERVERS USE NEUTRAL POLE FUSING. TO AVOID RISK OF SHOCK USE CAUTION WHEN WORKING WITH POWER SUPPLIES THAT USE NEUTRAL POLE FUSING.

CAUTION!

THE POWER SUPPLY IN THIS PRODUCT CONTAINS NO USER-SERVICEABLE PARTS. DO NOT OPEN THE POWER SUPPLY. HAZARDOUS VOLTAGE, CURRENT AND ENERGY LEVELS ARE PRESENT INSIDE THE POWER SUPPLY. RETURN TO MANUFACTURER FOR SERVICING.

CAUTION!

WHEN REPLACING A HOT-PLUG POWER SUPPLY, UNPLUG THE POWER CORD TO THE POWER SUPPLY BEING REPLACED BEFORE REMOVING IT FROM THE SERVER.

CAUTION!

TO AVOID RISK OF ELECTRIC SHOCK, TURN OFF THE SERVER AND DISCONNECT THE POWER CORD, TELECOMMUNICATIONS SYSTEMS, NETWORKS, AND MODEMS ATTACHED TO THE SERVER BEFORE OPENING IT.

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Power Cord Warnings

If an AC power cord was not provided with your product, purchase one that is approved for use in your country.

CAUTION!

TO AVOID ELECTRICAL SHOCK OR FIRE, CHECK THE POWER CORD(S) THAT WILL BE USED WITH THE PRODUCT AS FOLLOWS:

- Do not attempt to modify or use the AC power cord(s) if they are not the exact type required to fit into the grounded electrical outlets.
- The power cord(s) must meet the following criteria:

The power cord must have an electrical rating that is greater than that of the electrical current rating marked on the product.

CAUTION!

THE POWER CORD MUST HAVE SAFETY GROUND PIN OR CONTACT THAT IS SUITABLE FOR THE ELECTRICAL OUTLET.

CAUTION!

THE POWER SUPPLY CORD(S) IS / ARE THE MAIN DISCONNECT DEVICE TO AC POWER. THE SOCKET OUTLET(S) MUST BE NEAR THE EQUIPMENT AND READILY ACCESSIBLE FOR DISCONNECTION.

CAUTION!

THE POWER SUPPLY CORD(S) MUST BE PLUGGED INTO SOCKET-OUTLET(S) THAT IS /ARE PROVIDED WITH A SUITABLE EARTH GROUND.

System Access Warnings

CAUTION!

TO AVOID PERSONAL INJURY OR PROPERTY DAMAGE, THE FOLLOWING SAFETY INSTRUCTIONS APPLY WHENEVER ACCESSING THE INSIDE OF THE PRODUCT:

- Turn off all peripheral devices connected to this product.
- Turn off the system by pressing the power button to off.
- Disconnect the AC power by unplugging all AC power cords from the system or wall outlet.
- Disconnect all cables and telecommunication lines that are connected to the system.
- Retain all screws or other fasteners when removing access cover(s). Upon completion of accessing inside the product, refasten access cover with original screws or fasteners.
- Do not access the inside of the power supply. There are no serviceable parts in the power supply. Return to manufacturer for servicing.
- Power down the server and disconnect all power cords before adding or replacing any non hot-plug component.

When replacing a hot-plug power supply, unplug the power cord to the power supply being replaced before removing the power supply from the server.

CAUTION!

IF THE SERVER HAS BEEN RUNNING, ANY INSTALLED PROCESSOR(S) AND HEAT SINK(S) MAY BE HOT.

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CAUTION!

UNLESS YOU ARE ADDING OR REMOVING A HOT-PLUG COMPONENT, ALLOW THE SYSTEM TO COOL BEFORE OPENING THE COVERS. TO AVOID THE POSSIBILITY OF COMING INTO CONTACT WITH HOT COMPONENT(S) DURING A HOT-PLUG INSTALLATION, BE CAREFUL WHEN REMOVING OR INSTALLING THE HOT-PLUG COMPONENT(S).

CAUTION!

TO AVOID INJURY DO NOT CONTACT MOVING FAN BLADES. IF YOUR SYSTEM IS SUPPLIED WITH A GUARD OVER THE FAN, DO NOT OPERATE THE SYSTEM WITHOUT THE FAN GUARD IN PLACE.

Rack Mount Warnings

The following installation guidelines are required by UL for maintaining safety compliance when installing your system into a rack.

The equipment rack must be anchored to an unmovable support to prevent it from tipping when a server or piece of equipment is extended from it. The equipment rack must be installed according to the rack manufacturer's instructions.

Install equipment in the rack from the bottom up, with the heaviest equipment at the bottom of the rack.

Extend only one piece of equipment from the rack at a time.

You are responsible for installing a main power disconnect for the entire rack unit. This main disconnect must be readily accessible, and it must be labeled as controlling power to the entire unit, not just to the server(s).

To avoid risk of potential electric shock, a proper safety ground must be implemented for the rack and each piece of equipment installed in it.

Elevated Operating Ambient - If installed in a closed or multi-unit rack assembly, the operating ambient temperature of the rack environment may be greater than room ambient. Therefore, consideration should be given to installing the equipment in an environment compatible with the maximum ambient temperature (Tma) specified by the manufacturer.

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Reduced Air Flow - Installation of the equipment in a rack should be such that the amount of air flow required for safe operation of the equipment is not compromised.

Mechanical Loading - Mounting of the equipment in the rack should be such that a hazardous condition is not achieved due to uneven mechanical loading.

Circuit Overloading - Consideration should be given to the connection of the equipment to the supply circuit and the effect that overloading of the circuits might have on overcurrent protection and supply wiring. Appropriate consideration of equipment nameplate ratings should be used when addressing this concern.

Reliable Earthing - Reliable earthing of rack-mounted equipment should be maintained.

Particular attention should be given to supply connections other than direct connections to the branch circuit (e.g. use of power strips).

Electrostatic Discharge (ESD)

CAUTION!

ESD CAN DAMAGE DRIVES, BOARDS, AND OTHER PARTS. WE RECOMMEND THAT YOU PERFORM ALL PROCEDURES AT AN ESD WORKSTATION. IF ONE IS NOT AVAILABLE, PROVIDE SOME ESD PROTECTION BY WEARING AN ANTISTATIC WRIST STRAP ATTACHED TO CHASSIS GROUND -- ANY UNPAINTED METAL SURFACE -- ON YOUR SERVER WHEN HANDLING PARTS.

Always handle boards carefully. They can be extremely sensitive to ESD. Hold boards only by their edges. After removing a board from its protective wrapper or from the server, place the board component side up on a grounded, static free surface. Use a conductive foam pad if available but not the board wrapper. Do not slide board over any surface.

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Other Hazards

Battery Replacement

CAUTION!

THERE IS THE DANGER OF EXPLOSION IF THE BATTERY IS INCORRECTLY REPLACED. WHEN REPLACING THE BATTERY, USE ONLY THE BATTERY RECOMMENDED BY THE EQUIPMENT MANUFACTURER.

CAUTION!

DISPOSE OF BATTERIES ACCORDING TO LOCAL ORDINANCES AND REGULATIONS.

CAUTION!

DO NOT ATTEMPT TO RECHARGE A BATTERY.

CAUTION!

DO NOT ATTEMPT TO DISASSEMBLE, PUNCTURE, OR OTHERWISE DAMAGE A BATTERY.

Cooling and Airflow

CAUTION!

CAREFULLY ROUTE CABLES AS DIRECTED TO MINIMIZE AIRFLOW BLOCKAGE AND COOLING PROBLEMS. FOR PROPER COOLING AND AIRFLOW, OPERATE THE SYSTEM ONLY WITH THE CHASSIS COVERS INSTALLED. OPERATING THE SYSTEM WITHOUT THE COVERS IN PLACE CAN DAMAGE SYSTEM PARTS. TO INSTALL THE COVERS:

- Check first to make sure you have not left loose tools or parts inside the system.
- Check that cables, add-in cards, and other components are properly installed.

Attach the covers to the chassis according to the product instructions.

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Laser Peripherals or Devices

CAUTION!

TO AVOID RISK OF RADIATION EXPOSURE AND / OR PERSONAL INJURY:

- Do not open the enclosure of any laser peripheral or device.
- Laser peripherals or devices are not serviceable.

Return to manufacturer for servicing.

Use certified and rated Laser Class I for Optical Transceiver product.

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Regulatory and Compliance Information

Chapter 10

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10.1 Product Regulatory Compliance Markings

This product is marked with the following Product Certification markings:

Table 10-1: Product Regulatory Compliance Markings

REGULATORY COMPLIANCE	REGION	MARKING
cULus Listing Marks	USA / Canada	
CE Mark	Europe	
FCC Marking (Class A)	USA	This device complies with Part 15 of the FCC Rules. Operation of this device is subject to the following two conditions: (1) This device may not cause harmful interference, and (2) This device must accept any interference received, including interference that may cause undesired operation.

Table 10-1: Product Regulatory Compliance Markings (Continued)

VCCI Marking (Class A)	Japan	この装置は、クラス A 情報技術装置です。この装置を家庭環境で使用すると電波妨害を引き起こすことがあります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。VCCI-A
BSMI Certification Number & Class A Warning	Taiwan	 警告使用者： 這是甲類的資訊產品，在居住的環境中使用時，可能會造成射頻干擾，在這種情況下，使用者會被要求採取某些適當的對策
KCC	Korea	 A급 기기 (업무용 정보통신기기) 이 기기는 업무용으로 전자파적합등록을 한 기기이오니 판매자 또는 사용자는 이 점을 주의하시기 바라며, 만약 잘못판매 또는 구입하였을 때에는 가정용으로 교환하시기 바랍니다.

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Table 10-1: Product Regulatory Compliance Markings (Continued)

GOST R Marking	Russia	
ICES	Canada	This Class A digital apparatus complies with Canadian ICES-003. Cet appareil numérique de la classe A est conforme à la norme NMB-003 du Canada.
Recycling Package Mark	Other than China	

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10.1 Electromagnetic Compatibility Notices

FCC Verification Statement (USA)

This device complies with Part 15 of the FCC Rules. Operation is subject to the following two conditions: (1) this device may not cause harmful interference, and (2) this device must accept any interference received, including interference that may cause undesired operation.

This equipment has been tested and found to comply with the limits for a Class A digital device, pursuant to Part 15 of the FCC Rules. These limits are designed to provide reasonable protection against harmful interference in a residential installation. This equipment generates, uses, and can radiate radio frequency energy and, if not installed and used in accordance with the instructions, may cause harmful interference to radio communications. However, there is no guarantee that interference will not occur in a particular installation. If this equipment does cause harmful interference to radio or television reception, which can be determined by turning the equipment off and on, the user is encouraged to try to correct the interference by one or more of the following measures:

- Reorient or relocate the receiving antenna.
- Increase the separation between the equipment and the receiver
- Connect the equipment to an outlet on a circuit other than the one to which the receiver is connected.
- Consult the dealer or an experienced radio/TV technician for help.

Any changes or modifications not expressly approved by the grantee of this device could void the user's authority to operate the equipment. The customer is responsible for ensuring compliance of the modified product.

Only peripherals (computer input/output devices, terminals, printers, etc.) that comply with FCC Class A or B limits may be attached to this computer product. Operation with noncompliant peripherals is likely to result in interference to radio and TV reception.

All cables used to connect to peripherals must be shielded and grounded. Operation with cables, connected to peripherals, that are not shielded and grounded may result in interference to radio and TV reception.

Europe (CE Declaration of Conformity)

This product has been tested in accordance too, and complies with the Low Voltage Directive (73/23/EEC) and EMC Directive (89/336/EEC). The product has been marked with the CE Mark to illustrate its compliance.

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VCCI (Japan)

この装置は、情報処理装置等電波障害自主規制協議会（ＶＣＣＩ）の基準に基づくクラスＡ情報技術装置です。この装置を家庭環境で使用すると電波妨害を引き起こすことがあります。この場合には使用者が適切な対策を講ずるよう要求されることがあります。

English translation of the notice above:

This is a Class A product based on the standard of the Voluntary Control Council for Interference (VCCI) from Information Technology Equipment. If this is used near a radio or television receiver in a domestic environment, it may cause radio interference. Install and use the equipment according to the instruction manual.

BSMI (Taiwan)

The BSMI Certification Marking and EMC warning is located on the outside rear area of the product.

警告使用者：
這是甲類的資訊產品，在居住的環境中使用時，可能會造成射頻干擾，在這種情況下，使用者會被要求採取某些適當的對策

KCC (Korea)

Korean Communications Commission (KCC) Class A Statement:

A급 기기 (업무용 정보통신기기)

이 기기는 업무용으로 전자파적합등록을 한 기기이오니 판매자 또는 사용자는 이 점을 주의하시기 바라며, 만약 잘못판매 또는 구입하였을 때에는 가정용으로 교환하시기 바랍니다.

Regulated Specified Components

To maintain the UL listing and compliance to other regulatory certifications and/or declarations, the following regulated components must be used and conditions adhered to. Interchanging or use of other component will void the UL listing and other product certifications and approvals.

Updated product information for configurations can be found on the site at the following URL:

www.QuantaQCT.com

If you do not have access to the Web address, please contact your local representative.

- Add-in cards: must have a printed wiring board flammability rating of minimum UL94V-1. Add-in cards containing external power connectors and/or lithium batteries must be UL recognized or UL listed. Any add-in card containing

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modem telecommunication circuitry must be UL listed. In addition, the modem must have the appropriate telecommunications, safety, and EMC approvals for the region in which it is sold.

- Peripheral Storage Devices: must be UL recognized or UL listed accessory and TUV or VDE licensed. Maximum power rating of any one device is 19 watts. Total server configuration is not to exceed the maximum loading conditions of the power supply.

Restriction of Hazardous Substances (RoHS) Compliance

Quanta® Computer Inc. has a system in place to restrict the use of banned substances in accordance with the European Directive 2002/95/EC. Compliance is based on declaration that materials banned in the RoHS Directive are either (1) below all applicable threshold limits or (2) an approved / pending RoHS exemption applies.

RoHS implementation details are not fully defined and may change.

Threshold limits and banned substances are noted below:

- Quantity limit of 0.1% by mass (1000 PPM) for:
  - Lead
  - Mercury
  - Hexavalent Chromium
  - Polybrominated Biphenyls Diphenyl Ethers (PBDE)
- Quantity limit of 0.01% by mass (100 PPM) for:
  - Cadmium

End of Life / Product Recycling

Product recycling and end-of-life take-back systems and requirements vary by country. Contact the retailer or distributor of this product for information about product recycling and / or take-back.

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